REACH IS A PROPERTY OF THE HEAD: SCALE-COVARIANT ATTENTION FOR HIGH-RESOLUTION DIFFUSION TRANSFORMERS

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ABSTRACT

Diffusion transformers trained near one megapixel can generate 4K images without retraining once their attention is adapted. Each such method must set how far every attention head looks, its reach, at a size the model never saw. Current methods share one schedule across heads or search a setting per size. We show that reach is a property of the head: at the training resolution of FLUX.1-dev, the head explains 97.5% of the variance of reach and the denoising step 2.5%. We introduce Scale-Covariant Attention (SCA), which mixes a native-size window and a native-size grid over the whole image with one number per head. Reach emulation fits that number once at 1024^2, where full attention still works, in closed form. Inside HRDiT's sampler, SCA makes a 4096^2 image in 49.2 s, $4.1\times$ faster than AReA, and beats AReA on all ten per-image metrics at three seeds. It is statistically indistinguishable from HRDiT on three preference scores in 46% of HRDiT's time. Code: github.com/shreshthsaini/scale-covariant-attention.

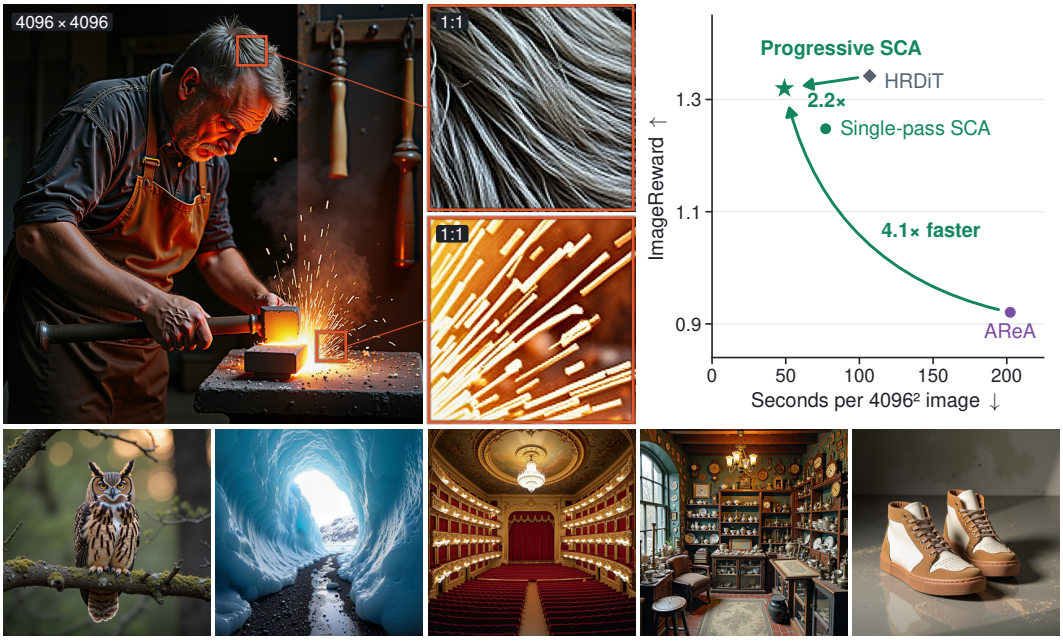


Figure 1: **Progressive SCA makes 4096^2 images with native detail, $4.1\times$ faster than AReA.** Images: FLUX.1-dev, one purpose-written prompt per category drawn with seed 0; red boxes mark native 1:1 crops at the highest edge energy. Plot: ImageReward ($n = 195$) against clean seconds per image on one GH200; arrows give clean-over-clean speedups. Figure L.3 shows more.

1 INTRODUCTION

Diffusion transformers (Peebles & Xie, 2022) such as FLUX.1 (Black Forest Labs, 2024) are trained near one megapixel. At 4096^2 an image has 16 times as many tokens, full attention costs about 256

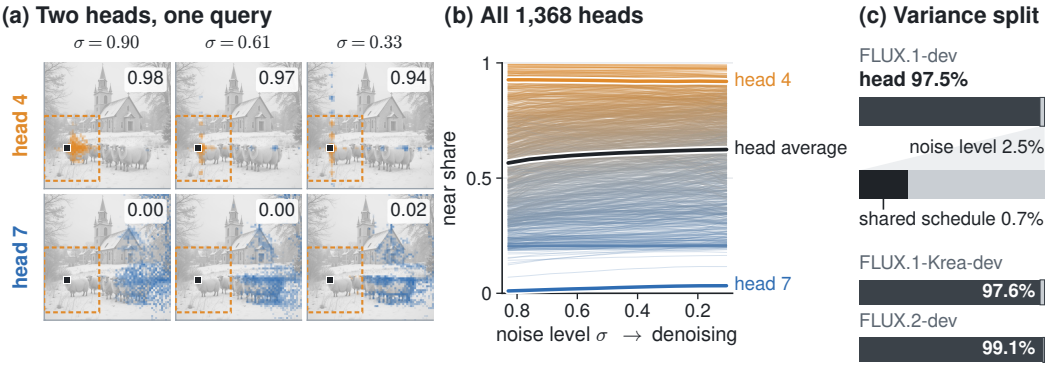


Figure 2: **Each head keeps its reach across denoising steps.** (a) Full attention of heads 4 and 7 of block 12 at one query of a 1024^2 image at three noise levels: amber within 16 tokens (dashed), blue beyond. (b) Near share of all 1,368 FLUX.1-dev heads after the first 8 steps, 32 prompts; black: head average. (c) Variance of the near share; a schedule shared by all heads can follow only the head average (Table B.1).

times as much, and every layer meets unseen position offsets. Full attention then returns stripes or a flat field (Figure L.2). Training-free methods therefore change the attention and must decide how far each head looks at the target size: its *reach*. AReA (Rajabi et al., 2026a) sees head-averaged attention narrow during denoising and gives all heads one schedule; HRDiT (Xue et al., 2026) searches per-head scopes for each target size; UltraImage (Zhao et al., 2025) types heads at run time. None of these choices can be checked at the target, where full attention fails. We ask whether reach can be measured where full attention works, at the training resolution, and reused at any size.

At the training resolution, reach is a property of the head (Figure 2). Over 32 prompts at 1024^2, we record each of FLUX.1-dev's 1,368 heads' share of image attention within 16 tokens of the query at every noise level after the first 8 steps. The head explains 97.5% of the variance of this share and the noise level 2.5%. The curve shared by all heads, all that one schedule for every head can follow, explains 0.7%. UltraImage noted that heads keep stable local and global roles (Zhao et al., 2025); we give the first decomposition of reach variance into heads and steps.

We introduce Scale-Covariant Attention (SCA; Figure 3). Each head mixes a native-size window around the query, for detail, and a native-size grid of keys over the whole image, for layout, with one number: its reach $\pi_h \in [0, 1]$. *Reach emulation* fits π_h at 1024^2: shrinking both views by the target's image-to-window ratio r reproduces the target's proportions, and the model's own full attention there is a valid reference. The best weight has a closed form, and one pass fills a *reach table* for every size; the table repeats on disjoint prompt halves (correlation 0.989). To our knowledge, no prior method measures per-head reach once at the training resolution and reuses it at any size.

Inside HRDiT's three-stage sampler, progressive SCA makes a 4096^2 FLUX.1-dev image in 49.2 s on one GH200, $4.1\times$ faster than AReA and $2.2\times$ faster than HRDiT (Figure 1). It beats AReA on all ten per-image metrics at each of three seeds and on HRGen-1K, with ImageReward rising from 0.92 to 1.32 ($+0.40$ $[0.32, 0.49]$, $n = 195$, seed 0). Against HRDiT, its ImageReward, HPSv2.1, and PickScore are statistically indistinguishable (every 95% interval includes zero) in 46% of HRDiT's time. A single-pass mode is $2.63\times$ faster than AReA and keeps the most crop-level detail; the progressive schedule costs patch-level fidelity, where HRDiT and single-pass SCA lead.

- We show that reach is a property of the head: at the training resolution, the head explains 97.5% to 99.1% of its variance in three FLUX models (Section 2).
- We introduce SCA, which mixes a native-size window and grid with one weight per head, and reach emulation, which fits that weight once at 1024^2 in closed form (Section 3).
- We show that progressive SCA makes a 4096^2 image in 49.2 s, $4.1\times$ faster than AReA and better on all ten per-image metrics at three seeds, and that per-head weights carry the gain (Section 4).

2 Reach is a property of the head

A head that attends near the query at one denoising step does so at the others, and a head that looks across the image keeps looking across it.

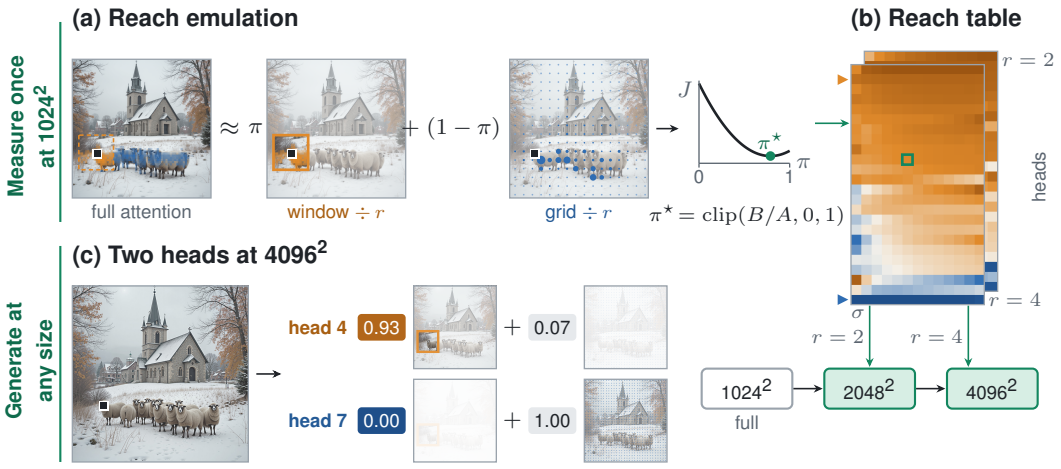


Figure 3: **Reach is measured once at 1024^2 and read at any size.** (a) Full attention of one head at 1024^2 is fitted by a mix of its window and grid views shrunk by r; the fit has a closed form. (b) The weight fills one cell of the reach table: heads by noise level, one sheet per r. (c) At 4096^2, head 4 of Figure 2 mixes mostly its window and head 7 only its grid; each sampler stage reads the table at its r. Figure C.1 zooms in.

Reach. FLUX.1 applies joint attention to text and image tokens in 57 blocks of 24 heads and is trained near 1024^2 pixels, a grid of 64×64 image tokens. For one head and a query i, split the keys into a near set around the query and a far set. The output splits exactly as $O_i^{\text{full}} = m_i\, O_i^{\text{near}} + (1 - m_i)\, O_i^{\text{far}}$, where O_i^{near} and O_i^{far} attend only to their own set and m_i is the near set's softmax mass (Lemma 1). This near share is the head's reach at that query: near 1 for a head that looks near, near 0 for one that looks far (Figure 2a). We record it within 16 tokens for 256 seeded queries per block and step, over 32 purpose-written prompts at 1024^2 outside the evaluation sets (Appendix K).

The head sets the reach. After the first 8 steps (noise level $\sigma < 0.89$), the head explains 97.5% of the variance of this share over all heads and noise levels. The noise level within a head explains 2.5% (Figure 2b,c). The head-averaged curve rises from 0.57 to 0.62, the narrowing AReA reports, yet explains only 0.7%; no schedule shared by all heads can explain more. FLUX.1-Krea-dev (Black Forest Labs & Krea AI, 2025) and FLUX.2-dev (Black Forest Labs, 2025) give 97.6% and 99.1% for the head (Table B.1). One number per head thus captures nearly all of the variation and, because it barely moves with the step, can be measured once.

3 Scale-Covariant Attention

3.1 Two native-size views

On its native grid the model saw position offsets of at most 63 tokens per axis and 64^2 image keys per query. SCA keeps both at any size with two views: a near view O_i^{loc}, a native-size window around the query, and a far view O_i^{glob}, a native-size grid over the whole image. An image query i of head h mixes them as $O_i = \pi_h O_i^{\text{loc}} + (1-\pi_h) O_i^{\text{glob}}$, with the head's reach $\pi_h \in [0, 1]$ (Figure 3c).

Setting. A multimodal diffusion transformer such as FLUX.1 concatenates T text tokens $\mathcal{T}$ with $N = HW$ image tokens on the grid $\mathcal{G} = [H] \times [W]$, where $[n] = \{0, \ldots, n-1\}$. At 1024^2 the grid is $n_0 \times n_0$ with $n_0 = 64$. Rotary embeddings (Su et al., 2021) give token j a position $p_j \in \mathbb{Z}^2$ (text tokens at 0) and a rotation R_p with $R_p^\top R_{p'} = R_{p'-p}$. For one head, a pre-rotation query q placed at u, and a set $\mathcal{K}$ of (key, value, key position) triples, attention is

$$\mathcal{A}(q, u; \mathcal{K}) \;=\; \textstyle\sum_{(k,v,u') \in \mathcal{K}} \operatorname{softmax}_{\mathcal{K}}\big(q^\top R_{u'-u}\, k/\sqrt{d}\big)\; v, \tag{1}$$

where d is the head dimension. Full attention O_i^{full} attends to all keys at their own positions; at a target grid $H = s_H n_0$, $W = s_W n_0$ (s when equal) it meets offsets up to s times the trained range.

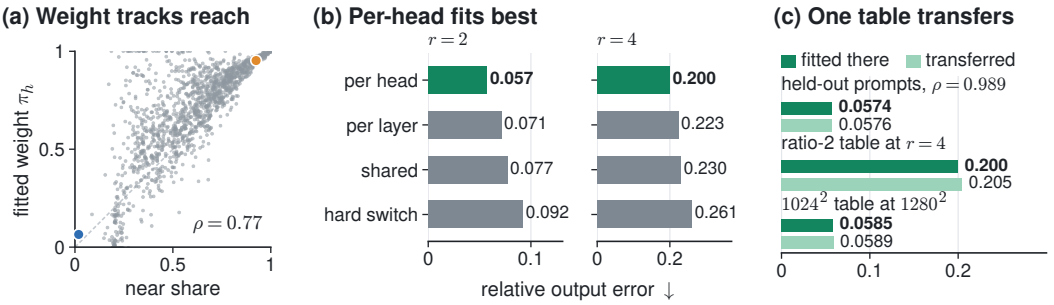


Figure 4: **The fitted weight measures reach, fits full attention best, and transfers.** (a) π_h against the near share of 1,368 heads at $\sigma = 0.62$; amber and blue: heads 4 and 7. (b) Relative output error of four mixing rules, 32 prompts. (c) A table used where it was not fitted (light) against the table fitted there (dark); held-out: each prompt half's table on the other half. Appendix B expands each panel.

Near view. Image queries form tiles of $t \times t$ tokens ($t = 8$), and the window $\mathcal{W}_w(i)$ of query i is the $w \times w$ block centred on its tile and clamped to the grid (Appendix C). Its keys are the text tokens, a bridge set $\mathcal{B}$ of image tokens, and the window, at their original positions:

$$\mathcal{K}_i^{\mathrm{loc}}(w) = \{(k_j, v_j, 0)\}_{j\in\mathcal{T}} \cup \{(k_j, v_j, p_j)\}_{j\in\mathcal{B}\cup\mathcal{W}_w(i)}, \qquad O_i^{\mathrm{loc}}(w) = \mathcal{A}\big(q_i,\, p_i;\, \mathcal{K}_i^{\mathrm{loc}}(w)\big), \quad (2)$$

where $\mathcal{K}_i^{\mathrm{loc}}(w)$ is the local key set; one FlexAttention call computes it (Dong et al., 2024). The bridge lets each window see the whole image. In each cell c of an 8×8 partition, it keeps the 16 tokens whose pre-rotation keys lie farthest from the mean image key (AReA's rule; Rajabi et al., 2026a):

$$\mathcal{B} = \bigcup_c \mathrm{top}_{16}\big\{\textstyle\sum_h \|k_j^h - \bar{k}^h\|^2 : j \in c\big\}, \quad (3)$$

where $\bar{k}^h$ is the mean image key of head h. With $w = n_0$ the window is an exact native-size sliding window, for content of fixed token size, such as edges and texture.

Far view. Let $\mathcal{G}_0 = [H/s_H] \times [W/s_W]$ be the $n_0 \times n_0$ grid of blocks. Block-centre subsampling keeps for each block b the token at $\kappa(b) = (s_H b_1 + \lfloor s_H/2 \rfloor,\, s_W b_2 + \lfloor s_W/2 \rfloor)$, and $c(p) = (\lfloor p_1/s_H \rfloor, \lfloor p_2/s_W \rfloor)$ maps a position to its block:

$$\mathcal{K}^{\mathrm{glob}} = \{(k_j, v_j, 0)\}_{j\in\mathcal{T}} \cup \{(k_{\kappa(b)}, v_{\kappa(b)}, b)\}_{b\in\mathcal{G}_0}, \qquad O_i^{\mathrm{glob}} = \mathcal{A}\big(q_i,\, c(p_i);\, \mathcal{K}^{\mathrm{glob}}\big), \quad (4)$$

where text queries sit at $c(p_i) = 0$. This is AReA's first phase at pooling factor s, for content that spans a fixed fraction of the image, such as layout. Every offset of the grid and of the native window stays in the trained range, and the cost is linear in N (Propositions 2 and 3). At 4096^2, SCA scores 6.9 times fewer query-key pairs than full attention (Table C.1).

3.2 Reach Emulation

Full attention fails at the target, so nothing there can score π_h; reach emulation scores it on a miniature of the target at 1024^2, against the model's own full attention (Figure 3a). On the native $n \times n$ grid ($n = 64$), emulated ratio r uses subsampling factor r, window side n/r, and tile side $\mathrm{round}(tn/(rn_0))$, so the image-to-window ratio is r, as at a target r times larger. The sampler runs unmodified full attention (Algorithm 1).

Closed-form fit. For block l, head h, noise bin β, and ratio r, we accumulate three sums over prompts, steps with $\sigma \in \beta$, image queries, and channels:

$$A = \sum \|O^{\mathrm{loc}} - O^{\mathrm{glob}}\|^2, \quad B = \sum \langle O^{\mathrm{full}} - O^{\mathrm{glob}},\, O^{\mathrm{loc}} - O^{\mathrm{glob}} \rangle, \quad E = \sum \|O^{\mathrm{full}} - O^{\mathrm{glob}}\|^2, \quad (5)$$

where O^{loc} and O^{glob} are the emulated views without the bridge. A weight π then has squared error $J(\pi) = E - 2\pi B + \pi^2 A$.

Proposition 1 (Closed-form fit and transfer cost)**.** *In each cell (l, h, β, r), J is minimized on $[0, 1]$ by $\pi^\star = \mathrm{clip}(B/A, 0, 1)$, with $\pi^\star = 1$ if $A = 0$. Any other weight $\pi' \in [0, 1]$ constant in the cell has $J(\pi') - J(\pi^\star) \geq A(\pi' - \pi^\star)^2$, with equality when $0 < \pi^\star < 1$. This covers weights fitted at another ratio or resolution, per layer, or shared, and a hard switch.*

One pass collects ratios 2 and 4 and fills the reach table $\pi_{lh}(\sigma; r)$, and its sums score every other rule without another pass. We assume that the weight depends only on r, which is scale covariance.

Algorithm 1: **Reach emulation**, once per checkpoint

```
# Input: emulation prompts, model at 1024^2
# Output: reach table π[r, l, h, β]; App. C.2
def emulate(q, k, v, l, sigma):  # attn hook
  O_full = attn(q, k, v)         # Eq. 1, all keys
  beta = noise_bin(sigma)        # 14 bins, App. B
  for r in (2, 4):               # r times larger
    O_loc = window(q, k, v, n // r)  # Eq. 2
    O_glob = grid(q, k, v, stride=r) # Eq. 4
    D, F = O_loc - O_glob, O_full - O_glob
    A[r, l, :, beta] += sum(D * D)   # Eq. 5
    B[r, l, :, beta] += sum(F * D)   # per head
  return O_full                  # sampler as is
pi = clip(B / A, 0, 1)           # Prop. 1
```

Algorithm 2: **SCA attention**, one call at any size

```
# Input: q, k, v (T text, H x W image), l, σ, π
# Output: all T + HW outputs; full code App. C.2
Kg = text + centres(k, v, H // n0)  # Eq. 4
O_glob = attn(q, Kg, at=c(p))  # coarse grid
if sigma > sigma_g:            # opening steps
  return O_glob                # global only
w = window_side(sigma)         # 128 to 80, n0
Kl = text + bridge(k) + window(w)  # Eq. 2, 3
O_loc = flex_attention(q, Kl)  # App. C
r = max(H, W) / w              # window ratio
pi = interp(table[l], sigma, r)  # Eq. 7
O = pi * O_loc + (1 - pi) * O_glob[img]  # Eq. 6
return [O_glob[txt], O]        # text: global
```

Table 1: **Main comparison at 4096^2.** FLUX.1-dev, 195 Aesthetic-4K prompts, seed 0; HPSv2.1 $\times 100$; M_c: MUSIQ on 1024^2 crops; FID_p: patch FID†. Time: clean batch-1 median on one GH200 (‡campaign mean) and its ratio to progressive SCA's of the same kind. $+/-$: per-image metrics, of ten, on which progressive SCA is significantly better / worse than the row after Holm's correction. Bold: best, underline: second per group; shaded: ours; Table E.1 has every metric. Key result: progressive SCA is the fastest method, and no baseline is significantly better on any per-image metric.

	Time per image		Preference ↑			Detail ↑	Set level ↓		Ten metrics
Method	s ↓	vs. ours	IR	HPSv2.1	Pick	M_c	FID	$FID_p^\dagger$	$+/-$
Single pass, AReA's sampler (28 steps)									
Full attention	186.5	3.8×	−2.18	7.60	17.94	33.1	312.4	252.2	10 / 0
DyPE	195.9‡	3.9×‡	−0.10	20.55	20.51	50.3	205.5	171.9	10 / 0
SEGA	190.4‡	3.8×‡	0.56	24.34	21.64	50.3	164.9	125.8	10 / 0
AReA	202.4	4.1×	0.92	27.16	22.36	50.4	145.8	102.9	10 / 0
Single-pass SCA (ours)	**77.1**	1.6×	**1.25**	**29.73**	**23.12**	**60.1**	**140.2**	**98.4**	7 / 2
Multi-stage, own recipes (UltraImage 30+30 steps; HRDiT and ours 30+17+10)									
UltraImage	308.5‡	6.1×‡	1.28	30.77	23.33	41.7	142.0	109.5	9 / 0
HRDiT	107.0	2.2×	**1.34**	30.92	**23.46**	53.5	**138.4**	**103.0**	2 / 0
Progressive SCA (ours)	**49.2**	1.0×	1.32	**31.02**	23.41	**54.1**	141.9	107.7	*ref.*

One table serves every size. Tables fitted on disjoint prompt halves correlate at 0.989 at ratio 2 and 0.986 at ratio 4, and the fitted weight tracks each head's measured near share ($\rho = 0.77$; Figure 4). In relative output error, $\sum J / \sum \|O^{\text{full}}\|^2$, the ratio-2 table applied at ratio 4 scores 0.205 against 0.200 for the ratio-4 table, below every coarser rule fitted at ratio 4. The 1024^2 table scored at 1280^2 gives 0.0589 against 0.0585, although the two resolutions emulate ratio 2 with windows of 32 and 40 tokens. The weights depend on the ratio, as scale covariance assumes.

3.3 Sampling with the reach table

Progressive SCA. A sampler reads the table at every size it visits (Algorithm 2). Our main configuration keeps HRDiT's three-stage sampler (Xue et al., 2026) and replaces only its four attention changes (NTK-scaled RoPE, a proportional attention scale, per-head key pruning, and position alignment) with SCA (Figure 3c). HRDiT denoises 30 steps at 1024^2 with the model's own attention, then re-noises the upsampled image and runs the last 17 steps of its schedule at 2048^2 and the last 10 at 4096^2 (Algorithm 5). These stages read the table at ratios 2 and 4, from $\sigma = 0.81$ and 0.61 on.

Single-pass mode. SCA also samples at the target in one pass of 28 steps. A window schedule $w(\sigma)$ sets the ratio $r(\sigma) = \max(H, W)/w(\sigma)$, and an image query takes

$$O_i \;=\; \pi\, O_i^{\text{loc}}\big(w(\sigma)\big) + (1-\pi)\, O_i^{\text{glob}} \;\text{ if } \sigma \le \sigma_g, \qquad O_i = O_i^{\text{glob}} \;\text{ if } \sigma > \sigma_g, \tag{6}$$

where $\pi = \pi_{lh}(\sigma; r(\sigma))$, $\sigma_g = 0.89$, and text queries take O_i^{glob}. Between ratios 2 and 4, π is linear in $\log r$, and in σ it is interpolated between occupied noise bins:

$$\pi_{lh}(\sigma; r) = (1-\lambda)\,\pi_{lh}(\sigma; 2) + \lambda\,\pi_{lh}(\sigma; 4), \qquad \lambda = \text{clip}(\log_2 r - 1,\, 0,\, 1), \tag{7}$$

where λ is the weight of the ratio-4 table. The global view runs alone for the first 8 of 28 steps, as in AReA, because output error cannot see layout (Section 4.3). Windows of 128, 112, 96, and 80

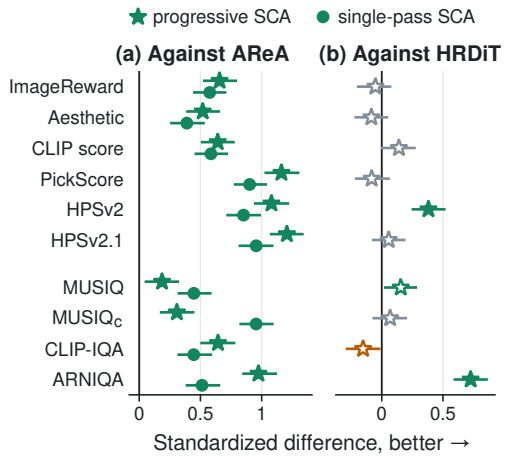


Figure 5: **Both modes beat AReA on all ten per-image metrics; progressive SCA is worse than HRDiT on none.** Paired differences ($n = 195$) over the SD of the per-prompt differences, 95% intervals: (a) both modes minus AReA, (b) progressive SCA minus HRDiT. Filled: significant after Holm's correction; open: before it only (Table E.4).

Table 2: **Generality.** Paired differences at 4096^2 (HPS: HPSv2.1×100), coloured as in Section 4. Holm +/−: per-image metrics, of ten, on which the first-named method is significantly better / worse. Table G.1 gives every metric with intervals. Key result: no setting is significantly worse on any metric.

		Paired difference ↑			Holm
Setting	n	IR	HPS	Pick	+/−
Progressive SCA − AReA					
Seed 0	195	+0.40	+3.86	+1.05	10/0
Seed 1	195	+0.37	+3.67	+1.00	10/0
Seed 2	195	+0.35	+3.61	+0.99	10/0
HRGen-1K	1000	+0.90	+6.59	+1.40	10/0
Single-pass SCA − AReA					
Seed 0	195	+0.33	+2.56	+0.76	10/0
Seed 1	195	+0.31	+2.49	+0.77	10/0
Seed 2	195	+0.26	+2.53	+0.74	10/0
HRGen-1K	1000	+0.76	+4.60	+1.09	10/0
FLUX.1-Krea	195	+0.02	+0.91	+0.17	7/0
Without the schedule − full attention					
FLUX.2-dev	50	+0.46	+1.84	+0.56	9/0

tokens then take one step each before the native $w = n_0$. The gate, the window sizes, and the 1,024 bridge tokens are set by hand; only the weights are measured (Table C.2). With $w = n_0$ right after the gate and no bridge, the operator is SCA without the schedule.

4 Experiments

Data. We follow the field's protocol at 4096^2: the 195-prompt Aesthetic-4K split with its 4K references (Zhang et al., 2025a), as in AReA, DyPE, TIDE, and SigMa (Rajabi et al., 2026a; Issachar et al., 2025; Liu et al., 2026; Zhao et al., 2026), and the 1000 HRGen-1K prompts. Seed labels 0, 1, and 2 are shared by all methods. Four prompts (boats, Moai, woman and bear, underwater surfer) were used in development, which favours SCA slightly; a fixed 46-prompt subset chosen before any ablation serves the ablations.

Models and baselines. The main model is FLUX.1-dev, and every baseline runs from its released code (Table 1). Single-pass SCA, full attention, AReA, DyPE, SEGA (Rajabi et al., 2026b), and SigMa share AReA's sampler, 28 Euler steps at guidance 4.5 (Appendix D.7); our AReA run reproduces its published FLUX.1-Krea-dev scores within 0.3 (Table D.3). Progressive SCA, UltraImage, and HRDiT use multi-stage recipes (Appendix D.8). One reach table per checkpoint (32 prompts for FLUX.1-dev, 16 for the others) serves every size and both modes (Table C.3).

Metrics and timing. The ten per-image metrics are ImageReward, HPSv2, HPSv2.1, PickScore, CLIP score, aesthetic score (Xu et al., 2023; Wu et al., 2023; Kirstain et al., 2023; Radford et al., 2021; Schuhmann, 2022), MUSIQ on the full image and over four 1024^2 crops (MUSIQ$_c$), CLIP-IQA, and ARNIQA (Ke et al., 2021; Wang et al., 2022; Agnolucci et al., 2024). FID and KID use all 195 references at 299^2, as in AReA (Heusel et al., 2017); patch FID and KID use the same crops, and patch FID is our own definition. Intervals are 95% paired percentile bootstraps over prompts (Appendix D.3). A method beats another on a metric when that interval excludes zero, which tables mark blue in the row's favour and orange against it. Holm's correction over the ten per-image metrics leaves every "all ten" claim intact (Table E.4). Clean latency is the batch-1 median of three prompts on one idle GH200 after a warm-up image, a point estimate, and a speed ratio divides two times of the same kind (Appendix D.4).

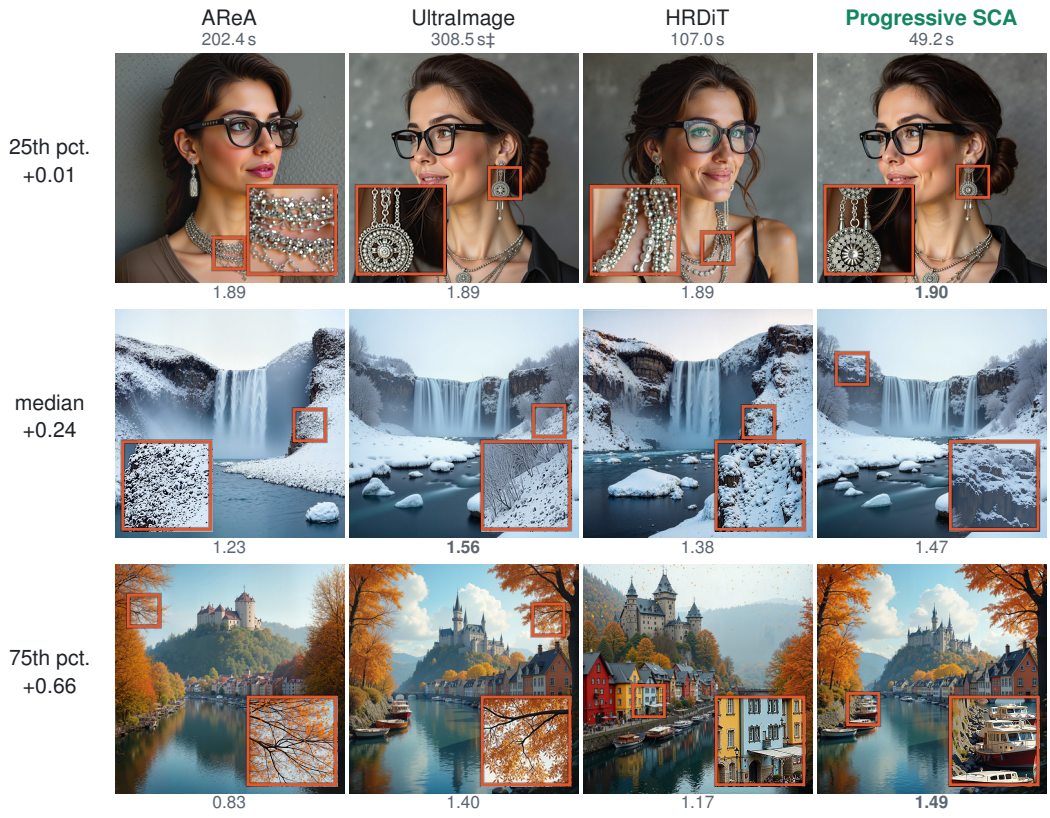


Figure 6: **Progressive SCA's gain over AReA is in the whole image, in a quarter of AReA's time, and it is level with HRDiT.** Rows: the evaluation prompts at the 25th, 50th, and 75th percentile of the paired ImageReward difference, progressive SCA minus AReA (seed 0, $n = 195$, mean $+0.40$). Numbers: each image's ImageReward, best bold; insets: each image's 512×512 window of highest edge energy at native pixels; heads: seconds per image (‡campaign mean). Over all prompts, ImageReward differs from HRDiT's by -0.02 $[-0.09, 0.03]$. Figures L.3 and L.4 show more.

4.1 Main results

Progressive SCA beats AReA on all ten per-image metrics at $4.1\times$ its speed. ImageReward rises from 0.92 to 1.32 ($+0.40$ $[0.32, 0.49]$), PickScore by $+1.05$ $[0.93, 1.19]$, and HPSv2.1 by $+3.86$ $[3.41, 4.30]$ ($n = 195$; Table 1, Figure 5). Seeds 1 and 2 repeat the result on all ten (ImageReward $+0.37$ $[0.29, 0.46]$ and $+0.35$ $[0.27, 0.45]$), and so does HRGen-1K ($+0.90$ $[0.84, 0.96]$, $n = 1000$; Table 2). Progressive SCA also beats DyPE, SEGA, and full attention on all ten and UltraImage on nine (Table E.2). Figure 6 shows that the gain lies in the whole image, on the prompts at the quartiles of the ImageReward difference.

Against HRDiT, progressive SCA is statistically indistinguishable on three preference scores in 46% of its time. Both follow HRDiT's schedule, guidance, and seeds and differ in attention, precision (fp16 for HRDiT's runner, bf16 for ours), and runner. The differences are -0.02 $[-0.09, 0.03]$ in ImageReward, $+0.10$ $[-0.14, 0.35]$ in HPSv2.1, and -0.05 $[-0.14, 0.04]$ in PickScore ($n = 195$); intervals that include zero do not establish equivalence. After Holm's correction, progressive SCA remains better on HPSv2 ($+0.38$ $[0.24, 0.51]$) and ARNIQA ($+2.9$ $[2.3, 3.4]$) and is worse on none. MUSIQ and CLIP-IQA differ only before the correction (Table E.4). Crop detail is statistically indistinguishable from HRDiT's (MUSIQ_c $+0.6$ $[-0.7, 1.8]$; Figure F.1).

Single-pass SCA is $2.63\times$ faster than AReA and keeps the most crop-level detail. It beats AReA on 12 of 15 metrics, all ten per-image ones among them, with ImageReward $+0.33$ $[0.25, 0.41]$ and FID from 145.8 to 140.2 (-5.6 $[-10.6, -0.4]$). KID, patch FID, and IS are unresolved (Table E.3). DyPE and SEGA trail it on all ten per-image metrics, and SigMa on every one it was scored on (Tables E.5 and H.3).

Table 3: **Ablations at 4096^2.** Each group opens with its shaded per-head reference (absolute values), and rows below give the variant minus it (views: minus the first shaded row), coloured as in Section 4. 46-prompt FID_p values are point estimates. $+/-$: per-image metrics, of ten, on which per-head weights are significantly better / worse after Holm's correction. Tables E.6 and E.7 give every metric. Key result: per-head weights beat coarser ones on per-image metrics in both samplers but have higher patch FID inside the progressive one.

	Preference ↑			Detail ↑	Set level ↓	Ten metrics
Variant	IR	HPSv2.1	Pick	M_c	FID_p	$+/-$
Reach weights, single pass without the schedule (46 prompts)						
per-head π_h (ours)	1.12	29.98	22.96	62.5	168.0	*ref.*
shared $\pi(\sigma)$	−0.09	−1.83	−0.38	−10.4	−1.3	9 / 0
per-layer $\pi_\ell(\sigma)$	−0.05	−0.70	−0.16	−2.4	−0.3	5 / 0
Reach weights, progressive SCA (195 prompts)						
per-head π_h (ours)	1.32	31.02	23.41	54.1	107.7	*ref.*
shared $\pi(\sigma)$	−0.01	−0.38	−0.07	−4.3	−7.3	10 / 0
per-layer $\pi_\ell(\sigma)$	−0.01	−0.39	−0.07	−2.8	−6.4	10 / 0
window only ($\pi = 1$)	−0.02	−0.36	−0.14	+0.5	+6.0	6 / 1
Views, single pass without the schedule (46 prompts; against the first shaded row)						
no opening	−1.33	−4.07	−2.18	+2.5	+34.6	6 / 2
local only	−0.36	−2.86	−1.20	−0.3	+24.2	7 / 0
global only	−0.40	−5.66	−1.44	−11.7	+69.0	10 / 0
mean pooling	−1.52	−9.75	−2.60	−4.2	+16.6	8 / 0

The progressive schedule trades patch-level fidelity for preference. Against single-pass SCA, it gains ImageReward ($+0.07$ $[0.01, 0.13]$) but loses MUSIQ_c (-6.0 $[-7.3, -4.6]$) and patch FID ($+9.2$ $[5.1, 13.2]$). Its patch FID also exceeds HRDiT's ($+4.6$ $[0.6, 7.7]$) and, at two of three seeds, than AReA's (Tables E.2 and F.1). Its MUSIQ_c still beats AReA's ($+3.7$ $[2.0, 5.4]$).

4.2 Generality, other sizes, and latency

Time grows linearly with pixel count. Single-pass SCA is faster than every baseline we timed at every size (Figure 7). Fitting clean seconds $\propto N^b$ gives $b = 0.99$ against 1.12 for AReA, and at 8192^2 SCA takes 309.7 s against AReA's 952.0 s (Table E.8). Quality depends on the aspect ratio: at 2048×6144 single-pass SCA beats AReA on seven of 15 metrics, but at 2048×4096 it wins three and loses three (Table G.3).

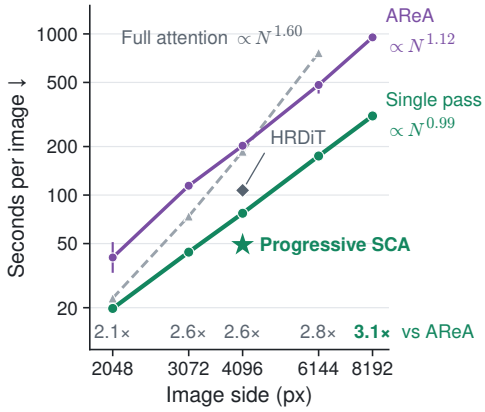


Figure 7: **Single-pass SCA's time grows linearly with pixel count.** Clean batch-1 seconds per image on one GH200; labels give b of seconds $\propto N^b$, the bottom row AReA's time over single-pass SCA's. Table E.8 lists every time.

Gains hold across seeds, prompt sets, and checkpoints. Single-pass SCA's gains over AReA hold at seeds 1 and 2 and on HRGen-1K (Table 2). Reach tables fitted on 4 to 16 prompts move ImageReward by at most 0.012 without the schedule (Table H.1). On FLUX.1-Krea-dev, AReA's own checkpoint, single-pass SCA wins seven of ten per-image metrics and loses none. Its ImageReward is unresolved ($+0.020$ $[-0.020, 0.063]$) and its patch FID is worse ($+3.4$ $[1.6, 7.1]$; Table G.2). On FLUX.2-dev, SCA without the schedule beats full attention on nine of ten.

4.3 The measured reach carries the gain

Per-head weights beat coarser ones in images (Table 3). At 4096^2, one shared weight per noise level changes ImageReward by -0.09 $[-0.17, -0.01]$ and MUSIQ_c by -10.4, and per-layer weights still lose five of ten per-image metrics. Inside the progressive sampler, per-head weights still win all ten per-image metrics against both after Holm's correction, by smaller margins, but have higher patch FID.

Output error shows the same order. In emulation at ratio 4, the per-head mix has relative error 0.200, per-layer weights 0.223, one weight per noise level 0.230, and the best hard switch 0.261. Ratio 2 keeps the order (Figure 4b). By Proposition 1, a coarser rule pays at least $\sum A(\pi' - \pi^\star)^2$, which grows with the spread of reach among merged heads.

Both views and the opening are needed. Local-only attention in the last 20 steps changes ImageReward by -0.36 $[-0.52, -0.21]$, global-only attention loses all ten per-image metrics, and mean pooling in place of block centres changes ImageReward by -1.52. Output error cannot see layout. Fitted on it alone, the best hard switch is local attention from the first step, yet dropping the opening changes ImageReward by -1.33 $[-1.60, -1.06]$. The global opening, or the 1024^2 stage of progressive SCA, therefore sets the layout. Without the hand-set schedule and bridge, MUSIQ_c changes by $+1.7$ and no other column moves significantly (Table E.6).

5 Related work

Training-free high-resolution sampling. Methods alter convolutions, spectra, or attention temperature (He et al., 2023; Zhang et al., 2023; Huang et al., 2024; Jin et al., 2023), fuse global and patch paths (Du et al., 2023; Yang et al., 2024) or split them by frequency (Qiu et al., 2024), add native-resolution guidance (Du et al., 2024; Bu et al., 2025; Koh et al., 2025), or adapt RoPE (Zhuo et al., 2024; Issachar et al., 2025; Zhao et al., 2026; Rajabi et al., 2026b; Liu et al., 2026). AReA (Rajabi et al., 2026a) sets one schedule for all heads by hand, in our notation a window weight of 0 and then 1 for every head; SCA borrows its opening, block-centre keys, and bridge selection (Table D.2). ResDiT also mixes position-scaled global and native-size patch attention (Ma et al., 2025).

Head-specific attention. HRDiT searches per-head scopes under a cost budget for each target size (Xue et al., 2026), and UltraImage types heads by entropy at run time and noted stable local and global head roles (Zhao et al., 2025). Sparse attention varies windows, spans, or key scales per layer or head in diffusion transformers (Yuan et al., 2024; Xi et al., 2025; Zhang et al., 2025b), language models (Fu et al., 2024; Xiao et al., 2024), and vision transformers (Pan et al., 2022). These settings are fitted against dense attention at the size where they run, which at 4096^2 is the failure. Head roles, two branches (our local branch is neighbourhood attention; Hassani et al., 2022), a global opening, and shrinking windows have precedents.

6 Limitations and conclusion

Limitations. Progressive SCA's patch FID is higher than HRDiT's and single-pass SCA's, and inside its sampler coarser weights have lower patch FID than per-head weights. In single-pass mode, small additional figures occasionally appear in scenes with people or animals; the progressive configuration reduces this to the level of the multi-stage baselines. The opening, window sizes, and bridge are set by hand, and the window sizes were chosen on the 46-prompt subset, part of the evaluation set. Progressive SCA depends on HRDiT's sampler, which does not support FLUX.1-Krea-dev, so the 6144^2, 8192^2, FLUX.2-dev, and sensitivity results use SCA without the schedule. Reach emulation omits the bridge and sees 16 times fewer keys per view than the target, so the deployed operator's output error is untested. All metrics are automatic.

Conclusion. Reach is a property of the head, so SCA fits it once per head at the training resolution, where full attention is still correct, and reuses it at any size. Inside HRDiT's sampler it is $4.1\times$ faster than AReA at 4096^2, better on all ten per-image metrics at three seeds, and statistically indistinguishable from HRDiT on three preference scores in 46% of HRDiT's time. Code: github.com/shreshthsaini/scale-covariant-attention.

REPRODUCIBILITY STATEMENT

Section 4 states the protocol, and Appendix D defines every metric, interval (Appendix D.3), and timing rule (Appendix D.4). Appendix B gives the reach-emulation geometry, statistics, and noise bins, and Appendix K the emulation prompts. Algorithms 1 and 2 give the procedures, and Appendix C.2 gives them in full together with progressive SCA (Algorithm 5). Appendix F gives the progressive protocol and every paired comparison of progressive SCA. Appendix C.5 lists every hyperparameter, including the window schedule, and Appendix C the kernel layout, numerical checks, and position rounding. Appendices D.7 and D.8 give each baseline's code revision, recipe, and seeds, and Appendix I the hardware and logged compute. Table I.1 maps every table and figure to the script and results file that produce it; code, reach tables, per-image metrics, and paired-difference files will be released with the code (link in the abstract).

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APPENDIX

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A PROOFS

This appendix proves the statements that Sections 2 and 3 use. Lemma 1 is the split behind the definition of reach. Propositions 2 and 3 give the trained position range and the cost of the two views (Section 3.1). The proof of Proposition 1 gives the closed-form fit and the price of any coarser rule (Section 3.2). Remark 1 treats text-image offsets.

Lemma 1 (Softmax split)**.** *Let the keys of a query be partitioned into a near set $\mathcal{K}^{\text{near}}$ and a far set $\mathcal{K}^{\text{far}}$, with logits ℓ_j. Then $O^{\text{full}} = m\,O^{\text{near}} + (1-m)\,O^{\text{far}}$ with $m = \sum_{j\in\mathcal{K}^{\text{near}}} e^{\ell_j} / \sum_j e^{\ell_j}$ and $O^{\text{near}}, O^{\text{far}}$ the softmax outputs restricted to $\mathcal{K}^{\text{near}}$ and $\mathcal{K}^{\text{far}}$.*

Proof. Write $Z^{\text{near}} = \sum_{j\in\mathcal{K}^{\text{near}}} e^{\ell_j}$, so that $Z^{\text{near}} O^{\text{near}} = \sum_{j\in\mathcal{K}^{\text{near}}} e^{\ell_j} v_j$, and likewise for the far set. Then $O^{\text{full}} = (Z^{\text{near}} O^{\text{near}} + Z^{\text{far}} O^{\text{far}})/(Z^{\text{near}} + Z^{\text{far}})$, and $Z^{\text{near}}/(Z^{\text{near}} + Z^{\text{far}}) = m$. □

The lemma motivates the two-branch form; SCA's key sets are not a partition. Block centres can fall inside the window, and text keys enter both branches. A bridge token inside the window appears twice in the local softmax, which doubles its weight there.

Proposition 2 (Trained position range)**.** *Assume exact integer positions and $t \le n_0 \le \min(H, W)$. Every query-key offset of the global operator, and every offset between an image query and a key of its native window $\mathcal{W}_{n_0}(i)$ (bridge keys and wider windows excluded), is at most $n_0 - 1$ per axis. Each global-operator query sees the native $T + n_0^2$ keys.*

Proof of Proposition 2. Global operator. Image queries sit at $c(p_i) \in \mathcal{G}_0 = [n_0]^2$ and text queries at $0 \in \mathcal{G}_0$; subsampled keys sit at $b \in \mathcal{G}_0$ and text keys at 0. Any two points of $[n_0]^2$ differ by at most $n_0 - 1$ per axis. The key count is $T + |\mathcal{G}_0| = T + n_0^2$.

Local operator. Fix one axis of length $H \geq w$ and a tile $[t\tau,\, t\tau + t)$ with $t \leq w$. The unclamped start $a' = t\tau + \lfloor t/2 \rfloor - \lfloor w/2 \rfloor$ satisfies $a' \leq t\tau$ because $\lfloor t/2 \rfloor \leq \lfloor w/2 \rfloor$, and $a' + w \geq t\tau + t$ because $w - \lfloor w/2 \rfloor = \lceil w/2 \rceil \geq \lceil t/2 \rceil = t - \lfloor t/2 \rfloor$. If $a' < 0$ the window is $[0, w)$, and $w \geq a' + w \geq t\tau + t$. If $a' > H - w$ the window is $[H - w, H)$, with $H - w < a' \leq t\tau$ and $H \geq t\tau + t$. In every case the window contains the tile, so it contains each query of the tile, and two positions in a window of side $w \leq n_0$ differ by at most $n_0 - 1$. The proposition makes no claim about the bridge keys, which keep their true positions. □

Remark 1 (Text-image offsets). FLUX.1 places all text tokens at position 0, so the offset between an image query at p_i and a text key is p_i. In the global operator the query sits at $c(p_i) \in [n_0]^2$, inside the trained range. In the local operator it keeps p_i, up to $sn_0 - 1$ per axis, exactly as in full attention and in AReA's windows at the target. Mapping text offsets through $c(\cdot)$ in the local operator is a possible refinement that we have not tested.

Proposition 3 (Cost)**.** *Per head and layer, full attention scores $(N+T)^2$ query-key pairs, the global operator $(N + T)(T + N_0)$ with $N_0 = |\mathcal{G}_0| = n_0^2$, and the local operator $N(T + R + w^2)$ with $R = |\mathcal{B}|$. For fixed n_0, w, R, and T the total is $O\big(N(T + R + w^2 + N_0)\big)$, linear in N.*

Proof of Proposition 3. Full attention scores every query against every key, $(N + T)^2$ pairs. The global operator scores $N + T$ queries against $T + |\mathcal{G}_0|$ keys with $|\mathcal{G}_0| = N/(s_H s_W) = n_0^2$. The local operator scores N image queries against T text keys, R bridge keys, and w^2 window keys. The sum is at most $(2N + T)(T + R + w^2 + N_0)$, which is $O(N(T + R + w^2 + N_0))$ for fixed n_0, w, R, and T. Value aggregation has the same pair counts. In the opening phase the local term is absent. The FlexAttention kernel visits whole blocks and masks the extra pairs. A query block of 8×16 tokens spans two tiles whose windows are 8 columns apart, so it visits a 64×72 key region, 12.5% more than one window, plus partial blocks at clamped borders. This changes the constant, not the order. □

Proof of Proposition 1. $J(\pi) = E - 2\pi B + \pi^2 A$ with $A \geq 0$. If $A = 0$ then $O^{\mathrm{loc}} = O^{\mathrm{glob}}$ on every summand, so $B = 0$ and J is constant; any π, in particular $\pi^\star = 1$, is optimal and the bound holds trivially. If $A > 0$, J is a strictly convex quadratic with unconstrained minimizer B/A, and its minimizer on $[0, 1]$ is the projection $\mathrm{clip}(B/A, 0, 1)$. For any π',

$$J(\pi') - J(\pi^\star) = A(\pi' - \pi^\star)^2 + 2(\pi' - \pi^\star)(A\pi^\star - B).$$

If $0 < \pi^\star < 1$ then $A\pi^\star = B$ and the second term vanishes. If $\pi^\star = 0$ then $B \leq 0$, so $A\pi^\star - B \geq 0$ and $\pi' - \pi^\star \geq 0$. If $\pi^\star = 1$ then $B \geq A$, so $A\pi^\star - B \leq 0$ and $\pi' - \pi^\star \leq 0$. In both clipped cases the second term is nonnegative.

Any rule that assigns one weight per cell is a feasible π' in every cell. Examples are a per-layer table, a weight shared per noise level, a hard switch, or a table fitted elsewhere. Its total error is therefore at least the per-head table's plus $\sum_{\text{cells}} A(\pi' - \pi^\star)^2$. A table interpolated in σ between the mean sampled noise levels of occupied bins is not constant within a bin. The statement applies to the bin-constant table used to score rules in Section 4.3. □

B Reach emulation: details and statistics

This appendix expands Section 2, Figure 2, and Section 3.2 with Figure 4. Appendix B.1 lists what one emulation pass records. Appendix B.2 gives the variance split and attention mass behind the observation, Appendix B.3 the fitted reach tables, and Appendix B.4 the output error of every mixing rule.

B.1 What one pass records

Statistics. For every emulated ratio, block, head, and noise bin, the pass stores A, B, E, the error of the local branch alone $\sum \|O^{\mathrm{full}} - O^{\mathrm{loc}}\|^2$, $\sum \|O^{\mathrm{full}}\|^2$, and the query count. These are float64 sums over image queries and channels. The sampler follows the trajectory of unmodified full attention; the emulated operators only observe. Statistics are written per prompt and merged by summation. Prompts can therefore be added later, and split-half fits need no extra pass. Text-query outputs are not in the objective, because SCA sends text queries through the global branch only.

Geometry. At every block and step both emulated views see the same queries, keys, and values as full attention. Ratio r reproduces the image-to-window ratio of a target r times larger, with r^2 times fewer keys per view. At 1024^2 the grid is 64×64. Ratio 2 uses a 32×32 window, 2×2 subsampling onto a 32×32 grid, and 4×4 query tiles. Ratio 4 uses a 16×16 window, 4×4 subsampling onto a 16×16 grid, and 2×2 tiles. At 1280^2 the grid is 80×80, and ratio 2 uses a 40×40 window, 2×2 subsampling onto a 40×40 grid, and 5×5 tiles. The code and the figures call the emulated ratio the emulated scale.

Noise bins. The latent at noise level σ is $x_\sigma = (1 - \sigma)x_0 + \sigma\epsilon$. With 28 steps and $\mu = 1.15$, σ runs from 1.0 to 0.105. The lowest of the 14 bins is therefore empty and copies its neighbour; 13 bins are populated. Under this schedule 8 steps have $\sigma > 0.89$ (the ninth has $\sigma = 0.888$) and 13 have $\sigma > 0.80$.

B.2 The head sets the reach

Table B.1 expands the variance split of Section 2 to two radii, three checkpoints, and two resolutions. Figure B.1 ties each head's measured share to its fitted weight.

Table B.1: **Reach is set by the head.** Variance of the share of each head's image attention within 8 or 16 tokens of the query (Chebyshev distance), over all (block, head, noise bin) cells. It is split into the head (between blocks plus between heads of one block) and the noise level within a head; the two sum to 100%. Avg. is the variance of the head-averaged curve over noise levels, the part one schedule shared by all heads could follow. 256 seeded queries per block and step, full attention at each row's resolution. AReA's statistic is the distance to dominant keys; ours is the mass share, which sets the output by Lemma 1. Key result: after the gate the head carries at least 97.0% of the variance in every model and size.

		within 8 tokens (%)			within 16 tokens (%)		
Model and size	Prompts	Head	Noise	Avg.	Head	Noise	Avg.
After the gate ($\sigma < 0.89$)							
FLUX.1-dev, 1024^2	32	97.0	3.0	1.02	97.5	2.5	0.65
FLUX.1-dev, 1280^2	16	97.4	2.6	0.75	97.8	2.2	0.46
FLUX.1-Krea-dev, 1024^2	16	97.2	2.8	1.03	97.6	2.4	0.79
FLUX.2-dev, 1024^2	16	99.0	1.0	0.10	99.1	0.9	0.04
All noise levels							
FLUX.1-dev, 1024^2	32	92.2	7.8	4.02	93.3	6.7	3.29
FLUX.1-dev, 1280^2	16	93.3	6.7	3.09	94.3	5.7	2.44
FLUX.1-Krea-dev, 1024^2	16	93.8	6.2	3.03	94.7	5.3	2.55
FLUX.2-dev, 1024^2	16	96.3	3.7	1.29	97.0	3.0	0.76

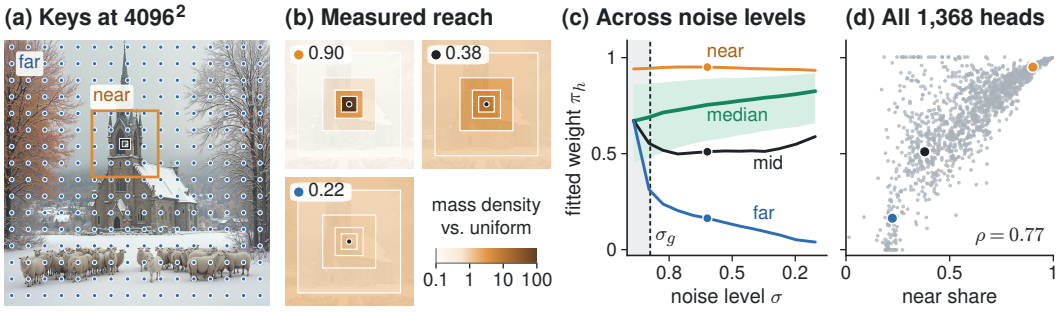


Figure B.1: **Measured reach tracks the fitted weight π_h.** (a) Near window (amber) and every fourth block-centre key per axis of the far view (blue) on a 4096^2 output. (b) Full-attention mass density of three heads at 1024^2 and $\sigma = 0.62$ (log scale, clipped at 0.1 and 100 times uniform), with each head's near share, its share of image attention within 16 tokens. (c) Their π_h with the median and interquartile range of all 1368 heads; σ falls left to right; dashed: the gate σ_g; grey band: the far-only opening. (d) π_h against the near share, Pearson $\rho = 0.77$.

Attention mass. For 256 seeded image queries per block and step, the pass also records the softmax mass on text keys and on image keys within Chebyshev distance 2, 4, 8, and 16. At 1024^2,

averaged over heads and queries, text receives 15 to 19% of the mass. Image keys within distance 16 receive 52% at $\sigma = 0.10$ and 40% at $\sigma = 0.97$. Keys outside that 33×33 neighbourhood still receive 29% and 41%. A purely local receptive field cannot account for any step, even at native resolution. Figure B.1b and d use the per-head shares in the noise bin at $\sigma = 0.62$.

B.3 The reach table

The reach table holds one weight per block, head, noise bin, and ratio. Figure B.2 shows the ratio-4 table that sampling reads and how well a table transfers to another ratio or resolution. Figure B.3 plots every head's weight against noise level with the split-half agreement, and Figure B.4 shows how widely the heads of one block differ.

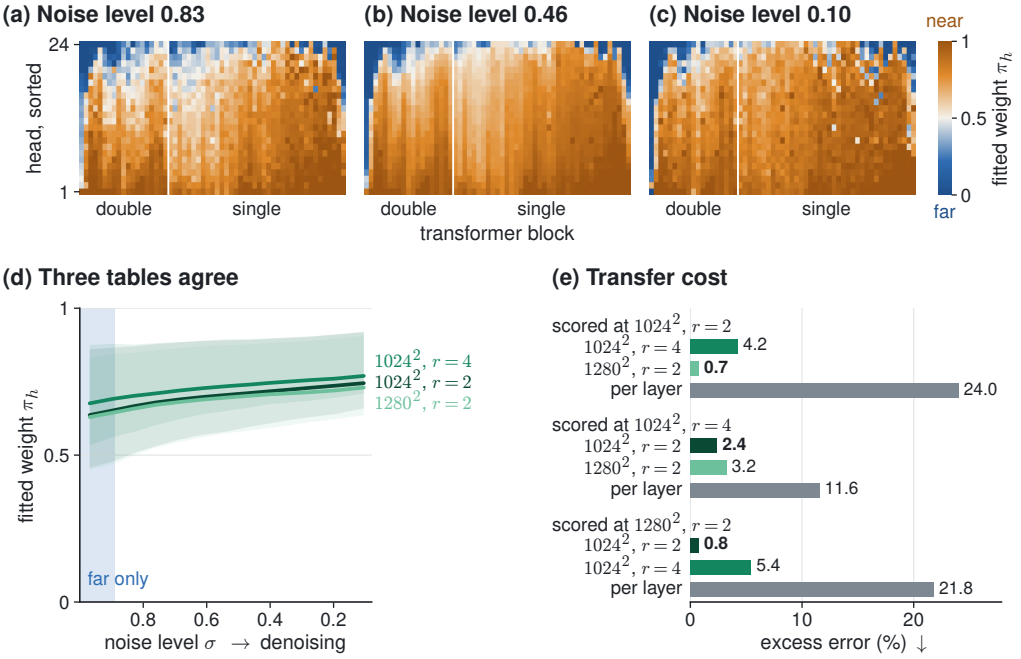


Figure B.2: **Heads differ widely, yet a reach table transfers across ratio and resolution.** (a to c) $\pi_h(\sigma)$ of the ratio-4 reach table at $\sigma = 0.83$, 0.46, and 0.10. (d) Mean and interquartile range across heads of three tables; shaded: far-only opening. (e) Excess error over each setting's own table: at most 5.4% transferred, 12 to 24% per layer.

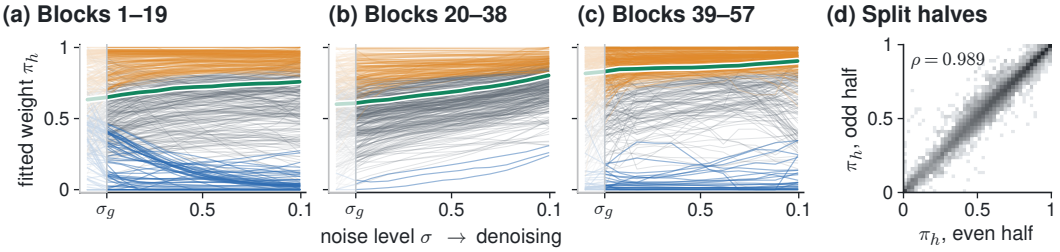


Figure B.3: **Heads use different mixes, and the mix repeats on new prompts.** (a to c) $\pi_h(\sigma)$ of all 1368 heads at ratio 2: blocks 1 to 19 (double-stream), 20 to 38, and 39 to 57 (single-stream); amber: mean above 0.8, blue: below 0.2, green: median; $\sigma > \sigma_g$ shaded. (d) Tables fitted on disjoint prompt halves agree (Pearson $\rho = 0.989$).

B.4 Output error of each mixing rule

Figure B.5 compares the relative output error of every mixing rule at both ratios, and Figure B.6 resolves it by noise level; Section 4.3 quotes these errors. A relative error is $\sum J / \sum \|O^{\text{full}}\|^2$, pooled over blocks, heads, and steps, on 32 prompts at 1024^2 and 16 at 1280^2. Tables fitted on

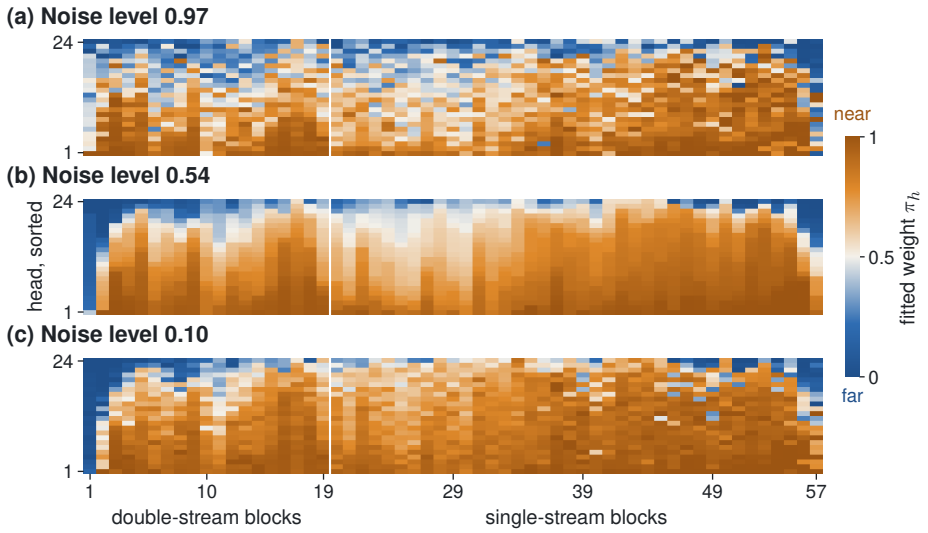


Figure B.4: **Heads in the same block differ widely at every noise level.** $\pi_h(\sigma)$ of the ratio-2 table at $\sigma = 0.97$ (a), 0.54 (b), and 0.10 (c): 57 blocks by 24 heads, 32 prompts, heads sorted per block by (b). The median within-block range is 0.66 to 0.84. Figure B.2 shows the ratio-4 table that sampling reads.

the first 4, 8, or 16 emulation prompts, merged from the stored statistics, are tested in generation in Appendix H.1.

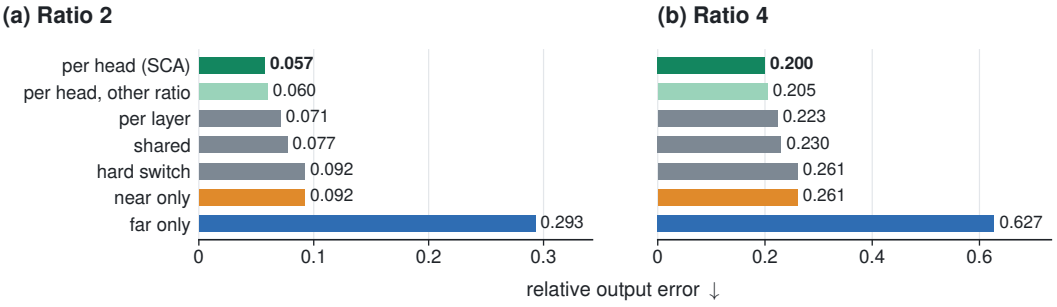


Figure B.5: **Per-head weights explain full attention best, and they transfer across ratios.** Relative output error of each mixing rule at emulated ratios 2 (a) and 4 (b), 32 prompts. The other-ratio rule applies the table fitted at the other ratio. The best hard switch ($\sigma^\star = 1$) equals the near view (local branch) alone. In-sample and held-out errors agree within 0.0003.

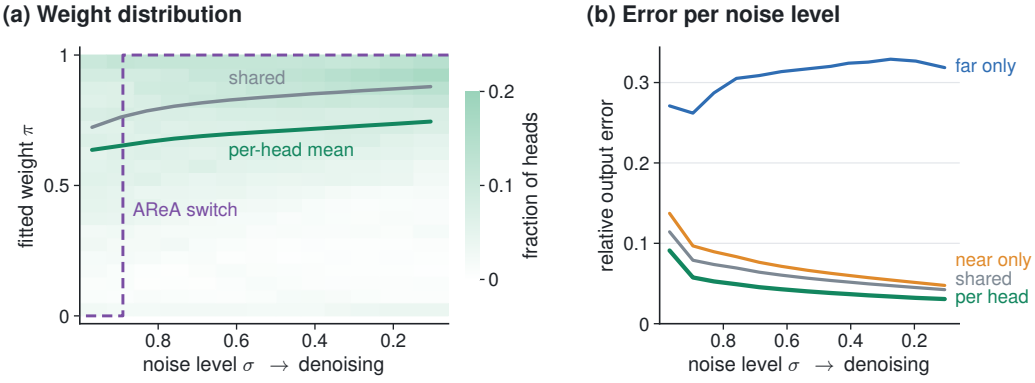


Figure B.6: **In output energy, no noise level prefers the far view.** (a) Fraction of heads per weight π and noise bin, with the per-head mean, the best shared $\pi(\sigma)$, and AReA's switch; ratio 2, 32 prompts; σ falls left to right, from pure noise at $\sigma = 1$. (b) Relative error per noise bin: the near view alone beats the far view alone everywhere, and the per-head mix beats both.

No global phase in output energy. The best single shared weight falls from 0.88 at $\sigma = 0.10$ to 0.72 at $\sigma = 0.97$, and the mean per-head weight from 0.74 to 0.64 (Figure B.6). The median within-block range of the per-head weights is 0.66 to 0.84 at every noise level (Figure B.4). Among hard switches from global to local, the best threshold is $\sigma^\star = 1$, which equals local only. A switch at $\sigma^\star = 0.89$ between our two branches raises the error from 0.092 to 0.152 at ratio 2 and from 0.261 to 0.376 at ratio 4. In the noisiest bin the global branch alone has relative error 0.27 at ratio 2, the local branch alone 0.14, and the per-head mix 0.09. Generation still needs the opening (Section 4.3), because output error cannot see layout.

C Implementation, pseudocode, and hyperparameters

This appendix expands Section 3, Figure 3, and Algorithms 1 and 2. Figure C.1 draws the operators at component level (Appendix C.1), and Appendix C.2 gives the full pseudocode. Appendix C.3 gives the kernel layout and the numerical checks, Appendix C.4 the attention cost, and Appendix C.5 every hyperparameter.

C.1 The operators in detail

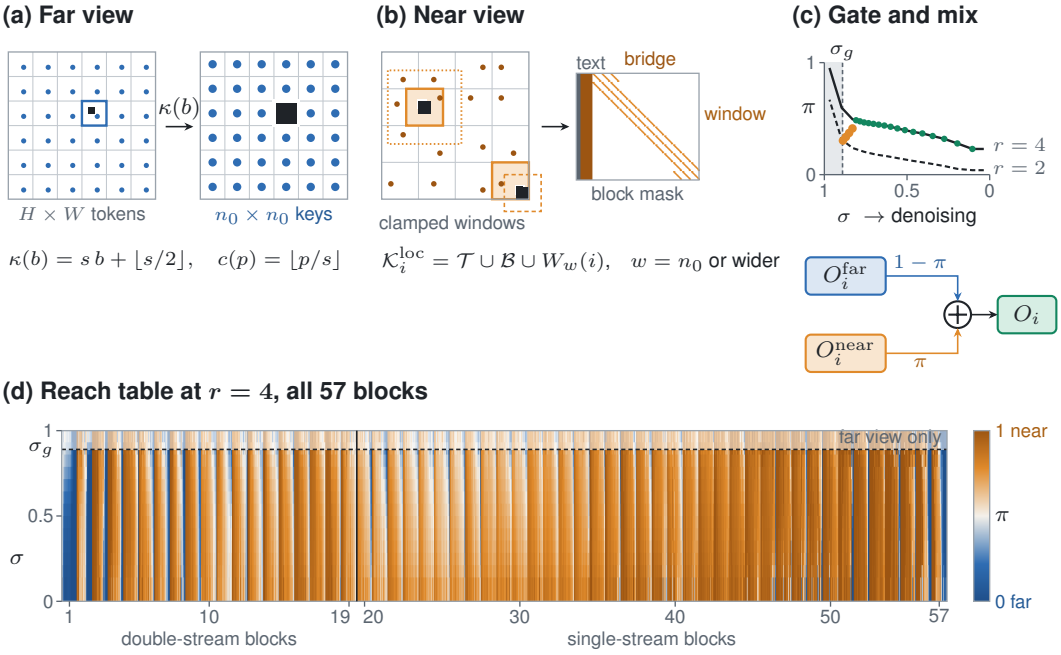


Figure C.1: **Both views keep the native geometry, and one weight per head and step mixes them.** The operators of Figure 3 at component level on a toy grid: $H = W = 24$ tokens, $s = 4$, $n_0 = 6$, $t = 2$. (a) The far view keys each block by its centre token (Equation 4). (b) A query tile's near view: text keys, bridge tokens (dots), and a border-clamped window, with its block mask (Equations 2 and 3). (c) Head 24 of block 12 at $r = 2$ (dashed) and $r = 4$ (solid); dots: the weight applied at each post-gate step at 4096^2 (amber: wide windows; σ falls left to right), and the mix (Equation 6). (d) The $r = 4$ table, 57×24 heads by 14 noise bins.

Design choices. The near view of the main text is the local branch here, and the far view the global branch, as in the code. A block centre is a real token, so the global branch attends to keys and values the model produced; mean pooling does worse (Section 4.3). Native windows keep offsets in the trained range, so SCA keeps FLUX.1's RoPE and softmax scale. Text keys sit at position 0 in both branches (Remark 1). Bridge keys keep their true positions, and windows wider than n_0 reach offsets up to $w - 1$. The local operator therefore leaves the trained range for its 1,024 bridge keys and during the four wide-window steps. FLUX.1 also rounds positions in bf16 beyond 256 tokens per axis (Appendix C.3).

Window placement. Image queries form tiles of $t \times t$ tokens; query i lies in tile $\tau = \lfloor p_i/t \rfloor$. The window $\mathcal{W}_w(i)$ of tile τ has side w, at most the grid side, and starts at $a(\tau) = \mathrm{clip}\left(t\tau + \lfloor t/2 \rfloor - \lfloor w/2 \rfloor,\ 0,\ H - w\right)$ per axis. It is the $w \times w$ block centred on the tile and clamped to the grid, and it contains the whole tile (proof of Proposition 2).

C.2 Pseudocode

Algorithms 3 and 4 expand Algorithms 1 and 2 to every step the code performs, and Algorithm 5 writes out progressive SCA (Section 3.3). Each comment names the equation or setting that its line implements. Appendix I.2 maps each algorithm to its source file.

C.3 Kernel and numerics

FlexAttention window. Image tokens are reordered tile-major, so that a sparse block is a small spatial tile. Query blocks cover 8×16 tokens (128 queries) and key blocks 8×8 (64 keys), shifted so that unclamped windows start on a block edge. The T text keys and the 1,024 bridge keys come first, as a prefix of $T + 1{,}024$ keys shared by every query; the bridge keys keep their true positions. The wider windows of the four steps after the opening use the same tiling with more key blocks per query block. The mask function states window membership per query-key pair, so the result does not depend on the block layout; only the amount of skipped work does. For the operator without the schedule on GH200, 8×16 query blocks with 8×8 key blocks were fastest at 2048^2, 4096^2, and 2048×6144. RoPE, subsampling, both branches, and the mix compile into one graph per layer, and the mix runs in fp32.

Numerical checks. The 18 CPU tests of the release run in float32. They cover full-mode agreement with stock FLUX attention, both branches against brute-force Equations 4 and 2 without the bridge, and full-grid and factor-1 identities. Reach emulation when the local branch is exact, the gate, and FlexAttention against an explicit gather reference complete them. Further CPU tests cover the bridge selection (Equation 3), the local branch with bridge keys against brute force, the window schedule, and the reduction to the operator without the schedule when these options are off. A separate GPU check compares the gather and FlexAttention implementations in bf16. The recorded maximum absolute difference is 0.015625, and both have the same maximum error against the fp32 reference on the tested grids. This is an observed absolute-error check; no per-element ULP bound was measured. Appendix I.2 names the test files.

Position ids in bf16. The FLUX.1 pipeline builds image position ids in bf16, which cannot represent every integer above 256. Our runs keep these rounded ids for RoPE. Passing the true grid from the sampler fixes window membership and subsampling geometry. It does not restore exact RoPE coordinates, because the coarse-query positions are also computed from the rounded ids. At a 384-token axis, index 383 rounds to 384, and its coarse coordinate at factor 6 becomes 64. At a 512-token axis, index 511 rounds to 512, and its coarse coordinate at factor 8 also becomes 64. A local 64-token window can likewise have a rounded endpoint difference of 64 instead of the ideal bound 63. Proposition 2 therefore applies exactly to the mathematical operator, not to these finite-precision positions at 2048×6144, 6144^2, or 8192^2. The 4096^2 grid uses indices 0 to 255, which bf16 represents exactly. FLUX.2 uses integer position ids. Rebuilding FLUX.1 positions in higher precision would need a separate controlled experiment; the reported runs are unchanged.

Non-divisible grids. When H/n_0 is not an integer, the subsampling factor per axis is the divisor of H nearest to H/n_0, and the tile side is the divisor nearest to t. Neither case arises at the sizes we evaluate.

C.4 Attention cost

Table C.1 counts the attention FLOPs of one image and times one attention layer. It expands the pair count of Section 3.1.

Algorithm 3: **Reach emulation**, in full: one full-attention pass at 1024^2 fills the table of every ratio

```
# Input: 32 prompts; FLUX.1-dev at 1024^2 (n = 64), 28 steps, guidance 4.5,
#   full attention
# Output: pi[r, l, h, beta] for emulated ratios r in {2, 4}; centres[beta]
def bin(sigma): return min(int(14 * sigma), 13)  # 14 bins of width 1/14
def on_step(sigma):                              # once per step
  S[bin(sigma)] += sigma; C[bin(sigma)] += 1
def emulate(q, k, v, l, sigma):                  # every block l
  O_full = attn(q, k, v)                         # Eq. 1, all keys
  beta = bin(sigma)
  for r in (2, 4):                               # emulation, App. B.1
    f, w = r, n // r                             # subsampling factor, window side
    t_c = round(t * w / n0)                      # tile sides 4 and 2
    O_glob = global_branch(q, k, v, f)           # Eq. 4
    O_loc = local_branch(q, k, v, w, t_c)        # Eq. 2, no bridge
    dl, df = O_loc - O_glob, O_full - O_glob
    A[r, l, :, beta] += sum(dl * dl)             # Eq. 5: per head, over
    B[r, l, :, beta] += sum(df * dl)             #   image rows and channels
    E[r, l, :, beta] += sum(df * df)
    F[r, l, :, beta] += sum(O_full * O_full)
  return O_full                                  # the sampler is unchanged
sample(prompts, on_step, emulate)
pop = C > 0                                      # 13 of 14 bins populated
centres = where(pop, S / C, (arange(14) + 0.5) / 14)
pi = clip(B / A, 0, 1); pi[A == 0] = 1           # Prop. 1
pi = fill_empty_bins(pi, pop)                    # copy the nearest populated bin
J = E - 2 * pi * B + pi**2 * A                   # error of the mix
rel_err = J.sum() / F.sum()                      # Sec. 3.2
```

Algorithm 4: **Sampling**, in full: one SCA attention call of single-pass SCA

```
# Input: q, k, v before RoPE (T text, then H x W image tokens), block l, sigma
# Output: O for all T + HW queries
# Per image: s_H, s_W = H // n0, W // n0; pi_s is the table entry nearest
#   max(s_H, s_W) in log. Per step, all 57 x 24 heads, table ends held:
w = n0                                           # Sec. 3.3
for sig_k, w_k in ((.88, 128), (.86, 112), (.84, 96), (.82, 80)):
  if sigma > sig_k: w = w_k; break
if w == n0: pi = interp(sigma, centres, pi_s)
else:                                            # Eq. 7, linear in log r
  r = max(H, W) / w; u = clip(log2(r / 2), 0, 1)
  pi = ((1 - u) * interp(sigma, centres, table[2])
        + u * interp(sigma, centres, table[4]))
kb = k_img[s_H//2::s_H, s_W//2::s_W]             # κ(b)
vb = v_img[s_H//2::s_H, s_W//2::s_W]
cp = p // (s_H, s_W)                             # c(p_i); text at 0
Q, K = cat(rope(q_txt, 0), rope(q_img, cp)), cat(rope(k_txt, 0), rope(kb, b))
O_glob = sdpa(Q, K, cat(v_txt, vb))              # Eq. 4
if sigma > sigma_g:                              # 0.89, Eq. 6
  return O_glob                                  # opening phase
d = ((k_img - k_img.mean(0))**2).sum(heads, ch)  # bridge, Eq. 3
B = top_k(d, 16) per cell, 8 x 8 cells           # 1,024, per block and step
kr = rope(k_img, p)                              # true positions
kl = cat(rope(k_txt, 0), kr[B], kr)
vl = cat(v_txt, v_img[B], v_img)                 # a bridge key in the window: twice
P = T + len(B)                                   # prefix, seen by all
w = (min(w, H), min(w, W))                       # window sides
a = clip(t*(p//t) + t//2 - w//2, 0, (H, W) - w)
keep(i, j) = j < P or a_i <= p_j < a_i + w       # Eq. 2
perm = tile_major(H, W)                          # 8 x 16 query blocks
mask = block_mask(keep, perm)                    # 8 x 8 key blocks
O_loc = flex_attention(rope(q_img, p)[perm], kl, vl, mask)[inverse(perm)]
O_img = pi[l] * O_loc + (1 - pi[l]) * O_glob[T:]  # fp32 mix, Eq. 6
return cat(O_glob[:T], O_img)                    # text queries take O_glob
```

Algorithm 5: **Progressive SCA**: HRDiT's schedule with SCA attention (Section 3.3, Appendix F)

```
# Input: prompt, seed; FLUX.1-dev in bf16; reach table at ratios 2 and 4
# Output: 4096^2 image
# one 30-step schedule for all stages, as in HRDiT:
sig = cat(shift(linspace(1, 1/30, 30), mu=1.15), [0])
x = noise(1024^2, seed)     # stage 1: stock dense attention, native RoPE
for i in range(30):                          # Euler; ref keeps each x0 estimate
  v = flux(x, sig[i], g=3.5); ref[i] = x - sig[i] * v
  x = x + (sig[i+1] - sig[i]) * v
for res, k, g, r, a in ((2048, 13, 4.5, 2, 1.0), (4096, 20, 6.0, 4, 0.25)):
  img = sharpen(bicubic(decode(x), res))     # 2 img - blur(img), blur width
                                             #   ~ U(0.1, 2.0)
  x = sig[k] * noise + (1 - sig[k]) * encode(img)  # re-enter at 0.81, 0.61
  # replaces NTK RoPE, attention scale, pruning, and alignment; w = n0 as
  # sig[k] < 0.82 (Sec. 3.3); no opening phase as sig[k] < sigma_g
  use_sca(table[r], n0=64, t=8, sigma_g=0.89, bridge=(8, 16))
  N = 30 - k                                 # 17, then 10 steps
  for n, i in enumerate(range(k, 30)):
    v = flux(x, sig[i], g); x0 = x - sig[i] * v   # SCA in every block
    r, ref[i] = ref[i], x0
    # low-band guidance; Butterworth, order 4, cut-off 0.2:
    x0 = x0 + a * (N - n) / N * (lowpass(up(r)) - lowpass(x0))
    # HRDiT's velocity momentum:
    u = momentum((x - x0) / sig[i], 0.5 * (N - n) / N)
    x = x + (sig[i+1] - sig[i]) * u
return decode(x)
```

Table C.1: **Single-pass SCA needs 6.9 times fewer attention FLOPs than full attention at 4096^2.** Top: analytic attention FLOPs per image (57 blocks, 24 heads, head dimension 128, 28 steps, $T = 512$), with full attention over each row in parentheses. Single-pass SCA runs the global branch alone for the 8 opening steps, local windows of side 128, 112, 96, and 80 for the next four, and the native window $n_0 = 64$ for the last 16. Every local window adds $T + 1{,}024$ text and bridge keys. Without the schedule the window is always n_0 and there is no bridge; "gated" skips the local branch in the first 8 steps. Bottom: time of one attention layer for SCA without the schedule (both branches and the mix) in bf16, measured once on one GH200 (minimum over interleaved runs; no archived timing file), with the speedup over full attention in parentheses; these per-layer times are indicative. Key result: the saving grows with the token count, from 3.5 to 6.9 times in FLOPs and from 1.4 to 4.7 times in measured layer time.

FLOPs per image grid, (s_H, s_W)	2048×4096 128×256, (2, 4)	4096×4096 256×256, (4, 4)	2048×6144 128×384, (2, 6)
Full attention	2.17×10^{16}	8.56×10^{16}	4.84×10^{16}
SCA without the schedule, no gate	5.97×10^{15} (3.6)	1.19×10^{16} (7.2)	8.93×10^{15} (5.4)
SCA without the schedule, gated	5.12×10^{15} (4.2)	1.02×10^{16} (8.4)	7.66×10^{15} (6.3)
Single-pass SCA	6.24×10^{15} (3.5)	1.24×10^{16} (6.9)	9.34×10^{15} (5.2)
One layer, bf16 (ms) token grid	2048^2 128×128	4096^2 256×256	2048×6144 128×384
Full attention (SDPA)	7.1	87.3	49.6
SCA without the schedule	5.2 (1.4)	18.7 (4.7)	14.0 (3.5)

C.5 HYPERPARAMETERS

Table C.2 separates the hand-set settings of single-pass SCA from its fitted weight, and Table C.3 lists every setting. Both expand the models-and-baselines paragraph of Section 4.

Table C.2: **Only the mixing weight is fitted.** Every other setting of single-pass SCA was chosen by hand. The four window bins hold one Euler step each only for the 28-step sampler at $\mu = 1.15$.

Setting	Value	Source
gate σ_g	0.89 (8 of 28 steps)	hand-set, AReA's boundary
window sizes	128, 112, 96, 80, then 64	hand-set on the 46-prompt subset
window σ bins	$(0.88, 0.89]$, $(0.86, 0.88]$, $(0.84, 0.86]$, $(0.82, 0.84]$	hand-set, tied to 28 steps
tile t	8	hand-set
bridge	8×8 cells, 16 tokens each (1,024)	hand-set, AReA's selection rule
emulated ratios	2 and 4	hand-set
noise bins	14 of width $1/14$	hand-set
interpolation	linear in $\log_2 r$ and in σ	hand-set
weight $\pi_{lh}(\sigma; r)$	per block, head, bin, and ratio	fitted (Proposition 1)

Table C.3: **Every setting of single-pass SCA; none is tuned per target size.** Progressive SCA uses the same attention settings inside HRDiT's schedule (Appendix F). Appendix D.8 gives the settings that differ for UltraImage, HRDiT, SigMa, and FLUX.2-dev, and Table I.1 the command that produces this configuration.

Setting	Value
Model	
checkpoint, precision	FLUX.1-dev, bf16 weights and activations
blocks, heads, head dim	19 double-stream + 38 single-stream, 24, 128
native grid n_0, text tokens T	64 (1024^2 pixels, 16 pixels per token), 512
Sampler	
steps, solver	28, flow-matching Euler
guidance, shift	4.5 (distilled guidance), $\mu = 1.15$ at every size
seed, decoding	seed label 0 (one prompt-derived seed per prompt, shared across methods); tiled VAE decoding
Local branch	
tile t	8
window $w(\sigma)$	128, 112, 96, 80 for σ in $(0.88, 0.89]$, $(0.86, 0.88]$, $(0.84, 0.86]$, $(0.82, 0.84]$, then 64; clamped at borders; original positions
bridge	16 tokens per cell of an 8×8 grid (1,024), farthest from the mean key, recomputed per block and step, true positions
kernel	FlexAttention; query blocks 8×16, key blocks 8×8, tile-major order
Global branch	
subsampling	block centre, factor (s_H, s_W) = target grid / n_0
positions	queries at $c(p)$, keys at block index, text at 0
Mixing	
table	per (block, head, σ bin, ratio), $57 \times 24 \times 14 \times 2$, ratios 2 and 4
ratio	$r = \max(H, W)/w(\sigma)$, weight linear in $\log_2 r$ between ratios 2 and 4, nearest outside
interpolation	linear in σ between the mean sampled noise levels of occupied bins, constant outside
gate	$\sigma_g = 0.89$ (8 of 28 steps global only, then 4 wide-window steps)
Emulation	
prompts, resolution	32 at 1024^2; prompts 1 to 16 also at 1280^2
emulated ratios	2 and 4, one pass; 14 bins of width $1/14$
hardware	one GH200; complete emulation runtime was not retained

D EVALUATION PROTOCOL AND BASELINES

This appendix expands the setup of Section 4. Appendix D.1 gives the protocol and the prompts, D.2 defines every metric, D.3 the intervals and table conventions, D.4 the timing protocol, and D.5 the rules behind every sample figure. Appendix D.6 compares how each method changes attention and expands Section 5. Appendices D.7 and D.8 say how each baseline was run.

D.1 PROTOCOL AND PROMPTS

Table D.1: **Every FLUX.1-dev attention comparison shares one sampler and one seed per prompt.** Evaluation protocol. UltraImage and HRDiT use their own multi-stage protocols, and progressive SCA uses HRDiT's (Appendix F); FLUX.2-dev uses its own native-resolution schedule (Appendix D.8). Prompt counts include the four development prompts. Table C.3 lists the hyperparameters, and Appendix D.2 defines the metrics.

Models	FLUX.1-dev (Black Forest Labs, 2024) (main results); FLUX.1-Krea-dev (Black Forest Labs & Krea AI, 2025) (195 prompts); FLUX.2-dev (Black Forest Labs, 2025) (50 prompts); bf16.
Sampler	28 Euler flow-matching steps, guidance 4.5, shift $\mu = 1.15$ at every size, seed label 0 per prompt, tiled VAE decoding; progressive SCA follows HRDiT's three stages (Appendix F), and the UltraImage, HRDiT, and FLUX.2-dev exceptions are in Appendix D.8.
Sizes	4096^2, 2048×4096, 2048×6144 (195 prompts each); 6144^2 (100-prompt subset); 8192^2 (12 exploratory paired prompts, metrics and qualitative examples).
Prompts	Aesthetic-4K evaluation set with 4K references (Zhang et al., 2025a), 195 prompts, fixed 46-prompt ablation subset; HRGen-1K (Rajabi et al., 2026a), all 1000 prompts.
Baselines	Full attention; AReA, released code (Rajabi et al., 2026a); DyPE (Issachar et al., 2025), SEGA (Rajabi et al., 2026b), UltraImage (two-stage defaults; Zhao et al., 2025), HRDiT (released progressive protocol; Xue et al., 2026); SigMa (Zhao et al., 2026) on 195 prompts.
Metrics	ImageReward, HPSv2/2.1, PickScore, CLIP, aesthetic; FID, KID, IS, patch FID/KID; MUSIQ (full, 1024^2 crops), CLIP-IQA, ARNIQA; 95% paired bootstrap intervals over prompts.
Hardware	One NVIDIA GH200 per process; verified timing logs identify the GH200 120 GB model. Clean latency uses batch 1; campaign timings are marked separately.

Prompts and references. The Aesthetic-4K evaluation split (Zhang et al., 2025a) has 195 prompts with 4K reference images. Four of them (boats, Moai, woman and bear, underwater surfer) were used to choose the opening phase and the native window, and the 46-prompt subset to set the four wide-window sizes. They stay in the evaluation set for every method, which favours both SCA configurations slightly. The 195 reference rows hold 168 distinct images, because 27 rows reuse an image under their own caption. Distribution metrics keep the official rows and weights, as Diffusion-4K's statistics do. The 46-prompt ablation subset was fixed before any ablation was run. Seeds hash the prompt id and a seed label shared by all methods; labels 0, 1, and 2 each cover all 195 prompts. HRGen-1K (Rajabi et al., 2026a) has 1000 prompts and no references; we use all of them. The 48 emulation prompts (Appendix K) were written for this purpose and share no prompt with Aesthetic-4K; the largest word-set Jaccard overlap with any evaluation prompt is 0.26.

D.2 METRICS

Implementations. torch-fidelity 0.3.0, pyiqa 0.1.16 (MUSIQ) and 0.1.15 (CLIP-IQA, ARNIQA), hpsv2 1.2.0, and image-reward 1.5.

Preference and alignment. ImageReward v1.0 (Xu et al., 2023): bicubic resize to 224, centre crop. HPSv2 and HPSv2.1 (Wu et al., 2023): cosine of normalized image and text features with the official validation transform (longest side 224, padded). PickScore (Kirstain et al., 2023): $\exp(\text{logit scale})$ times the cosine. CLIP score (Radford et al., 2021): $100 \max(\cos, 0)$ with ViT-L/14. Aesthetic: the LAION improved aesthetic predictor (sac+logos+ava1, linear MSE) on the L2-normalized CLIP ViT-L/14 image embedding (Schuhmann, 2022).

No-reference quality. MUSIQ (Ke et al., 2021), with pyiqa defaults, runs on the full image at native resolution. MUSIQ_c is its mean over four fixed 1024^2 crops per image; crop positions depend only on the prompt id, so they are identical across methods. CLIP-IQA (Wang et al., 2022) and ARNIQA (Agnolucci et al., 2024), with pyiqa defaults, run on the image downscaled with antialiased bicubic resampling to a long side of 1024 pixels, the scale both were trained on.

Distribution metrics. FID (Heusel et al., 2017) and KID compare the generated images with all 195 official reference rows, using Inception-v3 features (torch-fidelity) after bilinear resizing to 299^2. The generated sets hold 195 images on the main set of Table 1 and 195 on the panel of

Table H.3. The reference rows contain 168 distinct image contents; rows that share an image keep their official weights. Inception score uses full images at 299^2, with 10 splits.

Patch FID and KID. They are our own definition: the same Inception features on four fixed 1024^2 crops per generated image and per reference row (780 generated and 780 reference crops on the main set). AReA and SEGA report a patch FID without defining its crops, so patch FID is never compared across papers. Crop features are keyed by reference row, so rows that share an image keep their own crops. An earlier assembly keyed them by image path, so rows that share an image reused the last row's crop features. Every patch estimate reported here uses the corrected assembly; whole-image FID and KID do not depend on it.

D.3 Intervals and table conventions

Per-image metrics. Intervals are 95% paired percentile bootstraps over prompt ids. Summaries written by the evaluator use 1000 draws. Comparisons recomputed for the tables use 2000 draws with seed 0 (Table G.2: 1000), so they can differ slightly from the evaluator summaries. Resampling matched prompt ids never treats the crops of one image as independent images.

Set-level metrics. For FID, KID, and their patch versions, 1000 draws resample the generated prompt ids jointly across methods and the 168 distinct reference-image identities. Each draw keeps all rows and crops of a sampled identity together. This keeps the 195-row weighted point estimate and accounts for rows that share a reference image. For single-pass SCA against AReA, the FID difference is -5.6 $[-10.6, -0.4]$, patch FID -4.4 $[-7.9, 0.4]$, and patch KID ($\times 10^3$) -2.53 $[-4.57, -0.60]$ (Table E.3). Every FID, KID, and patch interval on a 195-prompt set resamples these clusters, including the comparisons at both non-square sizes, against every added baseline, and of the sampling-budget and progressive runs. The other whole-image FID and KID intervals are reference-row bootstraps, labelled historical; they were not recomputed with image clusters. Historical patch intervals are never attached to corrected estimates and are omitted where recomputation is incomplete. Inception-score intervals resample only generated images. FID's finite-sample bias can place a point estimate outside its percentile interval.

Sources. Every table is populated from archived evaluation files (Appendix I.2). Per-image means were checked against the per-image records. Distribution-level estimates use cached features and the corrected crop assembly, and set-level metrics use the whole set.

Conventions. A difference is the method minus the named reference. For the ten per-image metrics positive values favour the method; for FID, KID, their patch versions, and time, negative values do. Blue and orange mark intervals that exclude zero in favour of or against the method; black marks an interval that includes zero. The comparisons use 1000 HRGen-1K prompts, 195 prompts at each extra seed, 46 prompts in the sensitivity experiments, 50 for FLUX.2-dev, and 195 for the main baseline comparison. Intervals measure prompt and reference sampling uncertainty, not variability over independent training runs. They are not adjusted for multiple comparisons; Appendix E.3 gives Holm-adjusted counts for the headline pairs.

D.4 Timing

Clean latency follows one protocol on idle GH200s. At each size each method generates one excluded warm-up image and then three timed images in the same process at batch 1, and we report the median. Single-pass methods take 28 steps. Three runs use this protocol. Run T1 times full attention, AReA, and SCA without the schedule at every size. Run T2 times single-pass SCA and SCA without the schedule back to back on one GPU. Its medians for SCA without the schedule are within 0.8% of T1's at every size (71.1 s at 4096^2 in both). Run T3 times progressive SCA, HRDiT, and AReA back to back on one GH200 at 4096^2, and Tables 1 and E.1 take HRDiT's and progressive SCA's times from it. Reported time covers text encoding, every denoising step, and VAE decoding, and excludes model loading. For progressive SCA it is CUDA-synchronized and also covers the transitions between stages. Campaign means are the mean seconds per image of the evaluation runs on shared nodes. They are marked as such and never ranked against clean times. Appendix E.6 reports every timed image.

D.5 SAMPLE FIGURES

This subsection states how the prompts and crop windows of every sample figure were chosen. The full-page samples are in Appendix L. The others sit beside their results: Figures E.2 and E.3 in Appendix E, Figures F.1 to F.3 in Appendix F, Figures G.1 to G.4 in Appendix G, and Figure J.1 in Appendix J.

Selection. Each caption states how its prompts were chosen: a fixed seed over a stated pool of evaluation prompts, or, in Figure J.1, a metric rule. The pools exclude the development prompts. Figures that show single-pass SCA, SCA without the schedule, or the ablation variants draw only from the prompts that name no person, by a keyword rule on the prompt text. No prompt repeats across the 4096^2 FLUX.1-dev figures. The rule, the seed, the sources with their hashes, and the crop windows of every figure are archived beside it (Table I.1).

Defaults and crops. Unless a caption says otherwise, images are FLUX.1-dev outputs for Aesthetic-4K evaluation prompts at seed 0. Single-pass methods take 28 steps at guidance 4.5. HRDiT and progressive SCA take HRDiT's 30, 17, and 10 steps at 1024^2, 2048^2, and 4096^2, with guidance 3.5, 4.5, and 6.0. Every crop shows native pixels of its outlined window. The window is placed automatically: in a figure that compares methods, at the window whose smallest edge energy over the compared images is largest, so no method's detail is favoured; in Figure L.5, at the highest edge energy of each image. Figure F.1 instead takes, per category, the prompt and window on which the four methods agree most in content among high-detail windows.

Configurations. The row label of every image names its configuration: progressive SCA, single-pass SCA, or SCA without the schedule. Figures E.3, G.1, G.2, G.4, and J.1 show SCA without the schedule, the configuration of those experiments. Figure G.1 uses FLUX.2-dev with guidance 4.0 and a FLUX.2-dev reach table, and Figure G.3 the FLUX.1-Krea-dev setup of Appendix G.2.

D.6 HOW EACH METHOD CHANGES ATTENTION

Table D.2 compares the attention of the closest training-free methods. It expands Section 5.

Table D.2: **Only SCA fits per-head reach once at the training resolution.** Entries follow each paper's text or, where marked *, its mechanism. Offsets: whether image query-key offsets stay in the trained range (text keys aside, Remark 1); SCA stays in range in the native window but not for its bridge keys or four wide-window steps (Section 3.1). Cost: order in the image tokens N at a fixed native size. AReA's entry follows its code: its pooled steps at factor 2 attend to $N/4$ keys, and its first window is the largest whose cost stays below full attention. Fitted latency exponents are 1.12 for AReA and 0.99 for single-pass SCA (Section 4.2). †Also inside HRDiT's cascade, as progressive SCA. Key result: SCA is the only method with a measured per-head reach, and it runs at linear cost in a single pass or inside HRDiT's cascade.

Method	Local view	Global view	Per-head reach	Offsets	Cost	Passes
FreeScale	overlapping crops	full attention	×	n/a	$O(N^2)^*$	cascade
DyPE	×	full, scaled RoPE	×	×*	$O(N^2)^*$	single
SEGA	×	full, NTK, logit scale	×	×	$O(N^2)$	single
UltraImage	×	full, edited RoPE	entropy at run time	×*	$O(N^2)^*$	one or two
HRDiT	per-head scopes	×	budget, fitted per size	✓*	$O(N^2)^*$	cascade
ResDiT	native patches	full, interpolated	×	✓	$O(N^2)^*$	single
AReA	shrinking windows	pooled keys	×	×	$O(N^2)^*$	single
SCA (ours)	shrinking windows	block-centre keys	fitted at 1024^2	native	$O(N)$	single†

D.7 AREA

This subsection expands the AReA comparison of Section 4.1. It gives our protocol for AReA and sets our runs beside AReA's published numbers at 4096^2 (Table D.3) and on HRGen-1K (Table D.4).

Settings. We run AReA's released code (commit `9e38ec2`) with its FLUX.1 defaults: block-centre pooling, a first phase for 8 of 28 steps, halo width 8, and bridge ratio 0.5. We use the FLUX.1-dev checkpoint, guidance 4.5, 28 steps, μ at its maximum, and the seeds shared with the

other methods. Two settings therefore differ from the release defaults: the FLUX.1-dev checkpoint instead of FLUX.1-Krea-dev, and one seed per prompt shared across methods instead of one seed for all prompts. Our direct comparison of single-pass SCA with AReA holds these FLUX.1-dev settings fixed. UltraImage and FLUX.2-dev use separate protocols (Appendix D.8).

The released code. The release includes a cost cap on halo and bridge tokens and a jagged halo edge. Neither appears in AReA's paper, which does not say whether its Table 1 used them. Recomputed from the code with 512 text tokens, the first-phase pooling factors are 4, 4, 4, 4, 2, 2, 2, 2 at 4096×4096 and 2048×4096, and 8, 8, 8, 4, 4, 4, 2, 2 at 2048×6144. The second-phase windows shrink over 20 steps from 180, 151, and 192 tokens respectively to 64. At 2048×6144 no window is cheaper than full attention once the text tokens are counted. The first window therefore falls back to 192 tokens, and the cap drops the bridge and halo on the first three second-phase steps.

Published numbers. AReA's published numbers match FLUX.1-Krea-dev, the default checkpoint of its released code. Rerunning that code on FLUX.1-Krea-dev reproduces its published ImageReward, aesthetic score, and PickScore within 0.1 and its CLIP, HPS, and MUSIQ within 0.28 (Table D.3). On FLUX.1-dev the same code gives ImageReward 0.92 against the published 1.32. AReA's paper reports 151 s for AReA and 314 s for full attention at 4096^2 on an H100; our matched run measures 202.4 s and 186.5 s on one GH200 (medians of three images). Table D.4 expands the HRGen-1K block of Table D.3 to every row of AReA's Table 2.

FLUX.1-Krea-dev. On FLUX.1-Krea-dev we rerun AReA and fit a new reach table for SCA (16 prompts at 1024^2, ratio-4 table, other settings unchanged), with the same 28-step, guidance-4.5 sampler on all 195 prompts. DyPE and SEGA run the Krea transformer through our frozen baseline runner on the 100 prompts of rows 4 to 103. There single-pass SCA wins eight of the 15 metrics and loses only patch FID, at a third of AReA's time (Table G.2). Its FID is 1.0 higher, with an interval that spans zero ($[-3.6, 6.0]$). The progressive sampler does not support FLUX.1-Krea-dev.

Table D.3: **Our run of AReA's code on FLUX.1-Krea-dev reproduces six of its seven published Aesthetic-4K scores within 0.28.** Expands the AReA comparison of Table 1 with AReA's own published numbers. Aesthetic-4K, 195 prompts at 4096^2. Rows: AReA's Table 1 (HPS and MUSIQ from its Table 6; "FLUX.1"), our runs on FLUX.1-Krea-dev, which AReA's released code uses (28 steps, guidance 4.5), and our FLUX.1-dev protocol. The last block repeats the HRGen-1K comparison of Table D.4. Published numbers have no intervals, so every comparison here is between point estimates. HPSv2 $\times 100$; MUSIQ on the full image; FID at 299^2 against the 195 reference rows; patch FID is omitted because AReA does not state its crops; dashes: not published. **Bold** / underline: best / second best within each block (bold only in two-row blocks); †fewer than all prompts, not ranked; ours shaded. Key result: against our FLUX.1-Krea-dev run of AReA, the point estimates of single-pass SCA are better on CLIP, PickScore, HPSv2, and MUSIQ, within 0.1 on ImageReward and Aesthetic, and worse on FID; Table G.2 gives the paired intervals.

Method	n	ImageReward↑	Aesthetic↑	CLIP↑	PickScore↑	HPSv2↑	MUSIQ↑	FID↓
Published by AReA, single pass (FLUX.1)								
Full attention	195	−0.72	5.80	25.72	20.34	–	–	179.8
UltraImage	195	−0.14	6.11	26.77	20.85	–	–	169.4
DyPE	195	1.11	6.32	28.98	22.79	–	–	155.2
SigMa	195	1.01	6.16	28.36	22.47	–	–	156.8
SEGA	195	1.27	6.31	29.16	23.16	–	–	150.6
AReA	195	**1.32**	**6.37**	**29.71**	**23.31**	**29.69**	**45.2**	**148.5**
Our runs, FLUX.1-Krea-dev, same prompts and metrics								
AReA	195	1.28	6.32	29.49	23.21	29.41	45.0	**142.6**
DyPE	100†	0.96	6.41	28.73	22.50	28.39	42.7	171.6
SEGA	100†	1.13	6.31	29.19	23.01	29.15	46.0	164.2
Single-pass SCA (ours)	195	**1.30**	**6.37**	**29.64**	**23.38**	**29.81**	**46.9**	143.6
Our runs, FLUX.1-dev, shared sampler								
AReA	195	0.92	6.27	27.69	22.36	28.62	39.4	145.8
Single-pass SCA (ours)	195	**1.25**	**6.42**	**29.14**	**23.12**	**29.77**	**45.0**	**140.2**
HRGen-1K, 1000 prompts without references: AReA's Table 2 and our FLUX.1-Krea-dev run								
AReA, published	1000	0.60	5.87	29.89	21.44	27.50	–	–
Single-pass SCA (ours)	1000	**0.84**	**5.96**	**30.02**	**21.68**	**28.06**	44.5	–

Table D.4: **On HRGen-1K, single-pass SCA exceeds every score AReA publishes.** Expands the HRGen-1K block of Table D.3 to every row of AReA's Table 2. All 1000 HRGen-1K prompts at 4096^2, without reference images. Top: AReA's Table 2 ("FLUX.1", which its released code runs as FLUX.1-Krea-dev). Below: single-pass SCA on FLUX.1-Krea-dev with a reach table fitted on that checkpoint (28 steps, guidance 4.5), then our paired FLUX.1-dev runs, whose differences Table 2 reports. Published numbers have no intervals, so every comparison here is between point estimates. HPSv2 $\times 100$; IS: Inception score; MUSIQ on the full image; dashes: not published. **Bold** / underline: best / second best among the published rows and our FLUX.1-Krea-dev row; ours shaded. Key result: single-pass SCA on FLUX.1-Krea-dev exceeds AReA's published ImageReward, Aesthetic, CLIP, PickScore, HPSv2, and IS; on FLUX.1-dev its point estimates exceed our AReA run on all seven metrics.

Method	n	ImageReward↑	Aesthetic↑	CLIP↑	PickScore↑	HPSv2↑	IS↑	MUSIQ↑
Published by AReA (FLUX.1)								
Full attention	1000	−1.46	5.12	23.36	19.04	23.20	7.2	–
UltraImage	1000	−0.03	5.77	28.26	20.46	26.90	10.2	–
DyPE	1000	0.33	5.83	29.35	20.94	26.60	10.9	–
SigMa	1000	0.06	5.63	27.91	20.54	25.80	10.7	–
SEGA	1000	0.44	5.82	29.63	21.18	27.00	12.0	–
AReA	1000	<u>0.60</u>	<u>5.87</u>	<u>29.89</u>	<u>21.44</u>	<u>27.50</u>	<u>13.9</u>	–
Our run, FLUX.1-Krea-dev								
Single-pass SCA (ours)	1000	**0.84**	**5.96**	**30.02**	**21.68**	**28.06**	**16.4**	44.5
Our paired runs, FLUX.1-dev, shared sampler								
AReA	1000	−0.03	5.92	26.08	20.43	26.16	12.7	37.4
Single-pass SCA (ours)	1000	0.73	6.19	28.43	21.52	27.96	15.8	45.1

D.8 Other baselines and FLUX.2-dev

This subsection gives the settings of the other baselines in Tables 1 and H.3 and of the FLUX.2-dev check (Section 4.2). Appendix F gives those of progressive SCA.

DyPE and SEGA. Both run through our baseline runner, frozen for these runs, with FLUX.1-dev, 28 steps, guidance 4.5, maximum shift, tiled VAE decoding, and the same prompt-derived seeds as SCA. These settings isolate the attention and position treatment on the checkpoint of our main comparison. They do not reproduce the methods' reported FLUX.1-Krea-dev results; on Krea weights both trail AReA on ImageReward (Table G.2). Their low FLUX.1-dev scores should therefore not be read as a ranking under each method's preferred checkpoint.

UltraImage. We run a guided two-stage configuration from the FLUX branch of the released implementation. It uses a 1024^2 first stage and a 4096^2 second stage, 30 steps per stage, guidance 3.5 then 7, the `super_imax` guidance schedule, shifted-window patchification, proportional attention, RIFLEx positions, and attention decay. Each prompt also needs a 2+2-step block-average preparation pass. The reported campaign time includes the preparation when it is not cached and the two-stage call; main-stage and preparation times are kept separately in the per-image records. Each call reseeds both the generator and the global generator state, including the VAE sampling path, from the prompt id and seed label. Seed labels are therefore paired, but the two-stage method does not share the latent trajectory of the single-pass methods. The wrapper keeps, without tuning, an upstream scheduler behaviour: assigning `scheduler.config.shift` does not update the scheduler's internal shift in the installed diffusers version, so both stages keep shift 3.0. We report this system-level comparison separately from the shared-sampler attention comparisons.

HRDiT. We run the released progressive protocol of the official repository (commit `f3be935`) on FLUX.1-dev in fp16, with its shipped per-head scope plan and attention settings. It takes 30 steps at 1024^2 with guidance 3.5, then 17 steps at 2048^2 with guidance 4.5 and 10 steps at 4096^2 with guidance 6, 57 steps in all, with pixel-space upsampling between stages. Its sampling budget therefore differs from the shared protocol, and we report it with the multi-stage recipes. HRDiT's Gaussian blur draws its width from the global Torch generator, so we reset that generator per prompt. Its clean time, 107.0 s, is the median of run T3 (Appendix D.4). In run T1 three batch-1 HRDiT images gave a median of 108.6 s; the first of them, 144.4 s, includes warm-up. The 195-prompt HRDiT evaluation used the original crop assembly, under which patch FID is 109.9 for HRDiT and 105.9 for SCA without the schedule. Recomputed from the cached crop features with the corrected assembly of Table 1, HRDiT's patch FID is 103.0. The same assembly gives 107.7 for progressive SCA, 98.4 for single-pass SCA, 99.1 for SCA without the schedule, and 102.9 for AReA.

SigMa. The dev-matched profile runs the released code (commit `fbe3a86`) on FLUX.1-dev under the shared sampler: 28 steps, guidance 4.5, bf16, YaRN positions, and the shift capped at its maximum from the target sequence length. SigMa's paper states FLUX.1-dev with guidance 3.5, while its released command line hardcodes FLUX.1-Krea-dev with guidance 4.5; we ran only the profile that matches our protocol. The outputs are finite, fully decoded 4096^2 images. The images we inspected show a regular lattice of dots, as DyPE's do at 4096^2 on FLUX.1-dev. This may reflect our port or profile rather than the method, so the SigMa row should not be read as the method's best result. We did not run ResDiT (no verified official code) or UltraImage's direct-extrapolation variant.

Adapter for HRDiT and SigMa. SigMa and HRDiT run on all 195 prompts through one adapter with per-prompt CPU generators, resident models, and tiled VAE decoding. These differ from the upstream command-line seed, offload, and decode choices, so the runs are controlled adaptations, not reproductions of the authors' tables. Campaign time includes text encoding, decoding, and finite-value checks after every step, which add synchronization overhead. It excludes model loading and is not a clean latency benchmark.

FLUX.2-dev. The model has 56 attention blocks and 48 heads per block. We reuse the two branches and reach emulation with a native grid of 64^2 tokens, local window 64, and query tile 8. The four-axis positions keep the time and text axes and coarsen the two spatial axes. Reach emulation uses 16 prompts at 1024^2, 14 noise bins, and ratios 2 and 4; sampling uses the ratio-4 table. Full attention and SCA without the schedule both use 28 steps, guidance 4.0, seed label 0, and a shift fixed to the native sequence length, $\mu = 2.1514$. The pipeline's default extrapolated shift at 4096^2 would be about 11.55, so we do not read this comparison as a default-pipeline benchmark. The gate is 0.959, so that the first 8 steps use only global attention. The 50 evaluation prompts are Aesthetic-4K rows 4 to 53. Prompt embeddings are cached for both methods; reported times include denoising and tiled VAE decoding and exclude text encoding and model loading. At ratio 4 the fitted per-head relative output error is 0.1790, against 0.1889 for per-layer weights and 0.1924 for local-only attention. The split-half correlation is 0.950 and the cross-half error 0.1794. These are reach-emulation diagnostics, not generation ablations on FLUX.2.

E Extended results

This appendix expands the results of Section 4 to every metric, with intervals. Table E.1 expands Table 1 (Appendix E.1). Tables E.2 and E.3 expand the paired comparisons of Section 4.1 and Figure 5 (Appendix E.2), and Table E.5 pairs every baseline with single-pass SCA (Appendix E.4). Table E.6 expands Table 3 (Appendix E.5), and Table E.8 the latency of Figure 7 (Appendix E.6). Every table follows the conventions of Appendix D.3.

E.1 Every metric of Table 1

Table E.1 expands Table 1 to all 14 metrics and adds the two non-square sizes and 6144^2. Each row is marked against AReA; Appendix E.2 gives the intervals.

Table E.1: **Both SCA modes beat AReA on all ten per-image metrics at 4096^2; the non-square sizes are mixed.** Expands Table 1 to every metric and adds the non-square sizes and 6144^2. FLUX.1-dev, 195 prompts (100 at 6144^2, where the row is SCA without the schedule); HPS, CLIP-IQA, and ARNIQA $\times 100$; AReA runs its released code. Blue / orange: the 95% paired percentile-bootstrap interval against AReA excludes zero in favour of / against the row. Progressive SCA is paired over 2000 prompt draws, and for FID and FID_p over 1000 draws that also resample reference-image clusters. Full attention's FID and FID_p are unmarked; Table E.5 pairs DyPE, SEGA, UltraImage, and HRDiT with single-pass SCA. **Bold** / underline: best / second best per size (bold only in the two-row blocks); shaded: ours. Time: batch-1 median of three on an idle GH200 (HRDiT and progressive SCA: run T3, Appendix D.4); ‡campaign mean on shared nodes; °UltraImage's two-stage defaults. Key result: at 4096^2 progressive SCA is the fastest method (49.2 s) and best on CLIP, both HPS scores, and ARNIQA; single-pass SCA is best on patch FID and both MUSIQ scores.

	Time	Preference ↑						Image quality ↑				Set level ↓	
Method	s ↓	IR	Aesth.	CLIP	Pick	HPSv2	HPSv2.1	MUSIQ	MUSIQ_c	CLIP-IQA	ARNIQA	FID	FID_p
4096×4096, *scale factors (4, 4)*													
Full attention	186.5	−2.18	4.50	15.37	17.94	20.75	7.60	22.9	33.1	49.0	54.4	312.4	252.2
DyPE	195.9‡	−0.10	5.72	24.92	20.51	25.53	20.55	36.3	50.3	59.6	60.2	205.5	171.9
SEGA	190.4‡	0.56	6.02	26.95	21.64	27.13	24.34	37.3	50.3	66.7	64.8	164.9	125.8
UltraImage°	308.5‡	1.28	6.50	28.84	23.33	29.96	30.77	33.6	41.7	72.8	72.3	142.0	109.5
HRDiT	107.0	**1.34**	**6.53**	29.12	**23.46**	29.88	30.92	40.3	53.5	**75.8**	71.6	**138.4**	103.0
AReA	202.4	0.92	6.27	27.69	22.36	28.62	27.16	39.4	50.4	67.7	67.4	145.8	102.9
Single-pass SCA	77.1	1.25	6.42	29.14	23.12	29.77	29.73	**45.0**	**60.1**	72.3	70.7	140.2	**98.4**
Progressive SCA	**49.2**	1.32	6.51	**29.39**	23.41	**30.26**	**31.02**	41.6	54.1	74.9	**74.5**	141.9	107.7
2048×4096, *scale factors (2, 4)*													
AReA	112.4‡	**1.13**	**6.41**	**28.59**	**22.66**	28.36	**27.64**	51.9	61.6	**74.9**	**73.0**	144.6	**111.7**
Single-pass SCA	**41.1**‡	1.11	6.17	28.09	22.61	**28.57**	27.49	**56.2**	**65.3**	72.5	72.6	**142.5**	114.9
2048×6144, *scale factors (2, 6)*													
AReA	150.5‡	0.39	**6.14**	26.52	21.51	26.41	23.36	42.8	53.0	67.5	69.3	163.0	**105.7**
Single-pass SCA	**60.5**‡	**0.64**	5.88	**26.78**	**21.66**	**26.59**	**23.60**	**47.7**	**59.4**	**67.8**	**69.5**	**142.8**	109.7
6144×6144, *scale factors (6, 6), 100 prompts*													
AReA	507.9‡	−0.42	5.80	23.84	20.30	24.95	18.78	28.3	44.5	53.8	47.8	216.7	188.6
SCA w/o schedule	**164.7**‡	**0.84**	**6.24**	**27.12**	**21.89**	**27.48**	**25.16**	**37.5**	**48.3**	**60.1**	**62.6**	**169.9**	**122.4**

E.2 The headline comparisons with intervals

Table E.2 gives progressive SCA minus every 4096^2 baseline, and Table E.3 single-pass SCA minus AReA and full attention, each on every metric. They expand the paired comparisons of Section 4.1 and Figure 5.

Progressive SCA against AReA. It is significantly better on all ten per-image metrics. Its FID and KID differences span zero, while its patch FID ($+4.8$ $[0.1, 10.6]$) and patch KID are significantly higher. Seeds 1 and 2 and the 1000 HRGen-1K prompts give the same per-image result (Table 2); the patch FID gap is significant at seeds 0 and 2 but not at seed 1. Appendix F compares progressive SCA with HRDiT and with single-pass SCA.

Single-pass SCA against AReA. It is significantly better on all ten per-image metrics, on FID, and on patch KID. KID, patch FID, and Inception score are unresolved (Table E.3).

Table E.2: **Progressive SCA beats every single-pass baseline on all ten per-image metrics and pays in patch FID.** Expands the progressive row of Table 1. FLUX.1-dev at 4096^2, 195 Aesthetic-4K prompts, seed 0. Each cell is progressive SCA minus the row's method, with its 95% paired interval below it in grey: 2000 prompt draws for the per-image metrics, and for FID, KID, and their patch versions 1000 draws that also resample the 168 reference-image clusters. $^{\times}$: $\times 100$; *: $\times 1000$. Blue / orange: the interval excludes zero in favour of / against progressive SCA; for FID and KID lower is better, so a negative difference is a gain. Inception score has no paired interval (Table E.1); Table E.4 gives the Holm-adjusted calls. The point estimate lies outside its percentile interval for Full attn. patch FID, as it can for a biased statistic. Key result: progressive SCA is significantly better than AReA, DyPE, SEGA, and full attention on every per-image metric and than UltraImage on nine; its differences from HRDiT on ImageReward, HPSv2.1, and PickScore include zero, and its patch FID is higher than AReA's, HRDiT's, and single-pass SCA's.

(a) Preference

Progressive SCA −	IR	Aesthetic	CLIP	PickScore	HPSv2$^{\times}$	HPSv2.1$^{\times}$
AReA	+0.399 [0.317, 0.486]	+0.23 [0.17, 0.30]	+1.70 [1.33, 2.07]	+1.05 [0.93, 1.19]	+1.64 [1.42, 1.86]	+3.86 [3.41, 4.30]
HRDiT	−0.022 [−0.088, 0.034]	−0.03 [−0.07, 0.02]	+0.27 [−0.01, 0.54]	−0.05 [−0.14, 0.04]	+0.38 [0.24, 0.51]	+0.10 [−0.14, 0.35]
UltraImage	+0.037 [0.008, 0.067]	+0.00 [−0.02, 0.03]	+0.56 [0.38, 0.73]	+0.08 [0.03, 0.12]	+0.30 [0.21, 0.38]	+0.25 [0.14, 0.37]
DyPE	+1.417 [1.266, 1.561]	+0.78 [0.69, 0.87]	+4.47 [3.85, 5.10]	+2.90 [2.71, 3.10]	+4.73 [4.43, 5.05]	+10.47 [9.70, 11.26]
SEGA	+0.760 [0.632, 0.884]	+0.49 [0.41, 0.56]	+2.44 [1.98, 2.88]	+1.76 [1.61, 1.92]	+3.12 [2.87, 3.37]	+6.68 [6.07, 7.24]
Full attn.	+3.500 [3.402, 3.590]	+2.00 [1.91, 2.09]	+14.02 [13.21, 14.79]	+5.47 [5.27, 5.69]	+9.51 [9.18, 9.84]	+23.42 [22.64, 24.21]
Single-pass SCA	+0.072 [0.010, 0.134]	+0.09 [0.04, 0.13]	+0.25 [−0.02, 0.52]	+0.29 [0.20, 0.38]	+0.48 [0.32, 0.64]	+1.29 [0.99, 1.59]

(b) Image quality

Progressive SCA −	MUSIQ	MUSIQ$_c$	CLIP-IQA$^{\times}$	ARNIQA$^{\times}$
AReA	+2.3 [0.5, 3.9]	+3.7 [2.0, 5.4]	+7.2 [5.6, 8.7]	+7.1 [6.1, 8.2]
HRDiT	+1.3 [0.2, 2.4]	+0.6 [−0.7, 1.8]	−0.9 [−1.8, −0.1]	+2.9 [2.3, 3.4]
UltraImage	+8.1 [7.0, 9.1]	+12.4 [11.5, 13.2]	+2.1 [1.4, 2.7]	+2.1 [1.7, 2.6]
DyPE	+5.4 [3.6, 7.2]	+3.8 [2.0, 5.8]	+15.3 [13.4, 17.3]	+14.2 [12.9, 15.5]
SEGA	+4.3 [2.7, 5.9]	+3.9 [2.2, 5.6]	+8.2 [6.5, 10.0]	+9.7 [8.5, 10.9]
Full attn.	+18.7 [17.4, 20.0]	+21.0 [19.2, 22.8]	+25.9 [24.4, 27.4]	+20.0 [18.7, 21.3]
Single-pass SCA	−3.4 [−5.1, −1.6]	−6.0 [−7.3, −4.6]	+2.6 [1.6, 3.5]	+3.7 [3.2, 4.3]

(c) Set level

Progressive SCA −	FID↓	KID*↓	FID$_p$↓	KID$^{*}_p$↓
AReA	−3.9 [−8.5, 3.1]	−0.09 [−2.75, 2.50]	+4.8 [0.1, 10.6]	+3.35 [0.05, 6.61]
HRDiT	+3.4 [−1.8, 7.2]	+2.51 [0.84, 4.34]	+4.6 [0.6, 7.7]	+2.82 [1.00, 4.63]
UltraImage	−0.2 [−3.8, 2.5]	+0.70 [−0.50, 1.80]	−1.9 [−3.7, 0.5]	−1.66 [−2.85, −0.37]
DyPE	−63.6 [−74.6, −49.8]	−39.51 [−52.24, −29.83]	−64.2 [−73.8, −50.2]	−43.69 [−54.62, −33.89]
SEGA	−23.1 [−34.4, −15.8]	−3.81 [−9.34, 0.46]	−18.2 [−27.1, −9.8]	−8.33 [−14.21, −3.30]
Full attn.	−170.6 [−175.4, −144.2]	−157.07 [−175.94, −138.99]	−144.5 [−141.9, −113.8]	−156.26 [−173.40, −140.53]
Single-pass SCA	+1.6 [−2.3, 7.3]	+1.25 [−0.48, 3.10]	+9.2 [5.1, 13.2]	+5.88 [3.55, 8.25]

Table E.3: **Single-pass SCA beats AReA on all ten per-image metrics, FID, and patch KID at 4096^2.** Expands the single-pass row of Table 1. FLUX.1-dev, 195 prompts. Δ: the method minus AReA, with its 95% paired percentile-bootstrap interval in grey. Intervals use 1000 prompt draws for single-pass SCA (FID, KID, and their patch versions also resample reference-image clusters) and 2000 for full attention's per-image metrics; full attention's set-level differences and campaign time have none. Blue / orange: significantly better / worse than AReA; for FID, KID, and time lower is better. ‡Campaign mean (clean latency: Table E.8). Table E.2 pairs progressive SCA with every baseline. Key result: KID, patch FID, and Inception score are unresolved; every other metric favours single-pass SCA.

		Single-pass SCA		Full attention	
Metric	AReA	Value	Δ 95% interval	Value	Δ 95% interval
Preference ↑					
ImageReward	0.921	1.248	+0.327 [0.247, 0.405]	−2.180	−3.101 [−3.218, −2.984]
Aesthetic	6.27	6.42	+0.15 [0.10, 0.20]	4.50	−1.77 [−1.87, −1.67]
CLIP score	27.69	29.14	+1.45 [1.13, 1.82]	15.37	−12.32 [−13.12, −11.54]
PickScore	22.36	23.12	+0.76 [0.64, 0.88]	17.94	−4.42 [−4.63, −4.22]
HPSv2 (×100)	28.62	29.77	+1.15 [0.96, 1.34]	20.75	−7.87 [−8.24, −7.50]
HPSv2.1 (×100)	27.16	29.73	+2.56 [2.20, 2.97]	7.60	−19.56 [−20.41, −18.72]
Image quality ↑					
MUSIQ	39.4	45.0	+5.6 [3.7, 7.4]	22.9	−16.5 [−18.0, −14.9]
MUSIQ_c	50.4	60.1	+9.7 [8.2, 11.1]	33.1	−17.3 [−19.1, −15.5]
CLIP-IQA (×100)	67.7	72.3	+4.6 [3.1, 6.1]	49.0	−18.7 [−20.5, −16.9]
ARNIQA (×100)	67.4	70.7	+3.4 [2.5, 4.4]	54.4	−12.9 [−14.5, −11.3]
Set level and time					
Inception score ↑	6.66	6.44	−0.22 [−0.65, 0.39]	3.82	−2.84
FID ↓	145.8	140.2	−5.6 [−10.6, −0.4]	312.4	+166.6
KID ($\times 10^3$) ↓	7.17	5.82	−1.35 [−3.17, 0.49]	164.15	+156.98
FID_p ↓	102.9	98.4	−4.4 [−7.9, 0.4]	252.2	+149.3
KID_p ($\times 10^3$) ↓	11.40	8.87	−2.53 [−4.57, −0.60]	171.01	+159.61
Seconds per image‡ ↓	264.4	78.4	−186.0 [−199.6, −173.4]	196.2	−68.1

E.3 Multiple comparisons

Table E.4 counts, for each headline pair, the per-image metrics that stay significant after a Holm correction over the ten per-image metrics. Against AReA, progressive SCA stays significantly better on ten of the ten per-image metrics and single-pass SCA on ten of ten. Against HRDiT, two differences survive, HPSv2 and ARNIQA, both in favour of progressive SCA; MUSIQ (in its favour) and CLIP-IQA (against it) do not.

Table E.4: **Every win over AReA survives Holm correction; against HRDiT, two differences of four do.** FLUX.1-dev at 4096^2. (a) Paired differences, method minus reference, with two-sided bootstrap p-values from the same 2000 prompt draws (seed 0) that give every interval of the paper, before and after Holm's step-down adjustment over the ten per-image metrics of each comparison. Blue / orange: significantly better / worse after adjustment at 5%. $p \leq 0.001$ is the smallest value 2000 draws resolve. (b) Significantly better / worse metrics of ten for every paired comparison of Tables 1 to 3: unadjusted (the 95% interval excludes zero), Holm on the bootstrap p-values, and Holm on paired sign-flip p-values (2000 sign vectors, seed 0) as a check. Every archived interval is reproduced to six digits before a p-value is taken. Key result: progressive and single-pass SCA keep all ten wins over AReA at every seed and on HRGen-1K; against HRDiT, HPSv2 and ARNIQA remain better and MUSIQ and CLIP-IQA no longer differ.

(a) The headline comparisons, metric by metric

	Progressive − AReA			Progressive − HRDiT			Single-pass − AReA		
Metric	Δ	p	p_{Holm}	Δ	p	p_{Holm}	Δ	p	p_{Holm}
Preference									
ImageReward	+0.399	≤0.001	0.010	−0.022	0.453	1.000	+0.327	≤0.001	0.010
Aesthetic	+0.23	≤0.001	0.010	−0.03	0.230	1.000	+0.15	≤0.001	0.010
CLIP score	+1.70	≤0.001	0.010	+0.27	0.056	0.336	+1.45	≤0.001	0.010
PickScore	+1.05	≤0.001	0.010	−0.05	0.249	1.000	+0.76	≤0.001	0.010
HPSv2 (×100)	+1.64	≤0.001	0.010	+0.38	≤0.001	0.010	+1.15	≤0.001	0.010
HPSv2.1 (×100)	+3.86	≤0.001	0.010	+0.10	0.406	1.000	+2.56	≤0.001	0.010
Image quality									
MUSIQ	+2.3	0.009	0.010	+1.3	0.030	0.240	+5.6	≤0.001	0.010
MUSIQ_c	+3.7	≤0.001	0.010	+0.6	0.330	1.000	+9.7	≤0.001	0.010
CLIP-IQA (×100)	+7.2	≤0.001	0.010	−0.9	0.035	0.245	+4.6	≤0.001	0.010
ARNIQA (×100)	+7.1	≤0.001	0.010	+2.9	≤0.001	0.010	+3.4	≤0.001	0.010

(b) Wins / losses of ten for every paired comparison

Comparison	n	95% interval	Holm, bootstrap	Holm, sign-flip
Table 1: progressive SCA against each method (195 prompts, seed 0)				
AReA	195	10 / 0	10 / 0	10 / 0
HRDiT	195	3 / 1	2 / 0	2 / 0
UltraImage	195	9 / 0	9 / 0	9 / 0
DyPE	195	10 / 0	10 / 0	10 / 0
SEGA	195	10 / 0	10 / 0	10 / 0
full attention	195	10 / 0	10 / 0	10 / 0
single-pass SCA	195	7 / 2	7 / 2	6 / 2
Table 2: SCA against AReA, other seeds, prompts, checkpoints				
single-pass, seed 0	195	10 / 0	10 / 0	10 / 0
progressive, seed 1	195	10 / 0	10 / 0	10 / 0
progressive, seed 2	195	10 / 0	10 / 0	10 / 0
progressive, HRGen-1K	1000	10 / 0	10 / 0	10 / 0
single-pass, seed 1	195	10 / 0	10 / 0	10 / 0
single-pass, seed 2	195	10 / 0	10 / 0	10 / 0
single-pass, HRGen-1K	1000	10 / 0	10 / 0	10 / 0
single-pass, FLUX.1-Krea	195	7 / 0	7 / 0	7 / 0
w/o schedule − full attn., FLUX.2-dev	50	9 / 0	9 / 0	9 / 0
Table 3: per-head weights against each variant (reversed: variant losses are per-head wins)				
shared $\pi(\sigma)$, single pass	46	9 / 0	9 / 0	7 / 0
per-layer $\pi_\ell(\sigma)$, single pass	46	5 / 0	5 / 0	5 / 0
shared $\pi(\sigma)$, progressive	195	10 / 0	10 / 0	10 / 0
per-layer $\pi_\ell(\sigma)$, progressive	195	10 / 0	10 / 0	10 / 0
window only, progressive	195	8 / 1	6 / 1	6 / 1
no opening	46	6 / 2	6 / 2	6 / 2
local only	46	8 / 0	7 / 0	7 / 0
global only	46	10 / 0	10 / 0	10 / 0
mean pooling	46	9 / 0	8 / 0	8 / 0

E.4 EVERY BASELINE AGAINST SINGLE-PASS SCA

Table E.5 pairs every baseline of Tables 1 and H.3 with single-pass SCA on the same 195 prompts and seeds, and Figure E.1 plots four of its metrics. Per-image intervals use 2000 prompt draws with seed 0. FID, KID, and their patch versions are paired with SCA without the schedule, over 1000 draws that resample generated prompts and reference-image clusters. Every shared-sampler baseline trails single-pass SCA on all ten per-image metrics, and HRDiT and UltraImage lead on several preference metrics. Appendix H.3 discusses SigMa and adds the 56-step and guidance-7 runs of SCA without the schedule. Figure E.2 shows AReA, DyPE, SEGA, UltraImage, and single-pass SCA on three prompts. UltraImage runs its two-stage defaults (30+30 steps, guidance 3.5 then 7), so that comparison is not equal-cost.

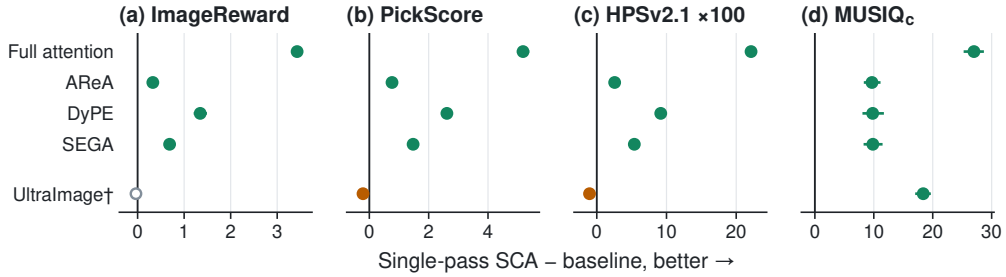


Figure E.1: **Single-pass SCA beats every shared-sampler baseline on all four metrics; against UltraImage the result is mixed.** Single-pass SCA minus each baseline on the same 195 prompts at 4096^2, seed 0; whiskers are 95% paired intervals over prompts (2000 bootstrap draws), not adjusted for multiplicity. Filled: the interval excludes zero (green: SCA better, orange: worse); open grey: it includes zero. Against UltraImage, SCA is better on MUSIQ_c and worse on PickScore and HPSv2.1; †UltraImage runs its own two-stage recipe, so this is not an equal-cost comparison. Table E.5 gives all metrics.

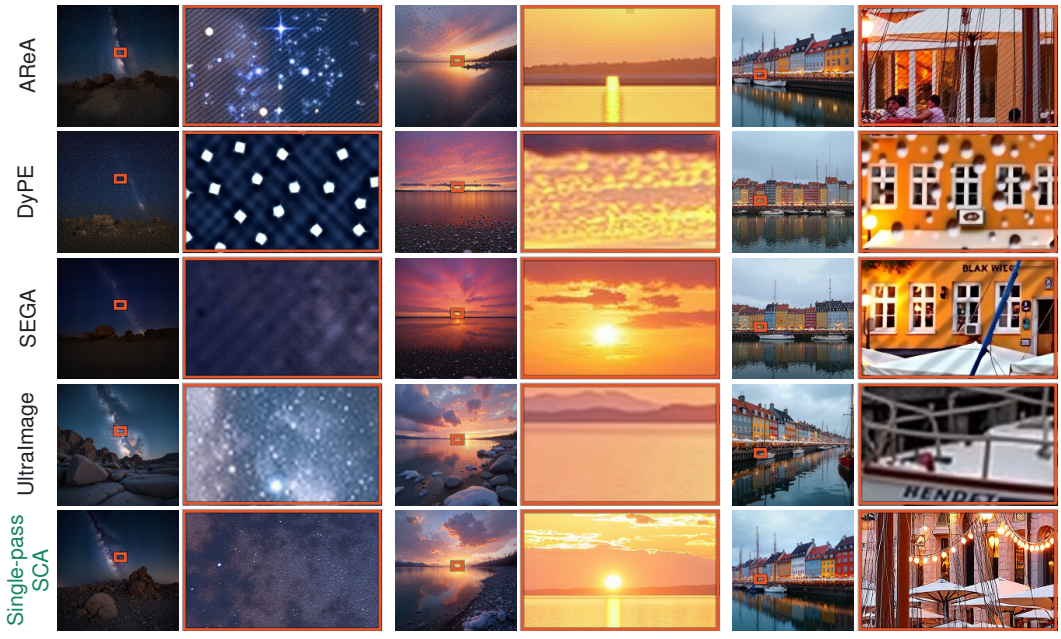


Figure E.2: **Single-pass SCA beside AReA, DyPE, SEGA, and UltraImage.** 4096^2, three evaluation prompts that name no person (keyword rule) not shown in an earlier sample figure, chosen by a fixed seed (seed 17); seed 0 for every image. UltraImage runs its two-stage defaults. Each crop shows native pixels of the window that maximises the smallest edge energy among the five methods. Over all 195 prompts, AReA, DyPE, and SEGA trail single-pass SCA on all ten per-image metrics; UltraImage is significantly better on aesthetic score, PickScore, and HPSv2.1 and worse on both MUSIQ scores (Table E.5).

Table E.5: **Every shared-sampler baseline trails single-pass SCA on all ten per-image metrics; HRDiT and UltraImage lead on several preference metrics.** Expands the baseline rows of Table 1. FLUX.1-dev at 4096^2, 195 prompts, seed 0: each baseline minus single-pass SCA in (a) and (b), and minus SCA without the schedule in (c), since set-level intervals across runs do not exist. Grey: the 95% paired percentile-bootstrap interval, over 2000 prompt draws per image and 1000 draws that also resample reference-image clusters for FID, KID, and their patch versions. HPS, CLIP-IQA, and ARNIQA $\times 100$; KID $\times 10^3$; time: campaign mean in seconds. Blue / orange: baseline significantly better / worse than SCA; in (c) lower is better. SigMa uses its dev-matched profile, UltraImage and HRDiT their own recipes (Appendix D.8). Full attention's FID_p lies outside its percentile interval, as a biased estimate can. Key result: UltraImage and HRDiT are significantly better on Aesthetic, PickScore, and HPSv2.1, worse on both MUSIQ scores, and higher in patch FID and patch KID.

(a) Preference: baseline − single-pass SCA

Baseline	ImageReward	Aesthetic	CLIP score	PickScore	HPSv2	HPSv2.1
Single pass, shared sampler						
Full attn.	−3.428 [−3.530, −3.327]	−1.92 [−2.00, −1.83]	−13.77 [−14.54, −13.01]	−5.18 [−5.39, −4.98]	−9.02 [−9.34, −8.69]	−22.13 [−22.92, −21.35]
DyPE	−1.345 [−1.490, −1.207]	−0.69 [−0.79, −0.60]	−4.22 [−4.83, −3.63]	−2.61 [−2.81, −2.43]	−4.25 [−4.55, −3.97]	−9.18 [−9.91, −8.46]
SigMa	−1.771 [−1.920, −1.624]	−0.82 [−0.92, −0.72]	−5.64 [−6.28, −4.98]	−3.11 [−3.32, −2.90]	−4.99 [−5.31, −4.71]	−11.08 [−11.83, −10.35]
SEGA	−0.688 [−0.803, −0.573]	−0.40 [−0.48, −0.33]	−2.19 [−2.62, −1.79]	−1.48 [−1.63, −1.33]	−2.64 [−2.88, −2.40]	−5.39 [−5.91, −4.82]
AReA	−0.327 [−0.405, −0.250]	−0.15 [−0.20, −0.09]	−1.45 [−1.80, −1.12]	−0.76 [−0.88, −0.66]	−1.15 [−1.34, −0.96]	−2.56 [−2.94, −2.18]
Multi-stage, own recipe						
UltraImage	+0.035 [−0.031, 0.102]	+0.08 [0.04, 0.12]	−0.31 [−0.60, −0.01]	+0.21 [0.12, 0.31]	+0.19 [0.03, 0.35]	+1.04 [0.75, 1.33]
HRDiT	+0.094 [0.032, 0.157]	+0.11 [0.07, 0.16]	−0.02 [−0.29, 0.27]	+0.34 [0.25, 0.43]	+0.11 [−0.05, 0.27]	+1.19 [0.92, 1.47]

(b) Image quality: baseline − single-pass SCA

Baseline	MUSIQ	MUSIQ_c	CLIP-IQA	ARNIQA
Single pass, shared sampler				
Full attn.	−22.1 [−23.7, −20.5]	−27.0 [−28.7, −25.2]	−23.3 [−24.8, −21.9]	−16.3 [−17.7, −14.9]
DyPE	−8.7 [−10.8, −6.6]	−9.8 [−11.7, −8.1]	−12.8 [−14.6, −11.0]	−10.5 [−11.9, −9.1]
SigMa	−9.5 [−11.5, −7.4]	−10.2 [−12.1, −8.5]	−14.3 [−16.3, −12.4]	−10.9 [−12.3, −9.5]
SEGA	−7.7 [−9.5, −5.9]	−9.8 [−11.5, −8.3]	−5.7 [−7.3, −4.2]	−6.0 [−7.2, −4.8]
AReA	−5.6 [−7.4, −3.9]	−9.7 [−11.1, −8.3]	−4.6 [−6.1, −3.2]	−3.4 [−4.3, −2.5]
Multi-stage, own recipe				
UltraImage	−11.4 [−13.3, −9.5]	−18.4 [−19.7, −17.0]	+0.5 [−0.6, 1.6]	+1.6 [0.9, 2.3]
HRDiT	−4.6 [−6.2, −2.9]	−6.6 [−7.9, −5.3]	+3.5 [2.6, 4.4]	+0.9 [0.2, 1.5]

(c) Set level and time: baseline − SCA without the schedule

Baseline	FID ↓	KID ↓	FID_p ↓	KID_p ↓	Time (s) ↓
Single pass, shared sampler					
Full attn.	+174.4 [150.2, 179.4]	+158.0 [140.3, 176.0]	+153.1 [122.3, 149.6]	+161.9 [146.6, 178.2]	+124.7 [120.8, 129.4]
DyPE	+67.4 [56.2, 80.3]	+40.4 [30.6, 52.9]	+72.8 [58.7, 81.6]	+49.3 [40.1, 60.0]	+124.4 [120.5, 128.6]
SigMa	+78.9 [66.7, 91.0]	+50.4 [39.1, 64.0]	+85.5 [70.2, 93.7]	+62.4 [52.0, 74.7]	+121.0 [119.0, 123.0]
SEGA	+26.9 [21.1, 39.1]	+4.7 [0.1, 10.5]	+26.7 [18.8, 35.3]	+14.0 [9.3, 20.0]	+118.9 [118.0, 119.7]
AReA	+7.7 [2.6, 13.1]	+1.0 [−1.5, 3.2]	+3.8 [−1.3, 7.5]	+2.3 [−0.1, 4.6]	+192.9 [180.3, 206.4]
Multi-stage, own recipe					
UltraImage	+4.0 [1.1, 10.2]	+0.2 [−1.8, 2.2]	+10.4 [6.2, 13.9]	+7.3 [5.1, 9.5]	+237.0 [233.5, 241.7]
HRDiT	+0.4 [−2.3, 6.4]	−1.6 [−3.6, 0.3]	+3.9 [0.7, 8.0]	+2.8 [0.9, 4.8]	+38.8 [37.2, 40.5]

E.5 Ablations

Table E.6 expands Table 3 to every metric for the single-pass variants, and Table E.7 expands its progressive rows. The single-pass variants run on the fixed 46-prompt subset and pair against SCA without the schedule. Figure E.3 shows the variants on prompts of that subset. Inside progressive SCA, the four runs share the three-stage schedule, the window, the bridge, and the per-prompt seeds. They differ only in the weights. Per-head π_h is the default; shared $\pi(\sigma)$ is the mean of the table over every layer and head at each noise level; per-layer $\pi_\ell(\sigma)$ is the mean over each layer's heads; and window only sets $\pi = 1$, which keeps the window and the bridge alone. Clean batch-1 time is the same for all four (49.3 to 49.4 s), so the comparison is one of quality.

Table E.6: **Every single-pass variant is significantly worse than per-head weights on five to ten per-image metrics.** Expands the single-pass rows of Table 3. FLUX.1-dev at 4096^2, the fixed 46-prompt ablation subset, SCA without the schedule throughout. The shaded row gives absolute values; every other cell is the variant minus that reference, with its 95% paired interval below it in grey (2000 prompt draws). $^\times$: $\times 100$. Blue / orange: the interval excludes zero in favour of / against the variant. (c) Patch FID is set-level and has no paired interval at 46 prompts; output error is the relative error of each weight rule at emulated ratio 4 (Section 4.3); mean pooling has none because reach emulation scores block-centre keys only, and the no-opening variant keeps the per-head table. Table E.4 gives the Holm-adjusted counts. Key result: shared and per-layer weights, which raise the output error, lose nine and five per-image metrics.

(a) Preference

Variant	IR	Aesthetic	CLIP	PickScore	HPSv2$^\times$	HPSv2.1$^\times$
per-head π_h	1.124	6.45	29.41	22.96	29.71	29.98
shared $\pi(\sigma)$	−0.090	−0.09	−0.38	−0.38	−0.96	−1.83
	[−0.173, −0.012]	[−0.18, −0.02]	[−0.78, 0.08]	[−0.51, −0.26]	[−1.19, −0.73]	[−2.24, −1.43]
per-layer $\pi_\ell(\sigma)$	−0.054	+0.01	−0.13	−0.16	−0.50	−0.70
	[−0.159, 0.030]	[−0.06, 0.07]	[−0.54, 0.28]	[−0.28, −0.05]	[−0.63, −0.37]	[−0.98, −0.41]
local only	−0.357	−0.21	−1.93	−1.20	−1.78	−2.86
	[−0.524, −0.207]	[−0.35, −0.07]	[−2.68, −1.20]	[−1.44, −0.99]	[−2.07, −1.50]	[−3.35, −2.36]
global only	−0.395	−0.76	−3.48	−1.44	−3.32	−5.66
	[−0.507, −0.295]	[−0.87, −0.64]	[−4.43, −2.59]	[−1.66, −1.22]	[−3.72, −2.94]	[−6.35, −4.98]
mean pooling	−1.518	−0.61	−3.59	−2.60	−4.04	−9.75
	[−1.847, −1.193]	[−0.77, −0.44]	[−4.76, −2.49]	[−3.00, −2.17]	[−4.66, −3.43]	[−11.30, −8.17]
no opening	−1.328	−0.08	−1.65	−2.18	−1.89	−4.07
	[−1.595, −1.064]	[−0.17, 0.01]	[−2.58, −0.84]	[−2.51, −1.85]	[−2.34, −1.45]	[−5.11, −3.04]

(b) Image quality

Variant	MUSIQ	MUSIQ$_c$	CLIP-IQA$^\times$	ARNIQA$^\times$
per-head π_h	46.1	62.5	71.3	70.5
shared $\pi(\sigma)$	−4.9	−10.4	−7.7	−6.5
	[−7.9, −1.9]	[−12.8, −8.3]	[−9.4, −6.0]	[−8.5, −4.8]
per-layer $\pi_\ell(\sigma)$	+0.7	−2.4	−0.9	−1.5
	[−1.6, 3.0]	[−3.6, −1.1]	[−2.0, 0.1]	[−2.3, −0.9]
local only	−3.5	−0.3	−4.6	−1.0
	[−6.5, −0.7]	[−2.5, 1.7]	[−6.3, −3.0]	[−2.0, 0.0]
global only	−3.5	−11.7	−27.0	−45.1
	[−6.7, −0.2]	[−13.6, −9.8]	[−30.2, −23.8]	[−47.6, −42.5]
mean pooling	−1.8	−4.2	−10.1	−4.8
	[−5.3, 1.9]	[−8.4, −0.5]	[−13.0, −7.1]	[−6.7, −2.6]
no opening	−5.6	+2.5	+2.9	+4.3
	[−9.0, −2.1]	[−0.4, 5.5]	[1.0, 4.9]	[2.7, 6.1]

(c) Patch FID and output error, absolute values

Variant	FID$_p$ ↓	Output error↓
per-head π_h	168.0	0.200
shared $\pi(\sigma)$	166.7	0.230
per-layer $\pi_\ell(\sigma)$	167.7	0.223
local only	192.2	0.261
global only	237.0	0.627
mean pooling	184.6	–
no opening	202.6	0.200

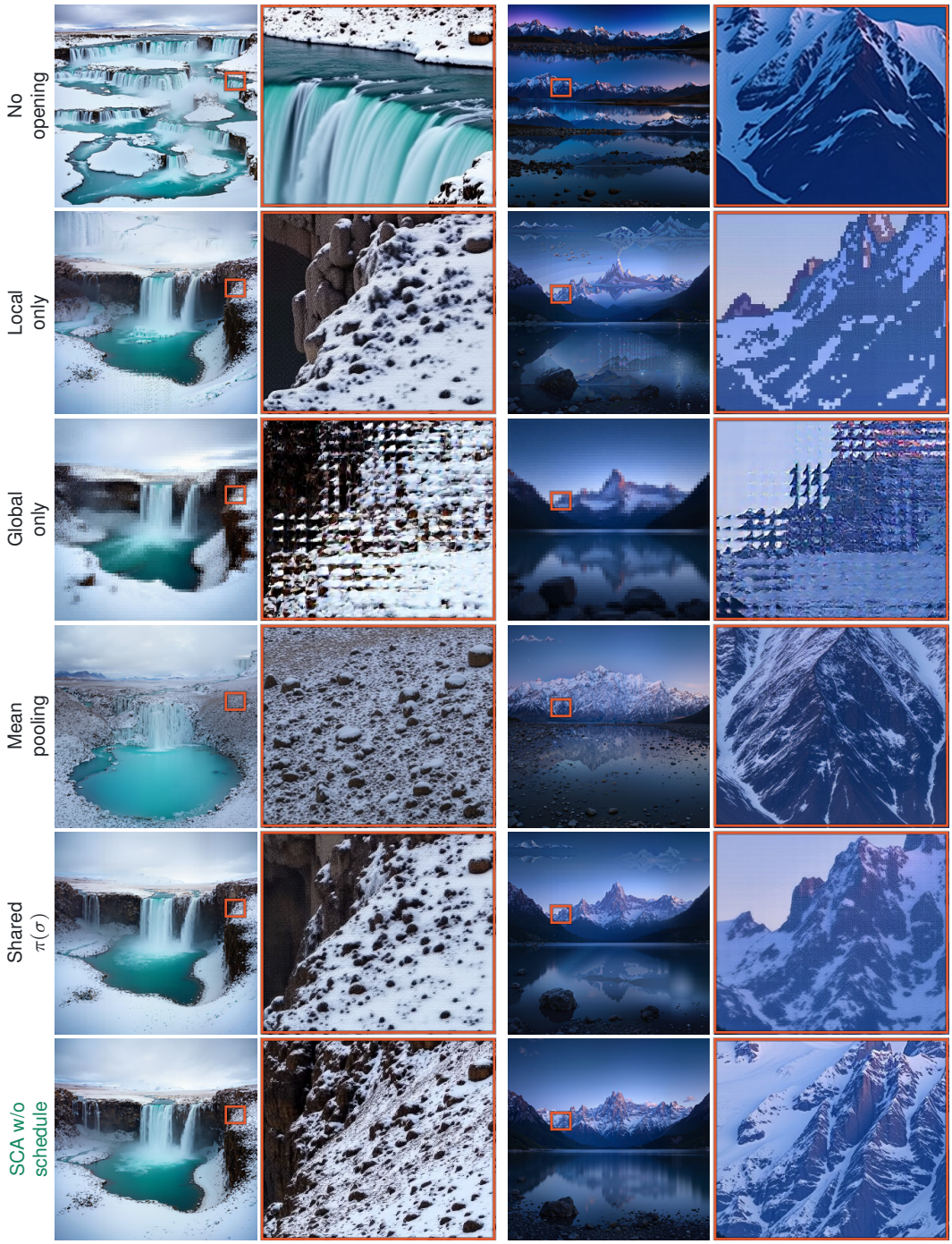


Figure E.3: **Without the opening phase, subjects shrink to a small scale.** Two of the 46 ablation prompts that name no person (keyword rule), chosen by a fixed seed (seed 46); 4096^2, seed 0. Each row changes one part of SCA without the schedule (Table 3): local only flattens texture into blocks, global only leaves a grid, and mean pooling moves the layout. Each crop shows native pixels of the window that maximises the smallest edge energy among the six variants. Over the 46 ablation prompts, dropping the opening phase changes ImageReward by -1.33 $[-1.60, -1.06]$ (Table E.6).

Table E.7: **Inside the progressive sampler, per-head weights win every per-image metric; coarser weights have lower patch FID.** Expands the progressive rows of Table 3. FLUX.1-dev, 195 Aesthetic-4K prompts at 4096^2, seed 0. Each cell is the variant minus progressive SCA with per-head weights, with its 95% paired interval below it in grey: 2000 prompt draws for the per-image metrics, and for FID, KID, and their patch versions 1000 draws that also resample the 168 reference-image clusters. $^{\times}$: $\times 100$; *: $\times 1000$. Blue / orange: the interval excludes zero in favour of / against the variant; for FID and KID lower is better. All four runs share the schedule, the window, the bridge, and the seeds, and differ only in the weights π. Key result: shared and per-layer weights lose all ten per-image metrics and lower patch FID; window-only attention raises it.

(a) Preference

Variant − per-head	IR	Aesthetic	CLIP	PickScore	HPSv2$^{\times}$	HPSv2.1$^{\times}$
shared $\pi(\sigma)$	−0.014 [−0.025, −0.004]	−0.08 [−0.09, −0.06]	−0.32 [−0.44, −0.20]	−0.07 [−0.09, −0.05]	−0.19 [−0.24, −0.13]	−0.38 [−0.45, −0.30]
per-layer $\pi_\ell(\sigma)$	−0.012 [−0.021, −0.003]	−0.04 [−0.06, −0.03]	−0.37 [−0.49, −0.25]	−0.07 [−0.09, −0.05]	−0.28 [−0.33, −0.22]	−0.39 [−0.46, −0.32]
window only ($\pi = 1$)	−0.020 [−0.038, −0.004]	−0.08 [−0.12, −0.03]	−0.57 [−0.82, −0.33]	−0.14 [−0.20, −0.09]	−0.28 [−0.36, −0.20]	−0.36 [−0.48, −0.25]

(b) Image quality

Variant − per-head	MUSIQ	MUSIQ$_c$	CLIP-IQA$^{\times}$	ARNIQA$^{\times}$
shared $\pi(\sigma)$	−3.8 [−4.6, −2.9]	−4.3 [−4.8, −3.8]	−3.6 [−4.1, −3.0]	−3.5 [−3.9, −3.2]
per-layer $\pi_\ell(\sigma)$	−2.2 [−2.8, −1.5]	−2.8 [−3.2, −2.4]	−1.8 [−2.1, −1.3]	−2.6 [−3.0, −2.3]
window only ($\pi = 1$)	−0.1 [−0.7, 0.4]	+0.5 [0.2, 0.9]	−1.6 [−2.2, −1.0]	−0.3 [−0.5, −0.01]

(c) Set level

Variant − per-head	FID↓	KID* ↓	FID$_p$ ↓	KID$^{*}_p$ ↓
shared $\pi(\sigma)$	−0.2 [−2.4, 2.0]	−0.28 [−1.31, 0.88]	−7.3 [−10.3, −5.2]	−4.70 [−6.90, −2.64]
per-layer $\pi_\ell(\sigma)$	+0.0 [−1.9, 1.9]	−0.02 [−0.85, 0.99]	−6.4 [−8.8, −4.8]	−4.20 [−5.92, −2.57]
window only ($\pi = 1$)	−0.4 [−2.4, 1.7]	−0.29 [−1.33, 0.58]	+6.0 [3.9, 9.9]	+4.38 [2.51, 6.53]

E.6 Latency

Table E.8 and Figure E.4 expand the latency of Figure 7 and Section 4.2. All times follow Appendix D.4.

Table E.8: **Single-pass SCA is the fastest method at every size.** Expands the timing of Section 4.2 and Figure 7. Seconds per image of full attention, AReA, and single-pass SCA: batch 1 on an idle GH200, 28 steps with text encoding and VAE decoding, median of three timed images after an excluded warm-up (Appendix D.4). N: image tokens; full attention was not run at 8192^2; bold: fastest per size. Key result: single-pass SCA is 2.6 times faster than AReA at 4096^2.

	Seconds per image ↓		
Size (N tokens)	Full attention	AReA	Single-pass SCA
2048^2 (16,384)	22.9	41.0	**19.8**
3072^2 (36,864)	73.9	114.3	**44.3**
4096^2 (65,536)	186.5	202.4	**77.1**
6144^2 (147,456)	762.7	482.3	**174.7**
8192^2 (262,144)	not run	952.0	**309.7**

Clean times. At 4096^2 the medians are 186.5 s for full attention (186.0, 186.6, 186.5), 202.4 s for AReA (192.7, 202.4, 204.3), and 77.1 s for single-pass SCA (77.3, 77.1, 77.1). At 2048^2 and 3072^2 they are 22.9 and 73.9 s for full attention, 41.0 and 114.3 s for AReA, and 19.8 and 44.3 s for single-pass SCA. At 6144^2 and 8192^2, AReA takes 482.3 and 952.0 s and single-pass SCA 174.7 and 309.7 s, a reduction of 63.8% and 67.5%. At 6144^2 full attention takes 762.7 s (763.5, 762.7, 762.5), $4.4\times$ the time of single-pass SCA. The schedule and the bridge add 8.4% at 4096^2 and 8 to 11% across sizes. In run T3 progressive SCA, HRDiT, and AReA take 49.2, 107.0, and 202.4 s.

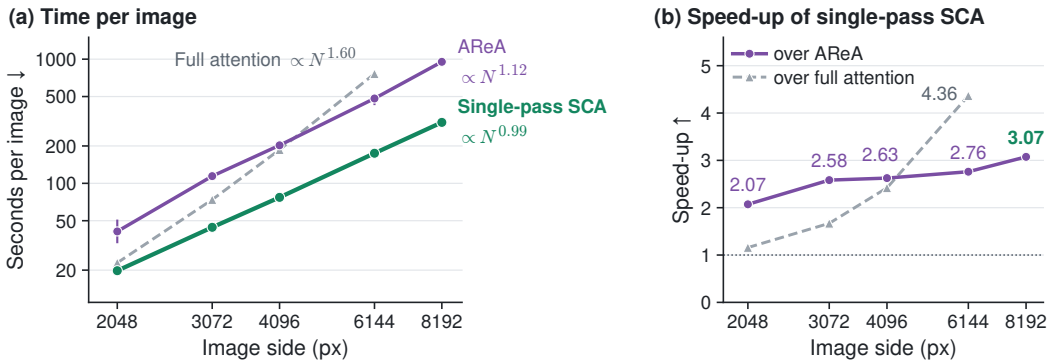


Figure E.4: **Single-pass SCA is the fastest method at every size; its lead over AReA grows to 3.07× at 8192^2.** (a) Median seconds per image of three timed images after one excluded warm-up (batch 1, one GH200, 28 steps including text encoding and VAE decoding), whiskers min to max; labels give b of seconds $\propto N^b$ (N image tokens). (b) Each baseline's median time over single-pass SCA's; dotted: parity. Full attention was not run at 8192^2. Table E.8 lists the times.

A separate 4096^2 run. A separate batch-1 run at 4096^2 gave 184.6 s for full attention (184.1, 184.7, 184.6), 212.6 s for AReA (207.3, 225.8, 212.6), and 71.0 s for SCA without the schedule (71.0, 83.1, 70.9).

Non-square sizes. Here we have only campaign means over 195 prompts on shared nodes. Single-pass SCA takes 41.1 s (38.1 s without the schedule) and AReA 112.4 s at 2048×4096, and 60.5 s (55.8 s) and 150.5 s at 2048×6144.

F Progressive SCA

This appendix expands Section 3.3 and the progressive rows of Table 1. Appendix F.1 gives the protocol, and Appendix F.2 compares progressive SCA with HRDiT and with single-pass SCA (Table F.1, Figures F.1 to F.3). Algorithm 5 writes out the sampler, and Table E.2 pairs it with every baseline; Figures L.3 to L.1 show more samples.

F.1 Protocol

Schedule. Progressive SCA is HRDiT's released three-stage schedule (Xue et al., 2026) with SCA attention. It keeps HRDiT's schedule, transitions, guidance, sharpening, and momentum, and changes the attention. Our runner reimplements the schedule with the stock diffusers FLUX components. One 30-step schedule with dynamic shift $\mu = 1.15$, from the 1024^2 sequence length, serves all stages and reproduces the recorded HRDiT noise levels to 10^{-6}. The first stage runs all 30 steps at 1024^2 with FLUX.1-dev's stock attention and native RoPE, at guidance 3.5. Each transition decodes the latent, upsamples it bicubically, and applies HRDiT's unsharp mask with a blur width drawn from the per-prompt global seed. It then re-encodes the image and re-noises it with the per-prompt generator. The 2048^2 stage re-enters at $\sigma = 0.81$ and runs the last 17 steps at guidance 4.5. The 4096^2 stage re-enters at $\sigma = 0.61$ and runs the last 10 at guidance 6.0.

Attention. In both high-resolution stages SCA replaces HRDiT's four attention changes: NTK-scaled RoPE, a proportional attention scale, per-head key pruning, and position alignment. It uses $n_0 = 64$, $t = 8$, the 1,024 bridge tokens, and the FLUX.1-dev table read at ratio 2 at 2048^2 and at ratio 4 at 4096^2. Both re-entry levels lie below the gate $\sigma_g = 0.89$ and below the wide-window steps ($0.82 < \sigma \leq 0.89$), so no step runs the opening phase or a wide window. Every high-resolution step therefore uses the native window, the bridge, and the fitted per-head mix. HRDiT's low-band structure guidance and velocity momentum are kept, and both methods use the same per-prompt seeds. The gate and every other setting were fixed before generation.

What else differs from HRDiT. Against HRDiT's own runs, three things change together: the attention, the precision (HRDiT's released runner uses fp16, ours bf16), and the runner. The com-

parison therefore does not isolate the attention. Our runner checks for finite values at the end of each stage instead of after every step. With bf16 and stock components the 1024^2 images, and hence the layouts, differ from HRDiT's for the same seed (Figure F.2). Reported time is CUDA-synchronized and covers text encoding, all stages, transitions, and the final decode, and excludes model loading.

F.2 Against HRDiT and single-pass SCA

Table E.1 gives the absolute values of progressive SCA, Table E.2 its difference from every baseline, and Table F.1 the per-image comparisons with HRDiT and with single-pass SCA. Set-level intervals use 1000 draws over prompts and reference-image clusters.

Table F.1: **Per image, progressive SCA is significantly worse than HRDiT only on CLIP-IQA, and on none after Holm correction.** Expands the progressive rows of Table 1. FLUX.1-dev at 4096^2, 195 prompts, seed label 0. Each cell is the first-named setting minus the second, with its 95% paired percentile-bootstrap interval over 2000 prompt draws in grey. Blue / orange: the interval excludes zero in favour of / against the first-named setting. Progressive SCA runs HRDiT's schedule with SCA attention (Section 3.3), and single-pass SCA takes 28 steps. The last two columns use SCA without the schedule, at 42 steps sized to HRDiT's campaign time and at 28. Intervals that include zero do not establish equivalence; Table E.4 gives the Holm-adjusted calls. Key result: after Holm correction progressive SCA stays better than HRDiT on HPSv2 and ARNIQA, and better than single-pass SCA on seven of ten metrics and worse on MUSIQ and MUSIQ_c.

	Progressive SCA −		SCA without the schedule	
Metric	HRDiT	single-pass SCA	42 − 28 steps	HRDiT − 42 steps
Preference ↑				
ImageReward	−0.022 [−0.088, 0.034]	+0.072 [0.010, 0.134]	+0.030 [−0.001, 0.059]	+0.083 [0.023, 0.150]
Aesthetic	−0.027 [−0.070, 0.017]	+0.087 [0.038, 0.130]	+0.020 [−0.007, 0.047]	+0.071 [0.022, 0.119]
CLIP score	+0.27 [−0.01, 0.54]	+0.25 [−0.02, 0.52]	−0.01 [−0.19, 0.20]	−0.02 [−0.31, 0.27]
PickScore	−0.05 [−0.14, 0.04]	+0.29 [0.20, 0.38]	+0.08 [0.03, 0.13]	+0.29 [0.19, 0.39]
HPSv2 (×100)	+0.38 [0.24, 0.51]	+0.48 [0.32, 0.64]	+0.23 [0.15, 0.31]	−0.10 [−0.26, 0.07]
HPSv2.1 (×100)	+0.10 [−0.14, 0.35]	+1.29 [0.99, 1.59]	+0.60 [0.44, 0.75]	+0.65 [0.38, 0.93]
Image quality ↑				
MUSIQ	+1.3 [0.2, 2.4]	−3.4 [−5.1, −1.6]	−0.4 [−1.5, 0.7]	−4.7 [−6.3, −2.9]
MUSIQ_c	+0.6 [−0.7, 1.8]	−6.0 [−7.3, −4.6]	+1.9 [1.3, 2.5]	−9.6 [−10.9, −8.4]
CLIP-IQA (×100)	−0.9 [−1.8, −0.1]	+2.6 [1.6, 3.5]	+0.4 [−0.1, 0.8]	+2.4 [1.5, 3.3]
ARNIQA (×100)	+2.9 [2.3, 3.4]	+3.7 [3.2, 4.3]	+1.1 [0.7, 1.4]	−0.7 [−1.4, −0.1]

Against HRDiT. Before correction for multiple comparisons, progressive SCA is significantly better on HPSv2, MUSIQ, and ARNIQA, worse only on CLIP-IQA, and within noise on the other six per-image metrics (Table F.1). After a Holm correction over the ten per-image metrics, HPSv2 and ARNIQA remain and no metric is worse (Table E.4). HRDiT has lower KID, patch FID (103.0 against 107.7; progressive minus HRDiT $+4.6$ $[0.6, 7.7]$), and patch KID, while the FID difference (141.9 against 138.4) spans zero. Progressive SCA takes 49.2 s per image and HRDiT 107.0 s in run T3, and 50.7 and 110.4 s as campaign means on shared nodes. Where the four methods draw the same content, progressive SCA's crop detail is level with HRDiT's (Figure F.1); over all 195 prompts the MUSIQ over four 1024^2 crops differ by $+0.6$ $[-0.7, 1.8]$.

Against single-pass SCA. Progressive SCA gains seven per-image metrics, ties on CLIP ($+0.25$ $[-0.02, 0.52]$), and loses both MUSIQ scores (Table F.1). Its ImageReward gain is the weakest of the seven: it survives the Holm correction with bootstrap p-values but not with the sign-flip test (Table E.4). Its patch FID ($+9.2$ $[5.1, 13.2]$) and patch KID are significantly higher, and its FID and KID differences span zero. It takes 27.7 s less per image in campaign means (50.7 against 78.4 s) and 27.9 s less in clean times (49.2 against 77.1 s). Figure F.2 shows that progressive SCA keeps a large subject, as HRDiT does, while single-pass SCA draws it smaller. Figure F.3 compares five methods crop by crop, and Figure L.1 sets the two modes side by side on six more prompts.

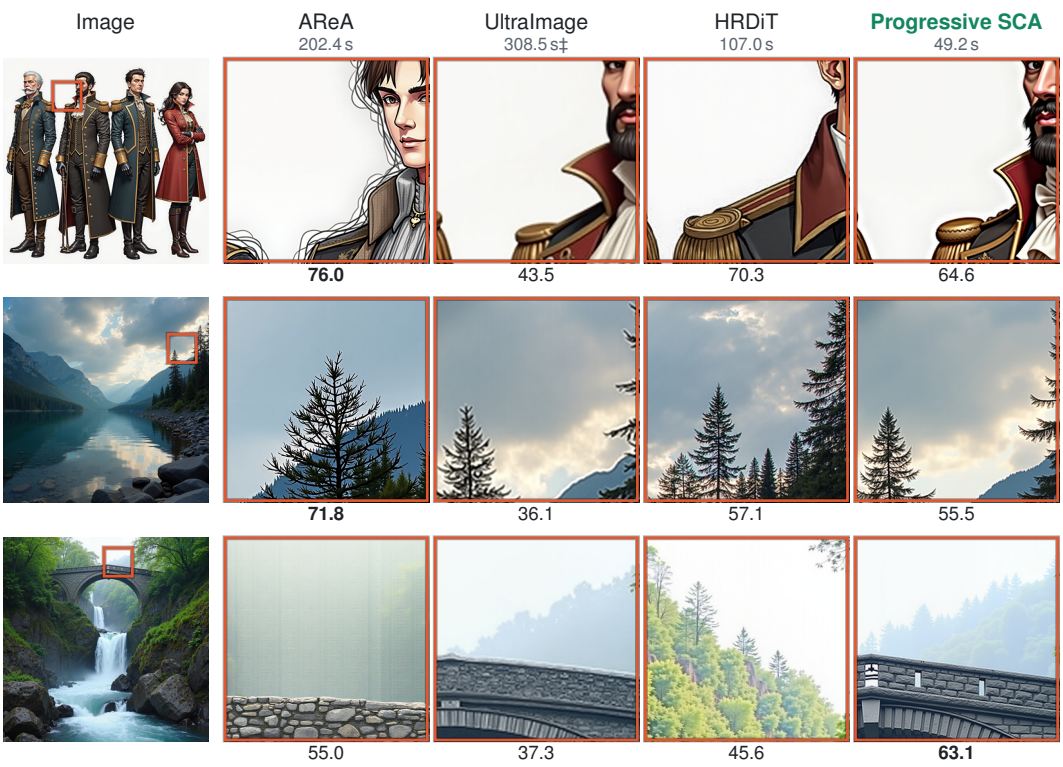


Figure F.1: **Where the four methods draw the same content, progressive SCA's crop detail is level with HRDiT's.** Per category, the evaluation prompt and 512×512 window on which the four methods agree most in content among high-detail windows, seed 0; AReA's and HRDiT's own layouts still differ in some rows. Numbers: MUSIQ of each crop, best bold; times: clean seconds per 4096^2 image (‡ campaign mean). On these three crops MUSIQ is lowest for UltraImage in every row and highest for AReA in two. Over all 195 prompts, progressive SCA's MUSIQ over four 1024^2 crops is indistinguishable from HRDiT's ($+0.6$ $[-0.7, 1.8]$) and exceeds AReA's by $+3.7$ $[2.0, 5.4]$ and UltraImage's by $+12.4$ $[11.5, 13.2]$ (Table E.2).

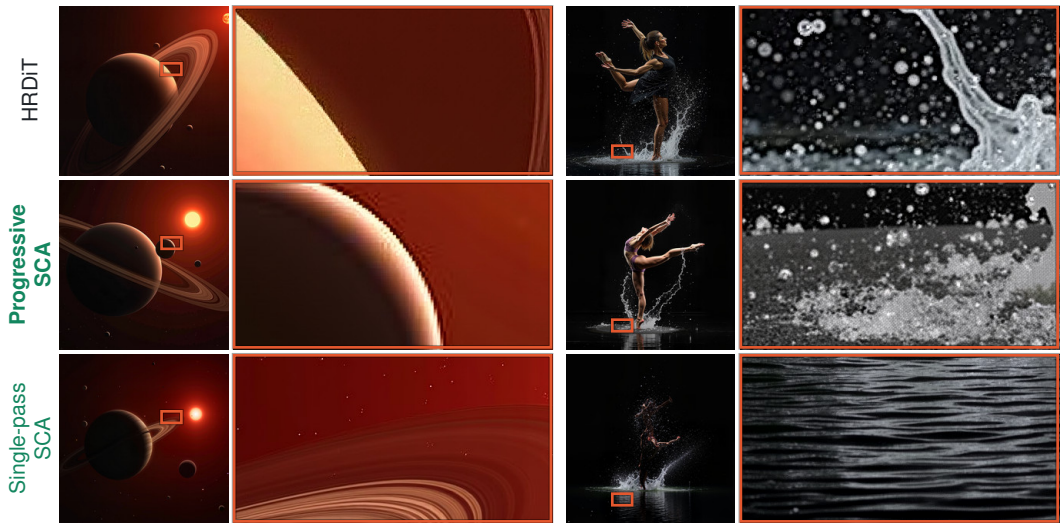


Figure F.2: **Progressive SCA keeps HRDiT's large subject; single-pass SCA draws it smaller.** Two prompts of the 46-prompt panel chosen by a fixed seed (seed 17); seed 0 for every image. Each crop shows native pixels of the window that maximises the smallest edge energy among the three methods. Over all 195 prompts, progressive minus single-pass SCA is -6.0 $[-7.3, -4.6]$ on MUSIQ over four 1024^2 crops and $+0.07$ $[0.01, 0.13]$ on ImageReward (Table F.1).

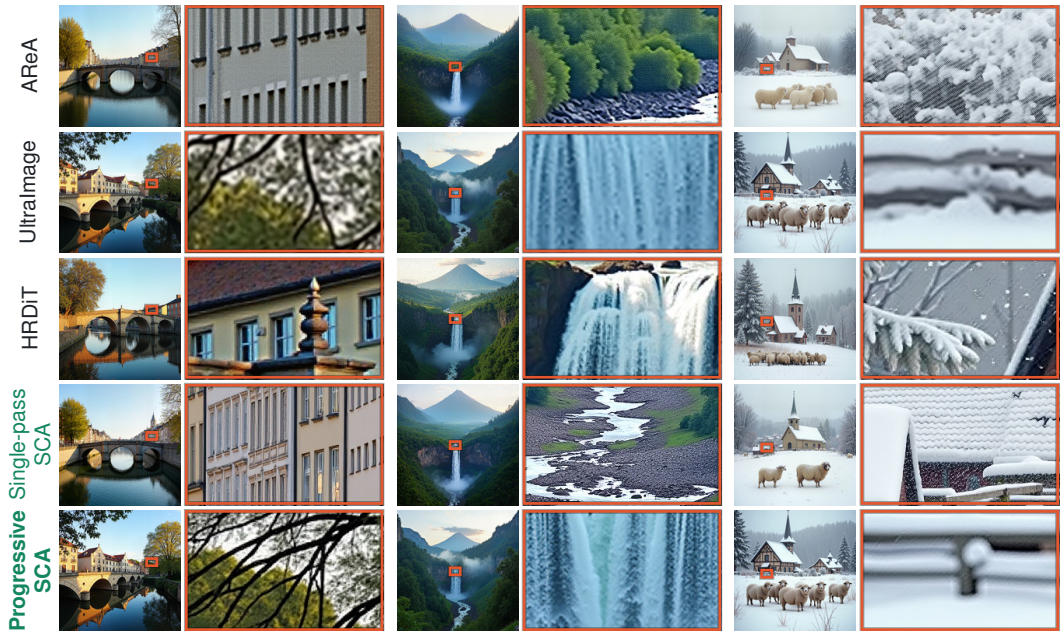


Figure F.3: **Five methods crop by crop at 4096^2.** Three evaluation prompts that name no person (keyword rule), chosen by a fixed seed (seed 4) outside the other progressive figures; seed 0 for every image. Each crop shows native pixels of the window that maximises the smallest edge energy among the five methods. Over all 195 prompts, progressive SCA's MUSIQ over four 1024^2 crops exceeds AReA's by $+3.7$ $[2.0, 5.4]$, is indistinguishable from HRDiT's ($+0.6$ $[-0.7, 1.8]$), and trails single-pass SCA's (-6.0 $[-7.3, -4.6]$; Tables E.2 and F.1).

G GENERALITY

This appendix expands Table 2 and Section 4.2. Table G.1 expands every row of Table 2 to all ten per-image metrics with intervals (Appendix G.1). Appendix G.2 covers 6144^2 and FLUX.1-Krea-dev, Appendix G.3 the non-square sizes, and Appendix G.4 the exploratory 8192^2 run.

G.1 SEEDS, HRGEN-1K, AND FLUX.2-DEV

Table G.1 gives every row of Table 2 on all ten per-image metrics with intervals: progressive SCA against AReA at three seeds and on HRGen-1K; single-pass SCA against AReA on HRGen-1K, at seeds 1 and 2, and on FLUX.1-Krea-dev; and SCA without the schedule against full attention on FLUX.2-dev (Appendix D.8). Table E.2 expands the progressive row at seed 0 to every metric. Figure G.1 shows four FLUX.2-dev prompts; FLUX.2-dev runs at guidance 4.0 with its own reach table.

G.2 6144^2 AND FLUX.1-KREA-DEV

Table G.2 gives every metric with its paired interval for the 100-prompt 6144^2 comparison, which uses SCA without the schedule, and for FLUX.1-Krea-dev. On FLUX.1-Krea-dev, DyPE and SEGA trail AReA on ImageReward, and single-pass SCA does not. The Krea reach table follows the FLUX.1-dev procedure with 16 prompts at 1024^2 (Appendix D.7). The progressive sampler does not support FLUX.1-Krea-dev, so every Krea row is single-pass SCA.

Figure G.2 shows four 6144^2 prompts, and Figure G.3 two FLUX.1-Krea-dev prompts with AReA, DyPE, SEGA, and single-pass SCA. The Krea images use the FLUX.1-Krea-dev transformer with FLUX.1-dev text encoders and VAE and a reach table fitted on FLUX.1-Krea-dev. Their DyPE and SEGA images come from the recorded FLUX.1-Krea-dev evaluation runs. In those runs seven SEGA outputs were regenerated from an archived runtime because their original source snapshots could not be recovered; neither prompt shown is among them. The inherited Krea baselines span several runtime revisions.

Table G.1: **Table 2 with all ten per-image metrics.** FLUX.1-dev at 4096^2 unless named; each cell is the paired difference (HPS, CLIP-IQA, ARNIQA $\times 100$) with its 95% interval below it in grey (2000 prompt draws, seed 0). Blue / orange: the interval excludes zero in favour of / against SCA. Seeds 0 to 2 each cover all 195 Aesthetic-4K prompts; HRGen-1K has 1000 prompts; FLUX.1-Krea-dev runs the same 195 prompts as seed 0; FLUX.2-dev uses 50. Table E.4 gives the Holm-adjusted calls. Key result: no comparison is significantly worse on any metric, and every comparison on FLUX.1-dev is significantly better on all ten.

(a) Preference

Setting	n	IR	Aesthetic	CLIP	PickScore	HPSv2 ($\times 100$)	HPSv2.1 ($\times 100$)
Progressive SCA − AReA							
Seed 0	195	+0.399 [0.317, 0.486]	+0.23 [0.17, 0.30]	+1.70 [1.33, 2.07]	+1.05 [0.93, 1.19]	+1.64 [1.42, 1.86]	+3.86 [3.41, 4.30]
Seed 1	195	+0.372 [0.292, 0.460]	+0.24 [0.18, 0.30]	+1.28 [0.90, 1.66]	+1.00 [0.88, 1.12]	+1.50 [1.31, 1.69]	+3.67 [3.26, 4.06]
Seed 2	195	+0.354 [0.267, 0.445]	+0.22 [0.16, 0.28]	+1.49 [1.13, 1.85]	+0.99 [0.87, 1.12]	+1.52 [1.31, 1.72]	+3.61 [3.20, 4.03]
HRGen-1K	1000	+0.898 [0.841, 0.956]	+0.41 [0.38, 0.45]	+2.80 [2.54, 3.05]	+1.40 [1.33, 1.46]	+2.38 [2.27, 2.49]	+6.59 [6.31, 6.88]
Single-pass SCA − AReA							
Seed 0	195	+0.327 [0.250, 0.405]	+0.15 [0.09, 0.20]	+1.45 [1.12, 1.80]	+0.76 [0.66, 0.88]	+1.15 [0.96, 1.34]	+2.56 [2.18, 2.94]
Seed 1	195	+0.306 [0.242, 0.378]	+0.11 [0.06, 0.16]	+1.22 [0.89, 1.54]	+0.77 [0.67, 0.88]	+1.17 [1.01, 1.33]	+2.49 [2.16, 2.83]
Seed 2	195	+0.264 [0.197, 0.338]	+0.09 [0.04, 0.14]	+1.37 [1.08, 1.66]	+0.74 [0.62, 0.86]	+1.18 [0.99, 1.38]	+2.53 [2.13, 2.90]
HRGen-1K	1000	+0.764 [0.714, 0.816]	+0.27 [0.24, 0.31]	+2.35 [2.12, 2.59]	+1.09 [1.04, 1.14]	+1.81 [1.71, 1.90]	+4.60 [4.37, 4.83]
FLUX.1-Krea	195	+0.020 [−0.020, 0.063]	+0.04 [0.01, 0.07]	+0.15 [−0.07, 0.36]	+0.17 [0.11, 0.24]	+0.39 [0.30, 0.49]	+0.91 [0.70, 1.10]
SCA without the schedule − full attention							
FLUX.2-dev	50	+0.456 [0.320, 0.598]	+0.30 [0.18, 0.42]	+0.38 [−0.38, 1.12]	+0.56 [0.29, 0.86]	+0.85 [0.50, 1.21]	+1.84 [1.10, 2.55]

(b) Image quality

Setting	n	MUSIQ	MUSIQ_c	CLIP-IQA ($\times 100$)	ARNIQA ($\times 100$)
Progressive SCA − AReA					
Seed 0	195	+2.3 [0.5, 3.9]	+3.7 [2.0, 5.4]	+7.2 [5.6, 8.7]	+7.1 [6.1, 8.2]
Seed 1	195	+3.0 [1.6, 4.4]	+4.1 [2.6, 5.8]	+7.4 [5.9, 8.8]	+6.8 [5.9, 7.9]
Seed 2	195	+3.8 [2.1, 5.5]	+5.9 [4.1, 7.7]	+8.1 [6.6, 9.4]	+6.7 [5.7, 7.7]
HRGen-1K	1000	+5.2 [4.5, 6.0]	+5.4 [4.6, 6.2]	+11.2 [10.5, 12.0]	+10.3 [9.8, 10.9]
Single-pass SCA − AReA					
Seed 0	195	+5.6 [3.9, 7.4]	+9.7 [8.3, 11.1]	+4.6 [3.2, 6.1]	+3.4 [2.5, 4.3]
Seed 1	195	+5.5 [3.8, 7.2]	+10.3 [8.9, 11.6]	+4.6 [3.2, 6.1]	+3.4 [2.5, 4.3]
Seed 2	195	+6.3 [4.4, 8.1]	+10.9 [9.3, 12.6]	+4.2 [3.0, 5.4]	+3.5 [2.6, 4.4]
HRGen-1K	1000	+7.7 [6.9, 8.5]	+12.6 [11.8, 13.3]	+9.0 [8.2, 9.7]	+6.5 [6.0, 7.0]
FLUX.1-Krea	195	+1.9 [1.0, 2.9]	+2.7 [2.0, 3.3]	−0.6 [−1.4, 0.3]	+0.8 [0.3, 1.2]
SCA without the schedule − full attention					
FLUX.2-dev	50	+16.0 [11.9, 19.9]	+21.4 [18.2, 24.8]	+6.6 [4.7, 8.6]	+7.1 [5.4, 8.8]

Table G.2: **At 6144^2 SCA without the schedule beats AReA on 14 of 15 metrics; on FLUX.1-Krea-dev single-pass SCA wins eight metrics and loses patch FID, in a third of the time.** Expands the FLUX.1-Krea-dev row of Table 2 and the 6144^2 result of Section 4.2. Each cell is the method minus AReA with its 95% paired percentile-bootstrap interval below it in grey. First row: FLUX.1-dev at 6144^2, 100 prompts. Other rows: FLUX.1-Krea-dev at 4096^2 with a reach table fitted on that checkpoint, on all 195 prompts for single-pass SCA and on the 100 that DyPE and SEGA ran. Per-image intervals use 1000 prompt draws; FID, KID, and their patch versions resample prompts and reference-image clusters. IS: Inception score; time: campaign mean on shared GH200 nodes. Blue / orange: significantly better / worse than AReA; for FID, KID, their patch versions, and time, lower is better. Key result: against AReA on FLUX.1-Krea-dev, DyPE wins one of the 15 metrics and loses eight, and SEGA wins one and loses four.

(a) Preference

Method − AReA	ImageReward	Aesthetic	CLIP score	PickScore	HPSv2 (×100)	HPSv2.1 (×100)
FLUX.1-dev, 6144^2						
SCA w/o schedule	+1.262 [1.051, 1.450]	+0.44 [0.31, 0.55]	+3.28 [2.32, 4.18]	+1.58 [1.31, 1.88]	+2.53 [2.12, 2.94]	+6.37 [5.39, 7.38]
FLUX.1-Krea-dev, 4096^2						
Single-pass SCA	+0.020 [−0.021, 0.060]	+0.04 [0.01, 0.07]	+0.15 [−0.06, 0.36]	+0.17 [0.11, 0.24]	+0.39 [0.30, 0.48]	+0.91 [0.70, 1.12]
DyPE	−0.282 [−0.424, −0.173]	+0.05 [−0.02, 0.12]	−0.95 [−1.47, −0.49]	−0.73 [−0.93, −0.55]	−1.02 [−1.24, −0.80]	−1.24 [−1.76, −0.76]
SEGA	−0.110 [−0.189, −0.030]	−0.05 [−0.11, 0.02]	−0.49 [−0.91, −0.07]	−0.23 [−0.36, −0.10]	−0.25 [−0.43, −0.09]	−0.25 [−0.61, 0.09]

(b) Image quality

Method − AReA	MUSIQ	MUSIQ_c	CLIP-IQA (×100)	ARNIQA (×100)
FLUX.1-dev, 6144^2				
SCA w/o schedule	+9.2 [6.9, 11.7]	+3.8 [1.4, 6.2]	+6.3 [4.3, 8.4]	+14.9 [13.0, 17.0]
FLUX.1-Krea-dev, 4096^2				
Single-pass SCA	+1.9 [1.0, 2.8]	+2.7 [2.1, 3.4]	−0.6 [−1.4, 0.3]	+0.8 [0.3, 1.2]
DyPE	−2.6 [−4.6, −0.4]	+3.2 [1.8, 4.6]	+0.9 [−0.7, 2.3]	−1.6 [−2.6, −0.7]
SEGA	+0.7 [−1.2, 2.8]	+1.3 [−0.1, 2.6]	+1.8 [0.4, 3.2]	−0.4 [−1.3, 0.4]

(c) Set level and time

Method − AReA	FID ↓	KID ($\times 10^3$) ↓	FID_p ↓	KID_p ($\times 10^3$) ↓	IS ↑	Time (s) ↓
FLUX.1-dev, 6144^2						
SCA w/o schedule	−46.8 [−58.6, −35.3]	−23.20 [−31.77, −16.84]	−66.2 [−70.2, −48.9]	−56.88 [−68.46, −47.50]	+0.45 [−0.06, 0.84]	−343.1 [−349.3, −337.4]
FLUX.1-Krea-dev, 4096^2						
Single-pass SCA	+1.0 [−3.6, 6.0]	−0.27 [−1.95, 1.16]	+3.4 [1.6, 7.1]	+0.66 [−0.57, 2.05]	+0.67 [0.14, 1.08]	−160.9 [−173.3, −149.4]
DyPE	+5.0 [−2.8, 16.3]	−0.81 [−4.63, 3.26]	+7.7 [5.0, 18.5]	+1.32 [−1.44, 4.94]	+0.25 [−0.22, 0.69]	−24.3 [−28.9, −19.6]
SEGA	−2.3 [−10.0, 7.4]	−0.85 [−4.00, 1.53]	+0.1 [−3.1, 6.7]	−0.81 [−2.48, 0.90]	+0.09 [−0.43, 0.56]	−20.1 [−25.0, −15.5]

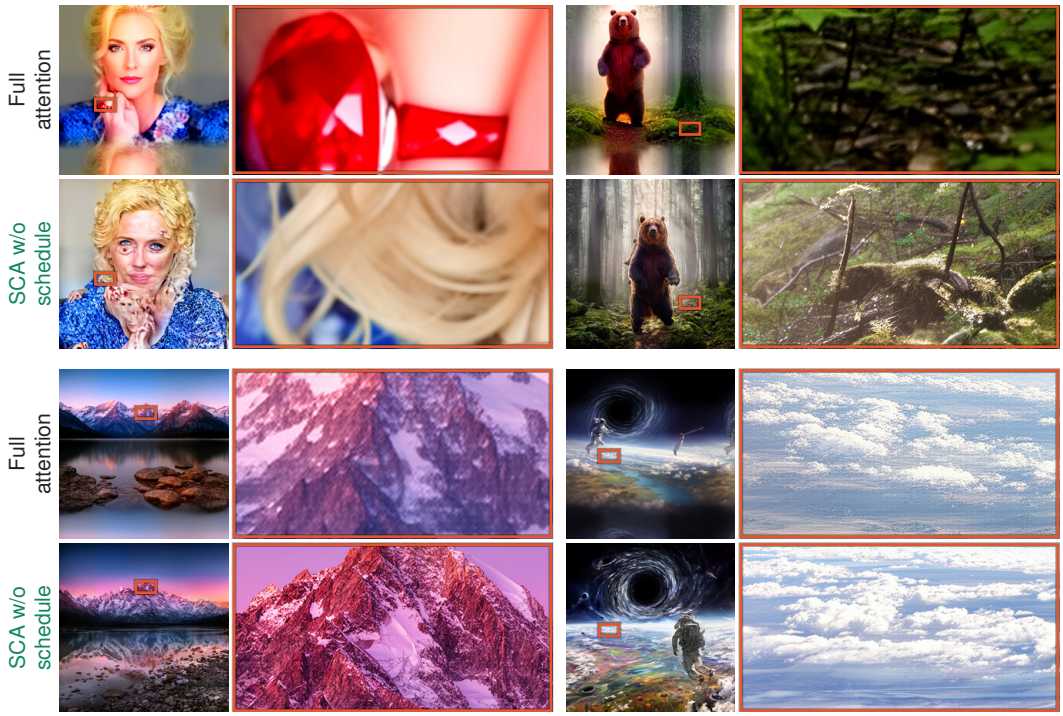


Figure G.1: **FLUX.2-dev at 4096^2: SCA without the schedule against full attention; in the portrait SCA distorts facial geometry and fine texture.** Four of 50 evaluation prompts chosen by a fixed seed (seed 17); seed 0 for every image. Each crop shows native pixels of the window that maximises the smallest edge energy among both methods. Over the 50 FLUX.2-dev prompts, SCA without the schedule is significantly better than full attention on nine of ten per-image metrics, both MUSIQ scores among them (Table G.1).

G.3 Non-square sizes

Table G.3 pairs single-pass SCA with AReA at 2048×4096 and 2048×6144. At 2048×4096 the window already covers half the image height, so the per-head mix gains little (Appendix J.1); Appendix H.2 checks which reach table that size should read.

G.4 Exploratory 8192^2 comparison

This subsection expands the 8192^2 sentence of Section 4.2. Table G.4 reports all 12 completed prompt pairs at 8192^2 under the shared FLUX.1-dev protocol, for SCA without the schedule against AReA. SCA without the schedule has higher ImageReward, CLIP, PickScore, both HPS variants, and ARNIQA, with paired intervals that exclude zero. Aesthetic score and the three remaining quality metrics have intervals that span zero, and MUSIQ, MUSIQ_c, and CLIP-IQA have point estimates that favour AReA. These mixed quality scores and the grid and stripe failures in Figure G.4 limit the evidence for reliable 8K generation. With only 12 prompts the comparison is exploratory, and we report no distribution metrics for it.

All 8K scores were computed on CPU (Table G.4 notes the HPS precision). For native-resolution MUSIQ, attention queries are processed in chunks of 256 while every key and native image patch is retained. This bounds intermediate memory without changing the attention context; equivalence checks with the pretrained model at three smaller image sizes reproduced the unchunked scores exactly.



Figure G.2: **At 6144^2 both methods show artifacts: dots for AReA, blocks for SCA without the schedule.** AReA and SCA without the schedule on four evaluation prompts that name no person (keyword rule), chosen by a fixed seed (seed 17) from the 100 completed ones; seed 0 for every image. Each crop shows native pixels of the window that maximises the smallest edge energy among both methods. Over the 100 prompts, SCA without the schedule is better than AReA on 14 of 15 metrics (Table G.2).

Table G.3: **At 2048×6144 single-pass SCA wins seven of 15 metrics; at 2048×4096 the result is mixed.** Expands the aspect-ratio paragraph of Section 4.2. FLUX.1-dev, 195 prompts each; AReA runs its released code. Δ: single-pass SCA minus AReA, with its 95% paired percentile-bootstrap interval in grey over 1000 prompt draws; for FID, KID, and their patch versions the draws also resample reference-image clusters. Blue / orange: significantly better / worse than AReA; for FID, KID, and time lower is better. ‡Campaign mean on shared nodes. Key result: at 2048×6144 single-pass SCA wins seven of 15 metrics and loses two, at 2048×4096 it wins three and loses three, and it is faster at both.

	2048×4096, factors (2, 4)			2048×6144, factors (2, 6)		
	Single-pass			Single-pass		
Metric	AReA	SCA	Δ 95% interval	AReA	SCA	Δ 95% interval
Preference ↑						
ImageReward	1.132	1.106	−0.026 [−0.107, 0.047]	0.391	0.638	+0.247 [0.125, 0.376]
Aesthetic	6.41	6.17	−0.23 [−0.29, −0.18]	6.14	5.88	−0.26 [−0.37, −0.15]
CLIP score	28.59	28.09	−0.49 [−0.89, −0.15]	26.52	26.78	+0.26 [−0.30, 0.79]
PickScore	22.66	22.61	−0.06 [−0.17, 0.06]	21.51	21.66	+0.15 [−0.03, 0.32]
HPSv2 (×100)	28.36	28.57	+0.21 [0.09, 0.32]	26.41	26.59	+0.17 [0.06, 0.31]
HPSv2.1 (×100)	27.64	27.49	−0.16 [−0.45, 0.12]	23.36	23.60	+0.23 [−0.14, 0.63]
Image quality ↑						
MUSIQ	51.9	56.2	+4.3 [2.9, 5.8]	42.8	47.7	+4.9 [3.1, 6.8]
MUSIQ_c	61.6	65.3	+3.7 [2.5, 4.8]	53.0	59.4	+6.4 [4.9, 8.0]
CLIP-IQA (×100)	74.9	72.5	−2.4 [−3.4, −1.5]	67.5	67.8	+0.3 [−1.1, 1.8]
ARNIQA (×100)	73.0	72.6	−0.4 [−1.1, 0.1]	69.3	69.5	+0.2 [−0.6, 1.0]
Set level and time						
Inception score ↑	6.27	6.82	+0.55 [−0.12, 1.04]	5.48	6.41	+0.93 [0.28, 1.39]
FID ↓	144.6	142.5	−2.1 [−6.9, 3.3]	163.0	142.8	−20.2 [−24.7, −11.9]
KID ($\times 10^3$) ↓	7.81	7.02	−0.79 [−3.25, 1.95]	17.92	7.01	−10.91 [−14.53, −7.26]
FID_p ↓	111.7	114.9	+3.2 [−0.5, 7.1]	105.7	109.7	+4.0 [1.0, 10.8]
KID_p ($\times 10^3$) ↓	14.26	15.20	+0.95 [−0.46, 2.59]	13.56	15.09	+1.54 [−2.01, 5.42]
Seconds per image‡ ↓	112.4	41.1	−71.3 [−73.3, −69.2]	150.5	60.5	−90.0 [−92.2, −87.8]

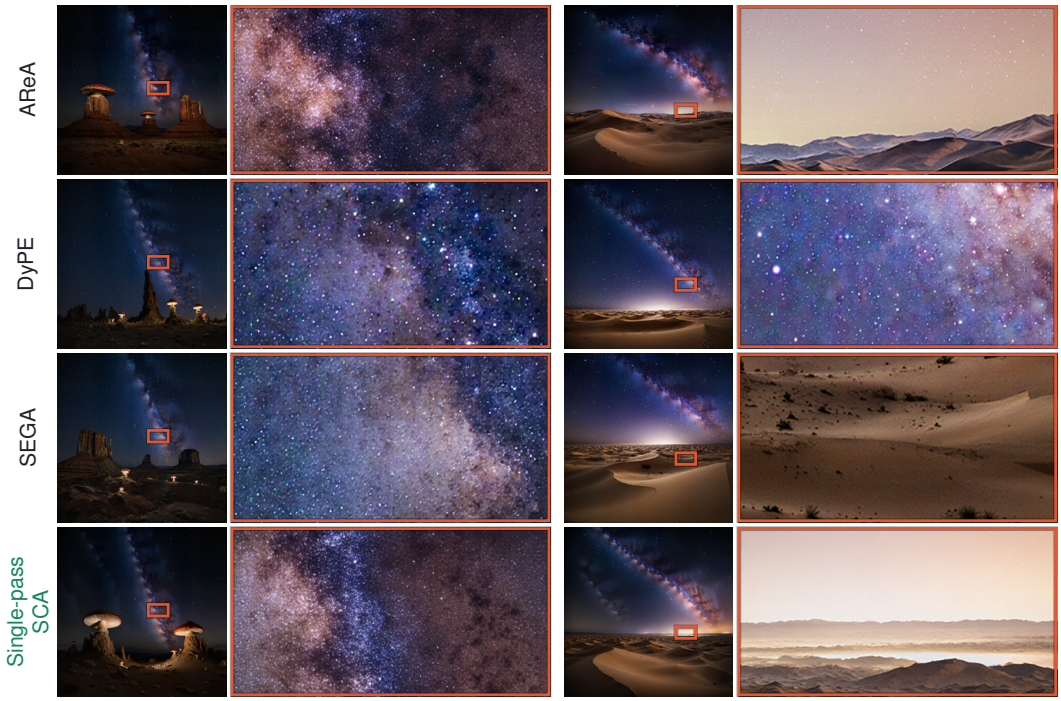


Figure G.3: **AReA, DyPE, SEGA, and single-pass SCA on FLUX.1-Krea-dev.** 4096^2, two evaluation prompts that name no person (keyword rule), chosen by a fixed seed (seed 17); seed 0 for every image, with a reach table fitted on FLUX.1-Krea-dev. Each crop shows native pixels of the window that maximises the smallest edge energy among the four methods. Over all 195 prompts, single-pass SCA is significantly better than AReA on seven of ten per-image metrics and worse on none; its ImageReward difference, $+0.020$ $[-0.020, 0.063]$, includes zero (Table G.2).

Table G.4: **Exploratory 8192^2 comparison: SCA without the schedule is significantly better than AReA on six of ten metrics.** FLUX.1-dev, 12 matched prompts, seed 0, both methods scored on CPU with the same settings. HPS runs in float32 here and in mixed precision in the main evaluation, so its values are not comparable across resolutions. All metrics are higher-is-better. Intervals are paired prompt-bootstrap 95% percentile intervals (2000 draws, bootstrap seed 0); blue: significantly better than AReA.

Metric	AReA	SCA w/o sched.	Difference [95% CI]
ImageReward	-0.376	0.675	$+1.051$ $[+0.386, +1.754]$
Aesthetic	5.64	5.72	$+0.08$ $[-0.26, +0.44]$
CLIP score	22.38	25.52	$+3.14$ $[+0.63, +6.11]$
PickScore	20.35	21.27	$+0.92$ $[+0.24, +1.75]$
HPSv2 ($\times 100$)	24.60	26.13	$+1.53$ $[+0.56, +2.66]$
HPSv2.1 ($\times 100$)	16.59	20.51	$+3.92$ $[+1.31, +6.85]$
MUSIQ	28.6	25.7	-2.9 $[-6.8, +0.8]$
MUSIQ_c	44.0	40.1	-3.8 $[-10.4, +1.6]$
CLIP-IQA ($\times 100$)	57.1	52.0	-5.1 $[-10.5, +0.6]$
ARNIQA ($\times 100$)	37.3	54.6	$+17.3$ $[+11.9, +22.6]$

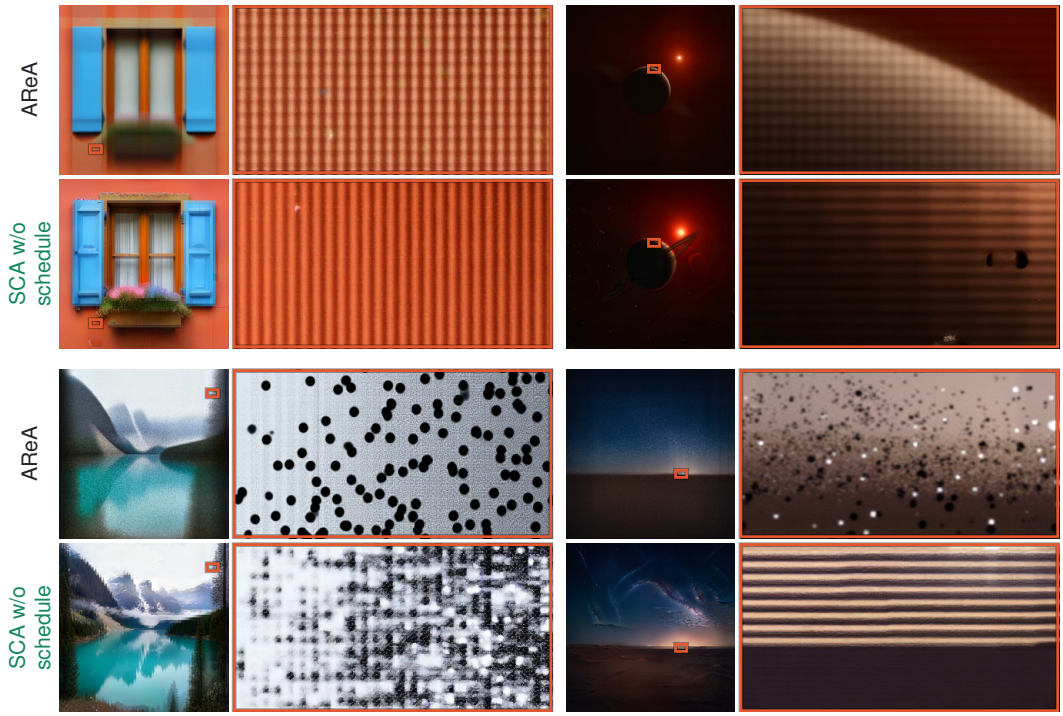


Figure G.4: **At 8192^2 both methods show artifacts: dots and grids for AReA, stripes and blocks for SCA without the schedule.** AReA and SCA without the schedule on four evaluation prompts that name no person (keyword rule), chosen by a fixed seed (seed 17) from the 12 completed ones; seed 0 for every image. Each crop shows native pixels of the window that maximises the smallest edge energy among both methods. Over the 12 prompts, SCA without the schedule is significantly better than AReA on six of ten metrics (Table G.4).

H Sensitivity and budgets

This appendix tests the settings that the main text fixes, and expands the sensitivity sentence of Section 4.2 and the SigMa comparison of Section 4.1. Appendix H.1 varies the number of emulation prompts and the gate, Appendix H.2 the table read at a non-square size, and Appendix H.3 the number of steps and the guidance.

H.1 Emulation budget and gate

We merge the stored sufficient statistics of the first 4, 8, or 16 emulation prompts and rerun the 46 ablation prompts with each resulting table. Table H.1 reports the generation-level differences from the 32-prompt table, all for SCA without the schedule, together with two other gates. The test reuses the emulation statistics but evaluates on separate prompts; it is not a held-out output-error benchmark. Tables fitted on 4 to 16 prompts change one of 30 differences significantly. A gate at $\sigma_g = 0.85$ lowers seven metrics, and a gate at 0.93 lowers CLIP score and PickScore and raises ARNIQA.

H.2 Reach table at non-square sizes

This subsection checks the ratio rule of Section 3.2 at a non-square size. At $(s_H, s_W) = (2, 4)$ the two axes have different factors, and the native phase reads the table of the ratio nearest the larger factor, ratio 4. On the 46 ablation prompts at 2048×4096, for SCA without the schedule, the ratio-2 table makes three significant changes, all at most 0.17 (Table H.2): HPSv2 $+0.09$, HPSv2.1 $+0.17$ (both $\times 100$), and PickScore $+0.05$. ImageReward ($+0.01$ $[-0.03, 0.05]$) and the other metrics are within noise. Changes this small cannot resolve the mixed results of single-pass SCA against AReA at this size, so we keep the default rule.

Table H.1: **Reach tables fitted on 4 to 16 prompts change one of 30 differences significantly; a gate at $\sigma_g = 0.85$ lowers seven metrics.** Supports the prompt-count check of Section 4.2. FLUX.1-dev at 4096^2, the 46 ablation prompts, SCA without the schedule throughout. Each row is the variant minus the default (the 32-prompt table, gate $\sigma_g = 0.89$). $^{\times}$: $\times 100$. Blue / orange: the 95% paired interval (2000 prompt draws) excludes zero in favour of / against the variant. Key result: tables fitted on 4 to 16 prompts move ImageReward by at most 0.012, and one of their 30 differences is significant.

(a) Preference

Variant − default	IR	Aesthetic	CLIP	PickScore	HPSv2$^{\times}$	HPSv2.1$^{\times}$
4-prompt table	+0.012 [−0.007, 0.029]	+0.01 [−0.00, 0.02]	+0.08 [−0.03, 0.19]	−0.01 [−0.03, 0.01]	+0.02 [−0.02, 0.05]	−0.02 [−0.07, 0.04]
8-prompt table	+0.010 [−0.004, 0.024]	+0.01 [−0.00, 0.02]	+0.05 [−0.04, 0.15]	+0.00 [−0.01, 0.02]	+0.02 [−0.00, 0.05]	+0.01 [−0.03, 0.05]
16-prompt table	−0.007 [−0.020, 0.005]	+0.00 [−0.01, 0.01]	+0.04 [−0.04, 0.13]	+0.01 [−0.01, 0.03]	+0.00 [−0.02, 0.03]	−0.01 [−0.05, 0.04]
gate $\sigma_g = 0.85$	−0.015 [−0.098, 0.064]	−0.19 [−0.26, −0.12]	−0.27 [−0.82, 0.26]	−0.12 [−0.27, 0.02]	−0.48 [−0.69, −0.28]	−0.85 [−1.22, −0.48]
gate $\sigma_g = 0.93$	−0.004 [−0.097, 0.069]	+0.00 [−0.05, 0.06]	−0.64 [−1.03, −0.29]	−0.22 [−0.38, −0.07]	+0.02 [−0.16, 0.20]	−0.15 [−0.56, 0.24]

(b) Image quality

Variant − default	MUSIQ	MUSIQ$_c$	CLIP-IQA$^{\times}$	ARNIQA$^{\times}$
4-prompt table	−0.1 [−0.4, 0.3]	+0.1 [−0.1, 0.3]	+0.2 [−0.1, 0.4]	+0.1 [−0.1, 0.3]
8-prompt table	−0.1 [−0.5, 0.2]	−0.0 [−0.2, 0.1]	+0.2 [−0.1, 0.4]	+0.0 [−0.1, 0.2]
16-prompt table	+0.1 [0.0004, 0.3]	−0.1 [−0.2, 0.0]	−0.1 [−0.4, 0.1]	−0.0 [−0.1, 0.1]
gate $\sigma_g = 0.85$	−1.9 [−3.5, −0.2]	−2.0 [−3.0, −1.1]	−3.2 [−4.8, −1.7]	−4.0 [−5.9, −2.3]
gate $\sigma_g = 0.93$	+0.4 [−1.0, 2.0]	+0.6 [−0.4, 1.9]	+0.0 [−1.2, 1.1]	+1.8 [0.7, 3.1]

Table H.2: **At 2048×4096 the ratio-2 table raises three metrics significantly, by at most 0.17, and lowers none.** Checks the ratio rule of Section 3.2 at a non-square size. FLUX.1-dev, the 46 ablation prompts, SCA without the schedule throughout. Each row is paired against the same operator with the default ratio-4 table: the operator with the ratio-2 table, and AReA (released code) for scale. $^{\times}$: $\times 100$. Blue / orange: the 95% paired interval (2000 prompt draws) excludes zero in favour of / against the row. Key result: the default ratio-4 table trails the ratio-2 table on three of ten metrics, by at most 0.17, and leads on none.

(a) Preference

Row − ratio-4 table	IR	Aesthetic	CLIP	PickScore	HPSv2$^{\times}$	HPSv2.1$^{\times}$
ratio-2 table	+0.012 [−0.025, 0.050]	−0.00 [−0.03, 0.02]	−0.02 [−0.22, 0.17]	+0.05 [0.002, 0.09]	+0.09 [0.04, 0.13]	+0.17 [0.08, 0.26]
AReA	+0.027 [−0.141, 0.192]	+0.29 [0.16, 0.43]	+0.19 [−0.53, 0.91]	+0.14 [−0.09, 0.39]	−0.19 [−0.44, 0.06]	+0.39 [−0.21, 1.01]

(b) Image quality

Row − ratio-4 table	MUSIQ	MUSIQ$_c$	CLIP-IQA$^{\times}$	ARNIQA$^{\times}$
ratio-2 table	+0.0 [−0.5, 0.5]	−0.0 [−0.4, 0.2]	−0.6 [−1.2, 0.0]	−0.2 [−0.6, 0.1]
AReA	−3.1 [−7.5, 1.2]	−4.1 [−6.7, −1.3]	+1.3 [−0.8, 3.4]	+0.4 [−1.5, 2.8]

H.3 Sampling budgets, guidance, and SigMa

Table H.3 adds SigMa and three budgets of SCA without the schedule to Table 1. Table H.4 gives the budget and guidance checks with intervals, and Figure H.1 plots ImageReward against time.

Table H.3: **SigMa trails single-pass SCA on every metric shown; 56 steps raise the ImageReward of SCA without the schedule by 0.03 at twice the time.** 195 prompts at 4096^2. s: campaign mean in seconds per image on shared nodes, which every row has; IR ImageReward, HPSv2.1 $\times 100$, Pick PickScore, M_c MUSIQ on four 1024^2 crops, $FID_p^\dagger$ our patch FID (Appendix D.2). Bold: best, underline: second within each group; shaded: ours. Colours mark each row against single-pass SCA at 28 steps; the budget rows use SCA without the schedule and are coloured against its 28-step run, which this table does not show (Table H.4). The FID cells of SigMa and single-pass SCA have no interval. Table 1 holds the multi-stage rows. Key result: 56 steps take 139.9 against 71.5 s per image.

	Cost		Preference ↑				Detail ↑	Set level ↓	
Method	Steps	s↓	IR	HPSv2.1	Pick	CLIP	M_c	FID	$FID_p^\dagger$
Single pass, AReA's sampler (28 steps, guidance 4.5)									
SigMa	28	192.5	−0.52	18.65	20.01	23.51	49.9	216.9	184.6
Single-pass SCA	28	**78.4**	**1.25**	**29.73**	**23.12**	**29.14**	**60.1**	**140.2**	**98.4**
SCA without the schedule, other budgets									
56 steps	56	139.9	**1.26**	**30.55**	**23.22**	**29.14**	**64.1**	**139.8**	102.9
guidance 7	28	**71.9**	1.16	29.07	22.82	28.94	61.4	140.4	**98.4**
42 steps	42	106.8	**1.26**	30.27	23.17	**29.14**	63.1	140.5	100.6

Table H.4: **Doubling the steps raises eight metrics; guidance 7 lowers seven and raises MUSIQ.** Expands Table H.3. FLUX.1-dev at 4096^2, 195 prompts, all with SCA without the schedule: the first two rows compare with its default 28-step run at guidance 4.5, the last compares its 56-step run with UltraImage. Each row is the first-named setting minus the second. $^\times$: $\times 100$. Blue / orange: the 95% paired interval (2000 prompt draws) excludes zero in favour of / against the first-named setting. Key result: at 56 steps SCA without the schedule is better than UltraImage on three metrics and worse on one.

(a) Preference

Comparison	IR	Aesthetic	CLIP	PickScore	HPSv2$^\times$	HPSv2.1$^\times$
56 vs. 28 steps	+0.032 [0.0002, 0.062]	+0.03 [0.01, 0.06]	−0.01 [−0.23, 0.22]	+0.13 [0.07, 0.18]	+0.32 [0.23, 0.40]	+0.88 [0.71, 1.04]
guidance 7 vs. 4.5	−0.068 [−0.115, −0.025]	−0.10 [−0.14, −0.07]	−0.21 [−0.45, 0.03]	−0.27 [−0.34, −0.20]	−0.25 [−0.35, −0.16]	−0.60 [−0.80, −0.41]
56 steps vs. UltraImage	−0.023 [−0.089, 0.040]	−0.03 [−0.07, 0.02]	+0.30 [−0.01, 0.62]	−0.11 [−0.21, −0.01]	+0.11 [−0.03, 0.26]	−0.22 [−0.51, 0.05]

(b) Image quality

Comparison	MUSIQ	$MUSIQ_c$	CLIP-IQA$^\times$	ARNIQA$^\times$
56 vs. 28 steps	−0.2 [−1.5, 1.0]	+2.9 [2.2, 3.6]	+0.9 [0.4, 1.4]	+1.5 [1.0, 1.9]
guidance 7 vs. 4.5	+1.4 [0.3, 2.6]	+0.2 [−0.4, 0.8]	−1.0 [−1.7, −0.4]	−1.2 [−1.5, −0.8]
56 steps vs. UltraImage	+11.6 [9.5, 13.6]	+22.4 [21.2, 23.6]	+1.1 [0.03, 2.1]	+0.4 [−0.4, 1.2]

SigMa. With its dev-matched profile (Appendix D.8), SigMa trails single-pass SCA on all ten per-image metrics (Table E.5) and takes 192.5 s per image against 78.4 s (campaign means).

More steps and higher guidance. On the same 195 prompts we compare the default 28-step run of SCA without the schedule with 56 steps at guidance 4.5 and with 28 steps at guidance 7 (Table H.4). The 56-step run keeps the noise threshold at 0.89, so it changes the discretization of both the opening and the mixed phase. It improves ImageReward by 0.032 $[0.0002, 0.062]$, HPSv2.1 by 0.88, PickScore by 0.13, and $MUSIQ_c$ by 2.9. It also raises corrected patch FID by 3.85 $[1.51, 6.33]$ under reference-cluster resampling. Guidance 7 alone lowers seven per-image metrics, ImageReward by 0.068 $[0.025, 0.115]$ among them, and raises only MUSIQ. The default settings therefore do not maximize every metric: 56 steps gain eight per-image metrics but raise patch FID by 3.85.

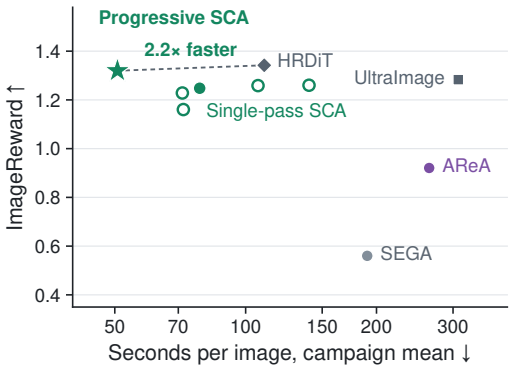


Figure H.1: **At 46% of HRDiT's time, progressive SCA is indistinguishable from HRDiT on ImageReward (difference -0.02 $[-0.09, 0.03]$).** ImageReward against campaign seconds per image on shared nodes, $n = 195$ prompts at 4096^2. Star: progressive SCA; filled circle: single-pass SCA; open circles: SCA without the schedule at 28, 42, and 56 steps and at guidance 7; the dashed guide carries the ratio of campaign times. Full attention, DyPE, and SigMa fall below the range (Table H.3).

Against UltraImage and HRDiT. On these 195 prompts, 56-step SCA without the schedule differs from UltraImage by -0.023 $[-0.089, 0.040]$ in ImageReward, -0.22 $[-0.51, 0.05]$ in HPSv2.1, and -0.11 $[-0.21, -0.01]$ in PickScore. The first two intervals span zero and do not establish equivalence; PickScore favours UltraImage. The MUSIQ advantage of 56-step SCA without the schedule is 11.6 $[9.5, 13.6]$, and its MUSIQ_c advantage 22.4 $[21.2, 23.6]$. Campaign means are 71.5 s for 28 steps, 139.9 s for 56 steps, and 308.5 s for UltraImage. This check matches neither FLOPs nor wall-clock time: UltraImage takes 308.5 against 139.9 s, runs two resolutions with different guidance, and needs per-prompt preparation. Against HRDiT on the same prompts (Table 1), the 56-step run stays lower on ImageReward (1.26 against 1.34), HPSv2.1 (30.55 against 30.92), and PickScore (23.22 against 23.46). It also takes longer (139.9 against 110.4 s). These are point estimates without paired intervals.

A 42-step run. Table F.1 adds a 42-step run of SCA without the schedule, sized to HRDiT's campaign time (106.8 against 110.4 s). The extra steps raise PickScore, HPSv2, HPSv2.1, ARNIQA, and MUSIQ_c over 28 steps but leave ImageReward unresolved. HRDiT stays significantly ahead of the 42-step run on ImageReward, PickScore, HPSv2.1, aesthetic score, and CLIP-IQA, and trails it on both MUSIQ scores and ARNIQA. With 195 generated images, 168 reference identities, and 1000 bootstrap draws, the 42-step run has significantly higher FID than the 28-step run ($+2.4$ $[0.1, 5.7]$); KID, patch FID ($+1.5$ $[-0.5, 3.6]$), and patch KID span zero. Appendix F runs SCA inside HRDiT's schedule instead.

I Compute and Reproducibility

This appendix expands the reproducibility statement. Appendix I.1 gives the hardware and the logged compute, and Appendix I.2 lists what the code release contains and where each reported number comes from.

I.1 Hardware and Compute

Generation and reach emulation used single NVIDIA GH200 GPUs on a shared cluster; the verified timing logs identify the GH200 120 GB model. Some metric evaluations ran on CPUs. The surviving logs do not establish the complete emulation runtime. At 4096^2 on shared nodes, campaign means are about 51 s per image for progressive SCA and 78 s for single-pass SCA (Table 1). They are 72 s for SCA without the schedule (Appendix H.3), 264 s for AReA, and 196 s for full attention. The main comparisons, ablations, and non-square sizes comprise more than 2000 generated images. Surviving worker logs establish a lower bound of 123.3 GH200 worker-hours up to 2026-10-17 06:59 UTC, including model loading, reach emulation, evaluation, and failed attempts. Later runs, among them single-pass and progressive SCA on all prompts, are not in this bound. The bound is the union of 581 closed task intervals per physical node, with timestamp precision treated conservatively. It measures occupied worker time, not GPU kernel activity. Missing logs, unmatched starts, and ongoing work prevent a complete total, and overlapping scheduler allocation time and per-image durations are not added to the bound.

I.2 RELEASE AND SOURCES

We will release the code: the attention processor, reach emulation, single-pass and progressive sampling, evaluation wrappers, the AReA wrapper, and tests. We will also release the FLUX.1-dev reach table, the per-image metric files, and the per-comparison summary and paired-difference files behind every table. The metric values and paired intervals of the paper are derived from those files. Each SCA generation task ran from a frozen copy of the code named in the task. Per-image records store the sampling settings and, where available, a code fingerprint; the inherited FLUX.1-Krea-dev baseline records do not establish one source revision for every image. Table I.1 maps each item to its file.

Table I.1: **Every configuration, check, and timing of this appendix maps to a released file.** Paths are relative to the repository root; $\star$ stands for a family of files.

Item	File
Single-pass SCA (Table C.3)	`scripts/sample.py --method sca --gate-sigma 0.89 --bridge-cells 8 --bridge-k 16 --win-sched 0.88:128,0.86:112,0.84:96,0.82:80 --win-pi-match --win-pi-interp --pi-table k1_1024.npz`, code commit `7bd64c0`
Progressive SCA (Algorithm 5)	`scripts/sample_progressive.py`
Reach emulation and SCA attention (Algorithms 3 and 4)	the statistics and attention modules in `sca/`; one attention call in `sca/attn.py`
FLUX.1-dev reach table and statistics	`calib/k1_1024.npz`, `calib/k1_1024.json`
Numerical checks (Appendix C.3)	`tests/test_sca.py` (18 CPU tests); `scripts/bench_attn.py` (GPU, bf16)
UltraImage wrapper	`scripts/run_ultraimage.py`
DyPE and SEGA runner	frozen at commit `c05ebdc`, recorded in `repro/README.md`
Corrected crop assembly (Appendix D.2)	`repro/evaluate_corrected.py`
Timing, run T1	`results/latency_curve_batch1.csv` (with the excluded warm-up images); `results/latency_curve_summary.csv` (every cell but full attention at 6144^2)
Timing, run T2	`results/latency_curve_v2/summary.csv`
Timing, run T3	`results/prog_latency/latency_summary.csv`
Separate 4096^2 run	`results/latency_4096_batch1.csv`
Table E.6	`results/ablnb_`$\star$ and `calib/k1_1024.json`
Table H.2	`results/ps2_`$\star$
Set-level intervals of progressive SCA	`results/prog195_vs_baselines/cfid`
Distribution metrics of the 42-step run	`results/a4k195_panel/corrected_fidp_ci/paired_deltas.csv`
Sample figures	`paper/figures/v6/*_provenance.json` beside each figure: selection rule and seed, sources, crop windows

J LIMITATIONS IN DETAIL

This appendix expands the limitations of Section 6. Appendix J.1 shows local failures and states the scope of each result, and Appendix J.2 gives the broader impact.

J.1 FAILURE CASES AND SCOPE

Figure J.1 shows the four prompts, among those that name no person, on which SCA without the schedule trails AReA most in ImageReward, each with AReA's image of the same prompt.

Hand-set settings. Only the mixing weight is fitted. The gate, the window sizes, the tile, the bridge, the emulated ratios, and the noise bins are set by hand (Table C.2), because output error cannot see layout (Appendix B.4). The window sizes were set on the 46-prompt subset, which is part of the evaluation set (Appendix D.1).

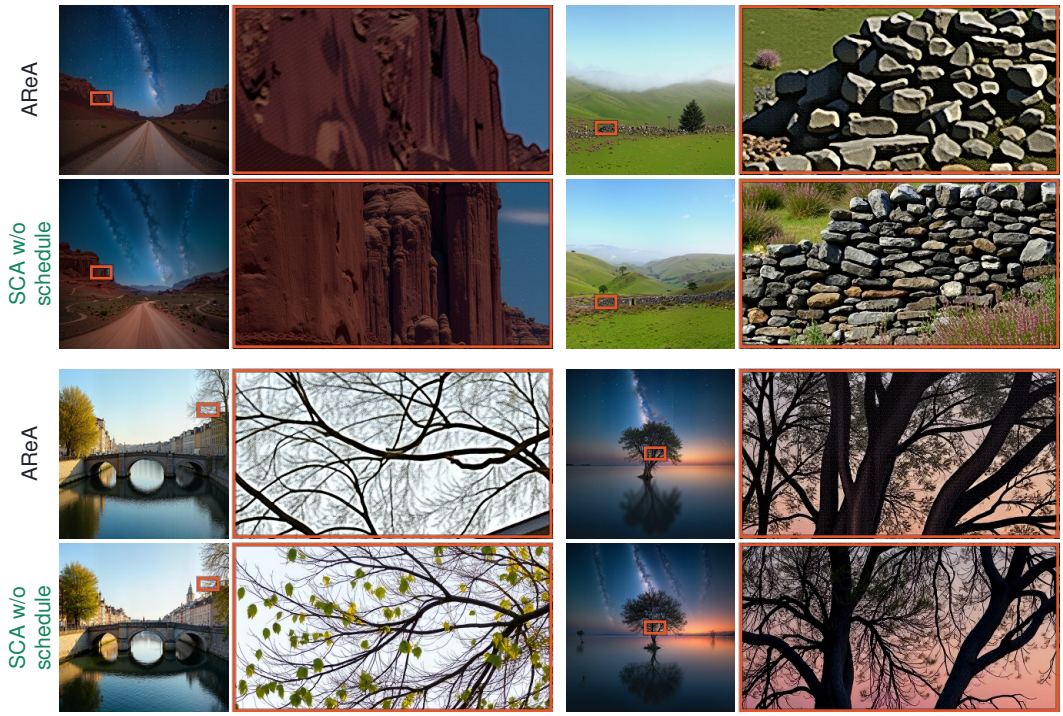


Figure J.1: **The four largest ImageReward deficits of SCA without the schedule against AReA.** The four evaluation prompts that name no person (keyword rule) with the largest ImageReward deficit against AReA; 4096^2, seed 0. Each crop shows native pixels of the window that maximises the smallest edge energy among both methods.

Patch fidelity. Progressive SCA's patch FID is higher than HRDiT's ($+4.6$ $[0.6, 7.7]$) and single-pass SCA's ($+9.2$ $[5.1, 13.2]$), and higher than AReA's at seeds 0 and 2 (Tables E.2 and F.1). Inside the progressive sampler the per-head weights win every per-image metric by small margins, while the shared and per-layer weights have lower patch FID (Table E.7).

Dependence on HRDiT's sampler. Progressive SCA uses HRDiT's sampler, which does not support FLUX.1-Krea-dev. The 6144^2, 8192^2, FLUX.2-dev, and sensitivity results use SCA without the schedule.

Small factors. At 2048×4096 the window covers half the image height, so the local branch alone already sees most of the image along that axis. There the per-head mix gains little over AReA's shrinking windows, and single-pass SCA is significantly worse on aesthetic score, CLIP score, and CLIP-IQA (Table G.3).

Larger scales and another backbone. Both use SCA without the schedule. The samples of the completed 12-prompt 8192^2 run still show artifacts in both methods, dots and grids for AReA and stripes and blocks for SCA without the schedule (Figure G.4); this exploratory set does not establish general 8K quality. In FLUX.2-dev, improved aggregate metrics coexist with distortions of facial geometry and fine texture in the portrait example of Figure G.1.

Emulation. Reach emulation omits the bridge and sees 16 times fewer keys per view than the target, so the deployed operator's error is untested.

Text-image offsets. The local branch keeps out-of-range text-image offsets (Remark 1).

Metrics. No-reference quality metrics reward sharpness even where the layout fails. The variant without the opening phase, which mixes both views with the per-head weights from the first step, scores higher on CLIP-IQA and ARNIQA (Table E.6). FID against 195 references at 299^2 resolution sees little of the 4K detail. We therefore report every metric and the qualitative galleries.

J.2 BROADER IMPACT

SCA shortens generation at 4096^2 relative to full attention under our measured protocol, which helps applications such as print, design, and scientific visualization. Energy use remains unmeasured. The same reduction lowers the cost of producing high-resolution synthetic images that could be used to mislead. SCA adds no capability beyond the base model's and does not change its safety filters or watermarking; the risks and mitigations of the base model apply unchanged.

K EMULATION PROMPTS

This appendix lists the 48 prompts written for reach emulation (Section 2 and Algorithm 1); none is an evaluation prompt (Appendix D.1). The 1024^2 table uses prompts 1 to 32, and the 1280^2, FLUX.1-Krea-dev, and FLUX.2-dev tables use prompts 1 to 16. Prompts 33 to 48 are spares that no reported table uses.

1. A candid street portrait of an elderly fisherman with a weathered face and a knitted wool cap, harbor in the background, overcast light
2. Two children laughing while running through a lawn sprinkler on a summer afternoon, backlit by the sun
3. A professional headshot of a young woman with curly red hair and freckles, soft studio lighting, neutral grey backdrop
4. A jazz trio performing in a dim basement club, saxophone player in the foreground, warm stage lights and haze
5. A chef plating a dessert in a busy restaurant kitchen, steam rising, stainless steel counters
6. A ballet dancer mid-leap in an empty rehearsal studio with tall windows and wooden floors
7. A red fox standing in fresh snow at the edge of a birch forest, early morning light
8. A macro photograph of a honeybee covered in pollen on a purple lavender flower
9. A herd of elephants crossing a dusty savanna river at sunset, long shadows
10. A tabby cat sleeping curled up on a stack of old books next to a window
11. A school of tropical fish swimming around a coral reef, sun rays through clear blue water
12. A bald eagle perched on a dead pine branch above a misty lake
13. A neon sign reading "OPEN 24 HOURS" in the window of a rainy diner at night, reflections on the wet sidewalk
14. A hand-painted wooden sign that says "Fresh Eggs" leaning against a farm fence, chickens nearby
15. A vintage movie theater marquee with the words "NOW SHOWING" lit by hundreds of bulbs
16. A chalkboard menu in a cafe listing coffee drinks and prices in neat handwriting
17. A highway road sign reading "Exit 42 Pine Valley" against a clear blue sky
18. A bookstore storefront with gold lettering "Second Chapter Books" on the glass, autumn leaves on the ground
19. A wide alpine valley with a winding river, snow-capped peaks and wildflower meadows, midsummer
20. Sand dunes in the desert at golden hour with sharp ridges and wind ripples
21. A rugged coastline with sea stacks and crashing waves under a stormy sky
22. A terraced rice field on a hillside in morning fog, farmers in conical hats
23. The northern lights over a frozen lake with a small wooden cabin, stars visible
24. A dense tropical rainforest with a waterfall and hanging vines, light filtering through the canopy
25. A cozy Scandinavian living room with a fireplace, sheepskin rug, and large windows showing snow outside
26. A modern open-plan kitchen with a marble island, pendant lights, and potted herbs
27. A cluttered artist studio with canvases, paint tubes, brushes in jars, and afternoon sunlight
28. A grand library reading room with long oak tables, green lamps, and arched ceilings
29. A small Japanese tatami room with a low table, sliding paper doors, and a garden view
30. A product photograph of a matte black wristwatch on a slate surface, dramatic side light
31. A pair of white running shoes floating against a pastel gradient background, studio product shot
32. A glass bottle of olive oil with a sprig of rosemary on a rustic wooden table, soft daylight
33. A flat lay of a leather notebook, fountain pen, glasses, and a cup of espresso on a desk
34. A ceramic teapot with a blue glaze and matching cups arranged on a linen tablecloth
35. A crowded night market in Taipei with food stalls, lanterns, and hundreds of people
36. A packed football stadium during a night match, floodlights and cheering fans
37. Commuters crossing a huge intersection in Tokyo during rush hour, seen from above
38. A music festival crowd at sunset with raised hands and confetti in the air
39. A busy train station concourse with travelers, departure boards, and a glass roof
40. A gothic cathedral facade with intricate stone carvings and a rose window, low angle
41. A row of colorful Victorian townhouses on a hilly street with a cable car
42. A brutalist concrete apartment block with repeating balconies under a pale sky
43. A glass skyscraper reflecting clouds, shot looking straight up from the street
44. A traditional Chinese pagoda beside a lake with cherry blossoms in full bloom
45. An old stone bridge over a canal in Venice with gondolas and laundry lines
46. An astronaut riding a horse across a red desert planet with two moons, cinematic
47. A detailed watercolor illustration of a lighthouse on a rocky island with seagulls
48. A steampunk airship floating above a Victorian city at dusk, intricate brass details

L SAMPLE PAGES

This appendix expands Figures 1 and 6 with full pages of samples; Appendix D.5 gives the selection and crop rules. Figure L.1 sets both modes of SCA beside AReA, UltraImage, and HRDiT on six prompts, and Figure L.2 shows eight methods on two more. Figures L.3 and L.4 set progressive SCA beside AReA, UltraImage, and HRDiT on 18 prompts. Figure L.5 shows single-pass SCA at three aspect ratios, and Figures L.6 to L.9 set it beside AReA on 16 prompts at 4096^2 and four at each wide size.

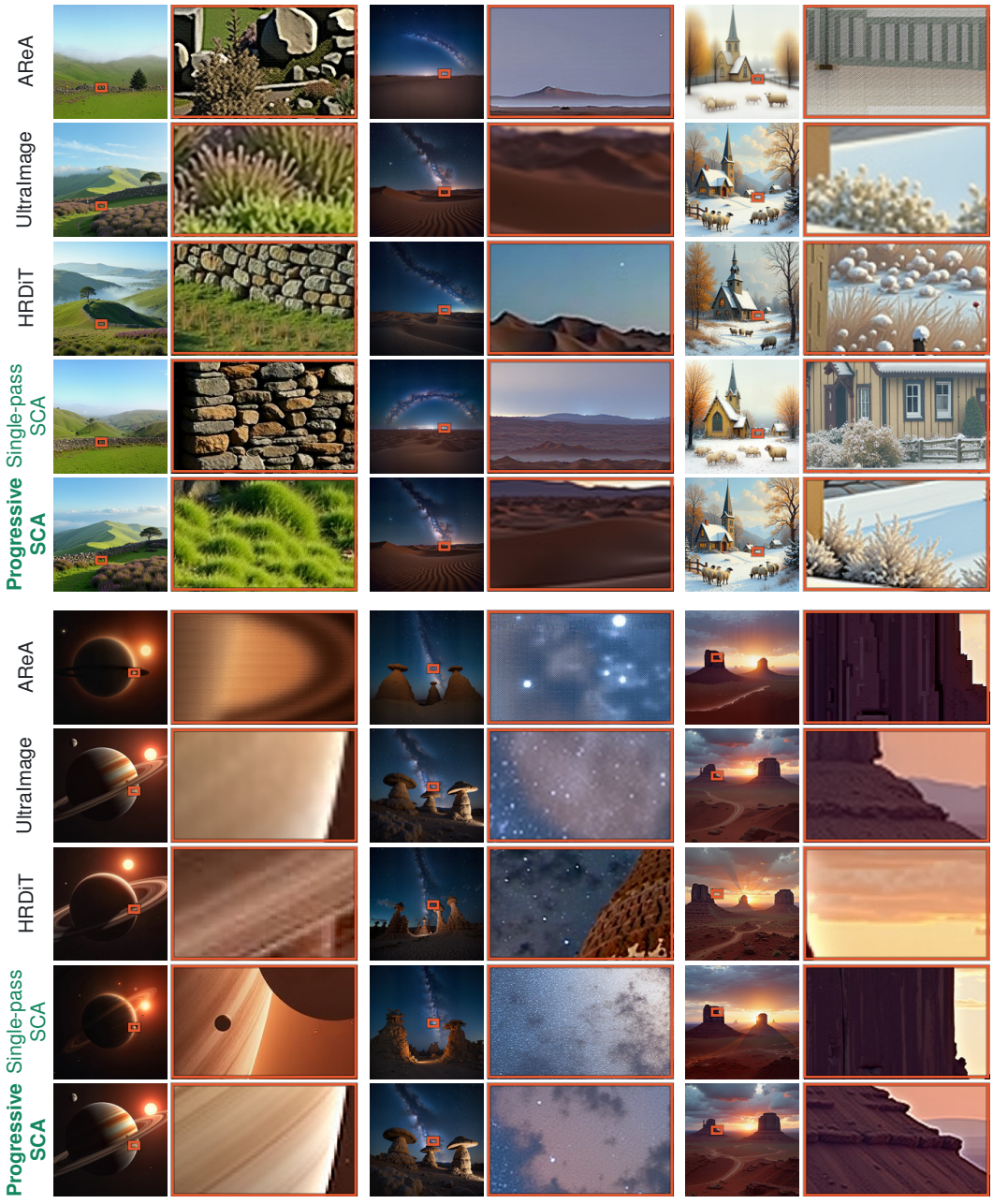


Figure L.1: **Both modes of SCA beside AReA, UltraImage, and HRDiT at 4096^2.** Six evaluation prompts that name no person (keyword rule) outside Figure 6, chosen by a fixed seed (seed 0); seed 0 for every image, 4096^2. Each crop shows native pixels of the window that maximises the smallest edge energy among the five methods. Over all 195 prompts, progressive minus single-pass SCA is -6.0 $[-7.3, -4.6]$ on MUSIQ over four 1024^2 crops and $+0.07$ $[0.01, 0.13]$ on ImageReward (Table F.1).

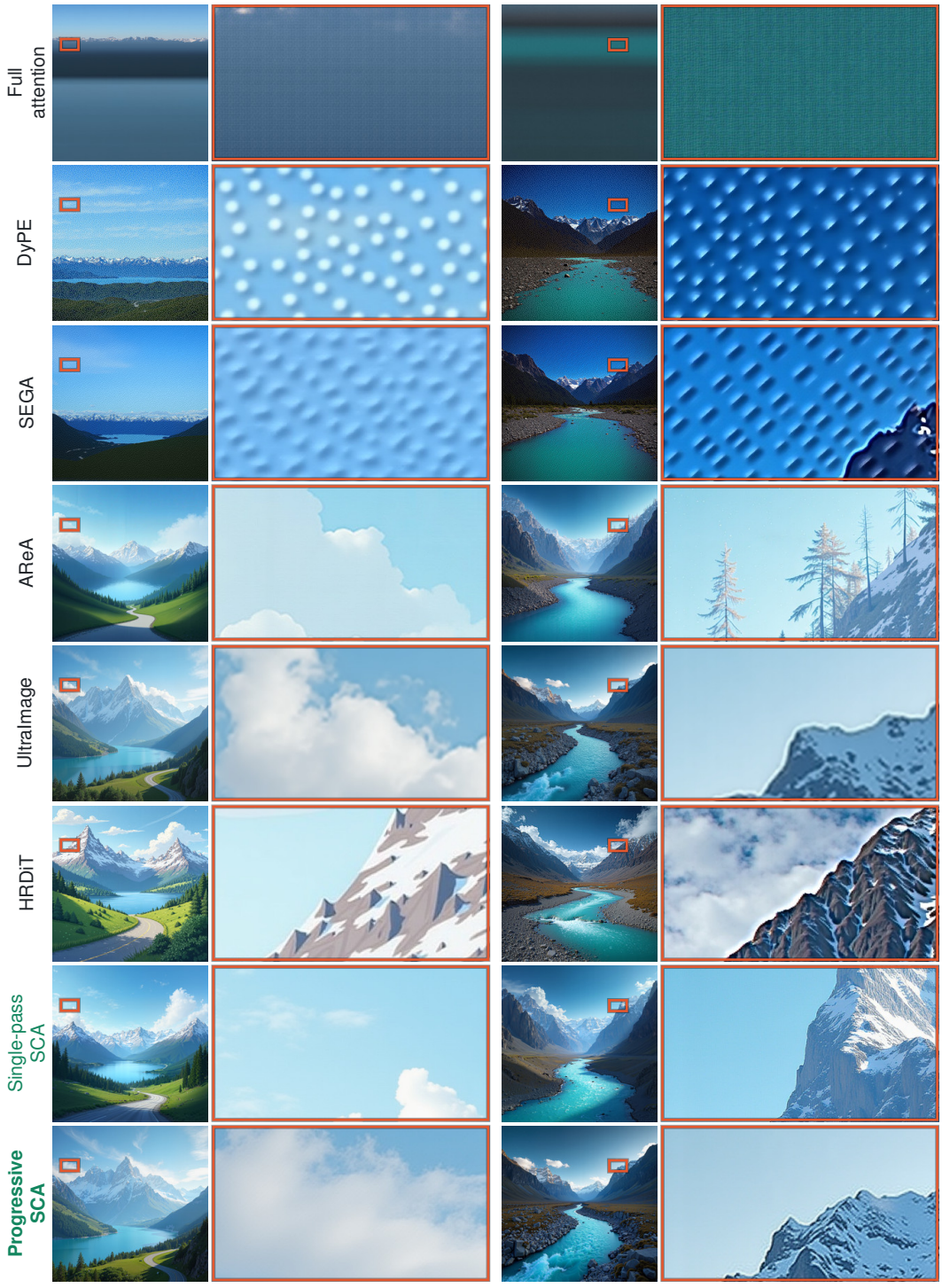


Figure L.2: **Full attention loses the scene and DyPE and SEGA overlay a lattice; the other five keep it.** 4096^2, evaluation prompts that name no person (keyword rule) not shown in an earlier sample figure, chosen by a fixed seed (seed 8); seed 0 for every image. Each crop shows native pixels of the window that maximises the smallest edge energy among the eight methods. Over all 195 prompts, full attention, DyPE, and SEGA each trail progressive SCA on all ten per-image metrics (Table E.2).

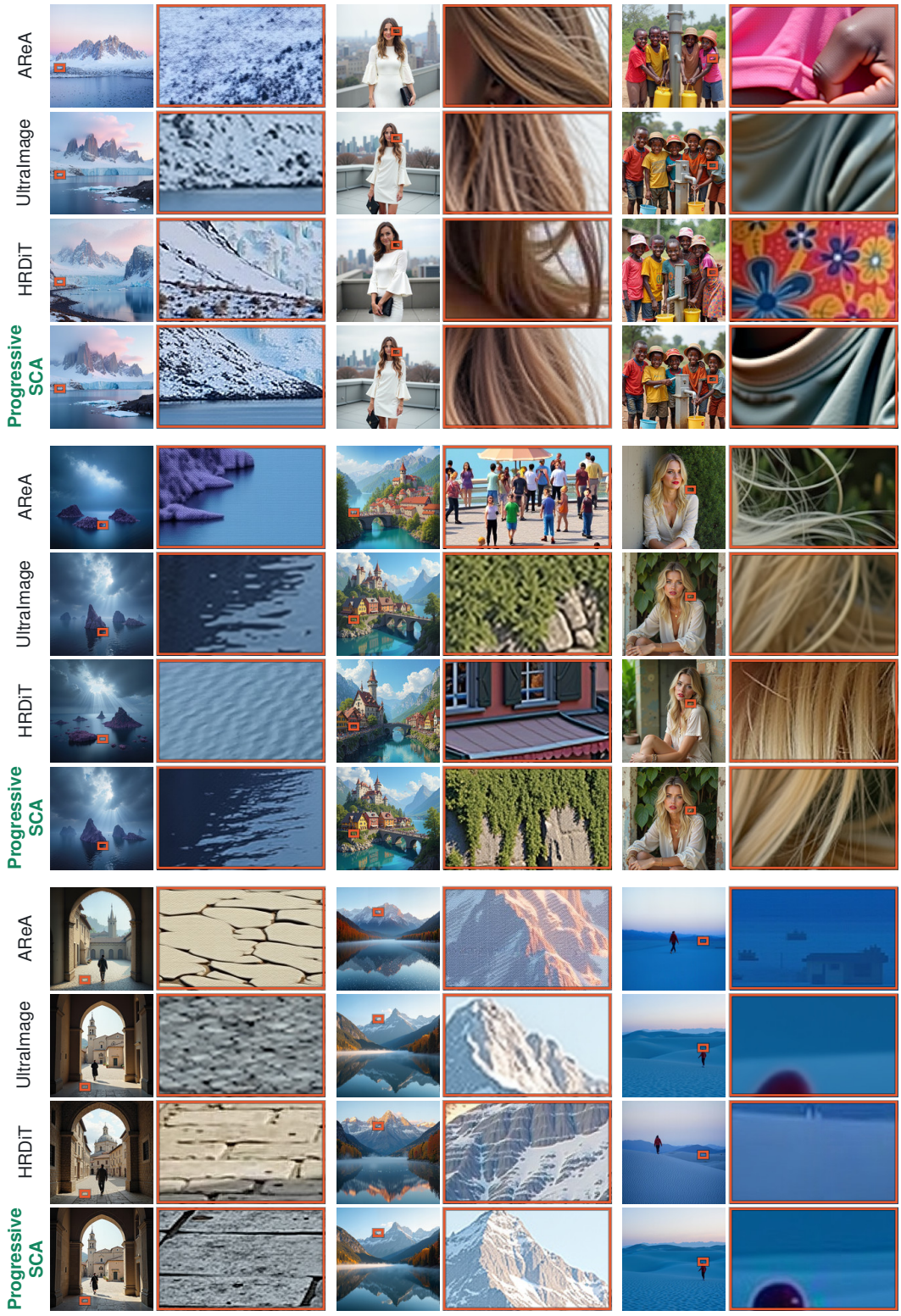


Figure L.3: **Progressive SCA beside AReA, UltraImage, and HRDiT at 4096^2, prompts 1 to 9 of 18.** Evaluation prompts outside Figures 6 and L.1, chosen by a fixed seed (seed 16); seed 0 for every image. Each crop shows native pixels of the window that maximises the smallest edge energy among the four methods. Figure L.4 continues. Over all 195 prompts, progressive SCA's MUSIQ over four 1024^2 crops exceeds AReA's by $+3.7$ $[2.0, 5.4]$ and UltraImage's by $+12.4$ $[11.5, 13.2]$, and is indistinguishable from HRDiT's ($+0.6$ $[-0.7, 1.8]$; Table E.2).

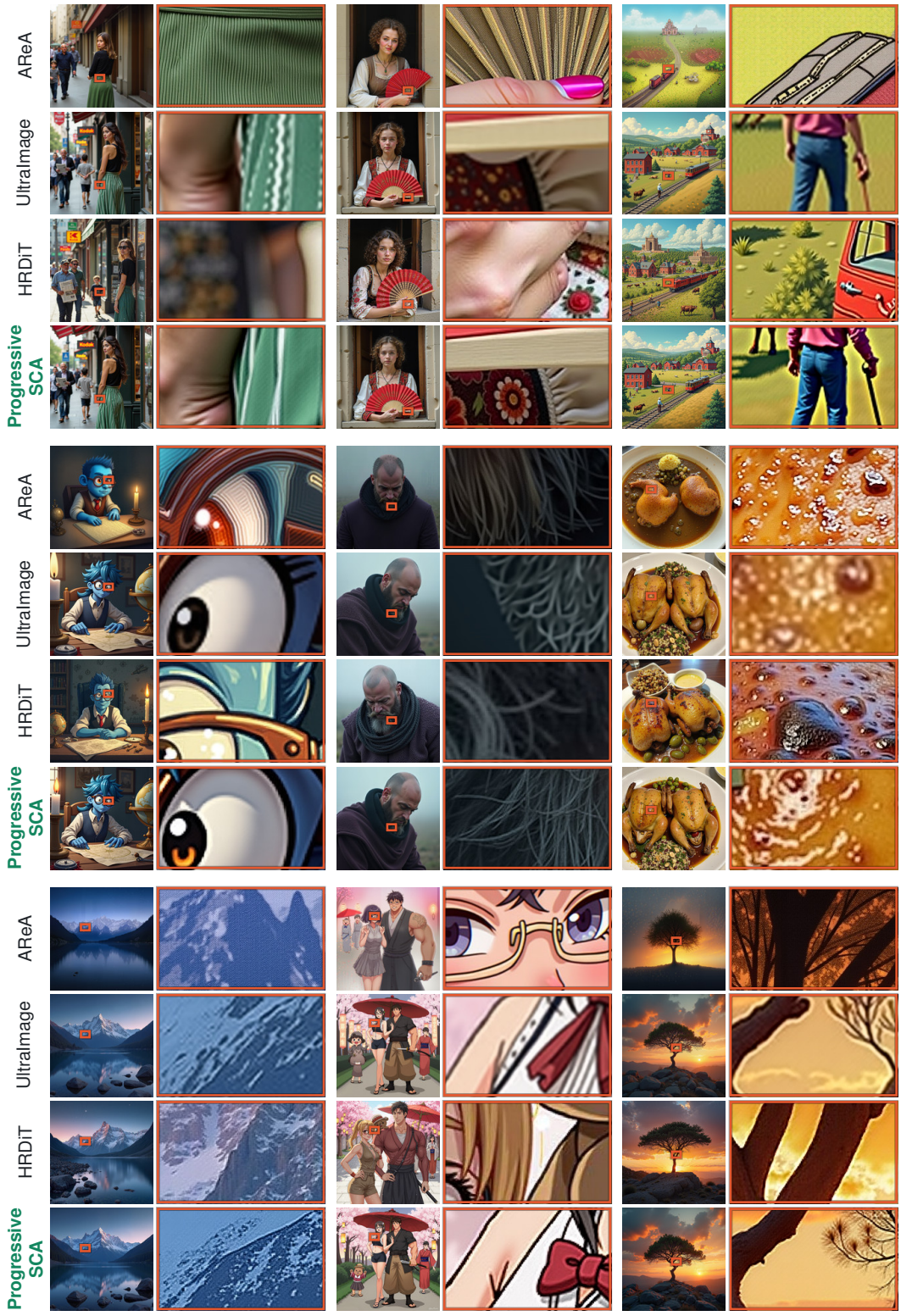


Figure L.4: **Progressive SCA beside AReA, UltraImage, and HRDiT at 4096^2, prompts 10 to 18 of 18.** Evaluation prompts outside Figures 6 and L.1, chosen by a fixed seed (seed 16); seed 0 for every image. Each crop shows native pixels of the window that maximises the smallest edge energy among the four methods.

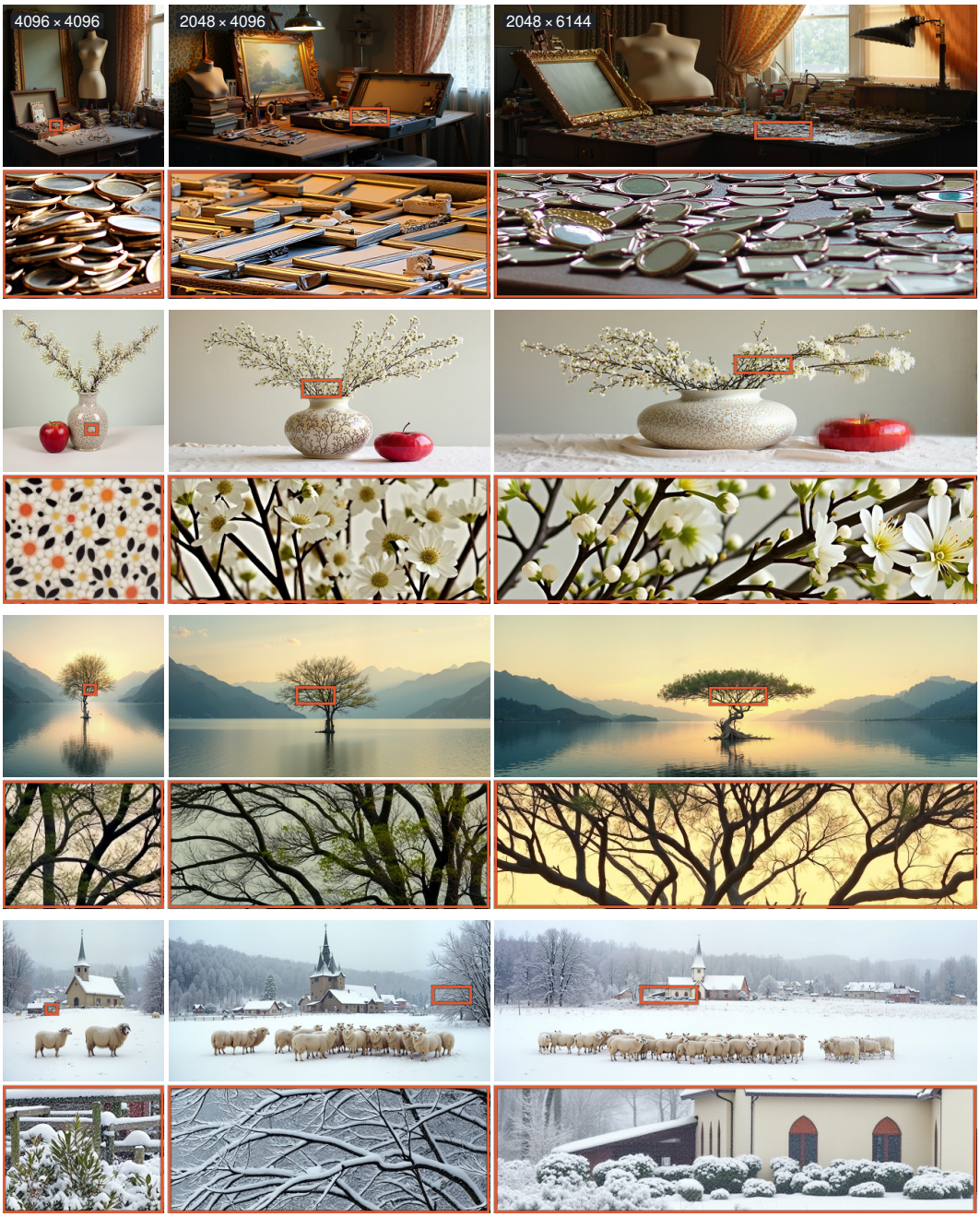


Figure L.5: **Single-pass SCA at three aspect ratios with one reach table.** Single-pass SCA on FLUX.1-dev at 4096^2, 2048×4096 and 2048×6144 for four evaluation prompts that name no person (keyword rule) and no quoted text, chosen by a fixed seed (seed 3); seed 0 for every image. Each crop shows native pixels of the red box above it, at the highest edge energy of that image. Against AReA, single-pass SCA wins seven of 15 metrics at 2048×6144, and wins three and loses three at 2048×4096 (Table G.3).

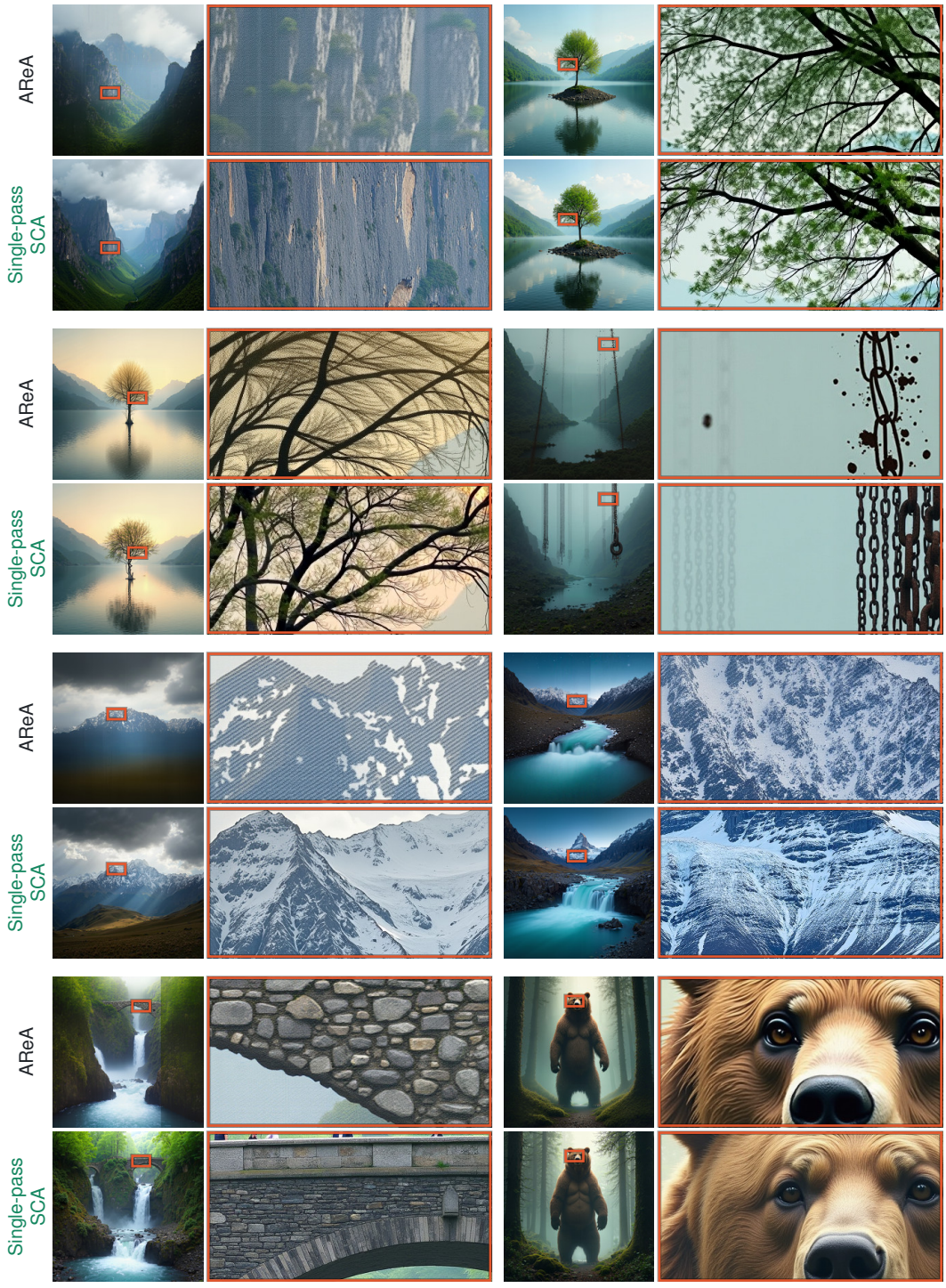


Figure L.6: **Single-pass SCA beside AReA at 4096^2, prompts 1 to 8 of 16.** Evaluation prompts that name no person (keyword rule) not shown in an earlier sample figure, chosen by a fixed seed (seed 0); seed 0 for every image. Each crop shows native pixels of the window that maximises the smallest edge energy among both methods. Over all 195 prompts, single-pass SCA is significantly better than AReA on all ten per-image metrics, both MUSIQ scores among them (Table E.3).

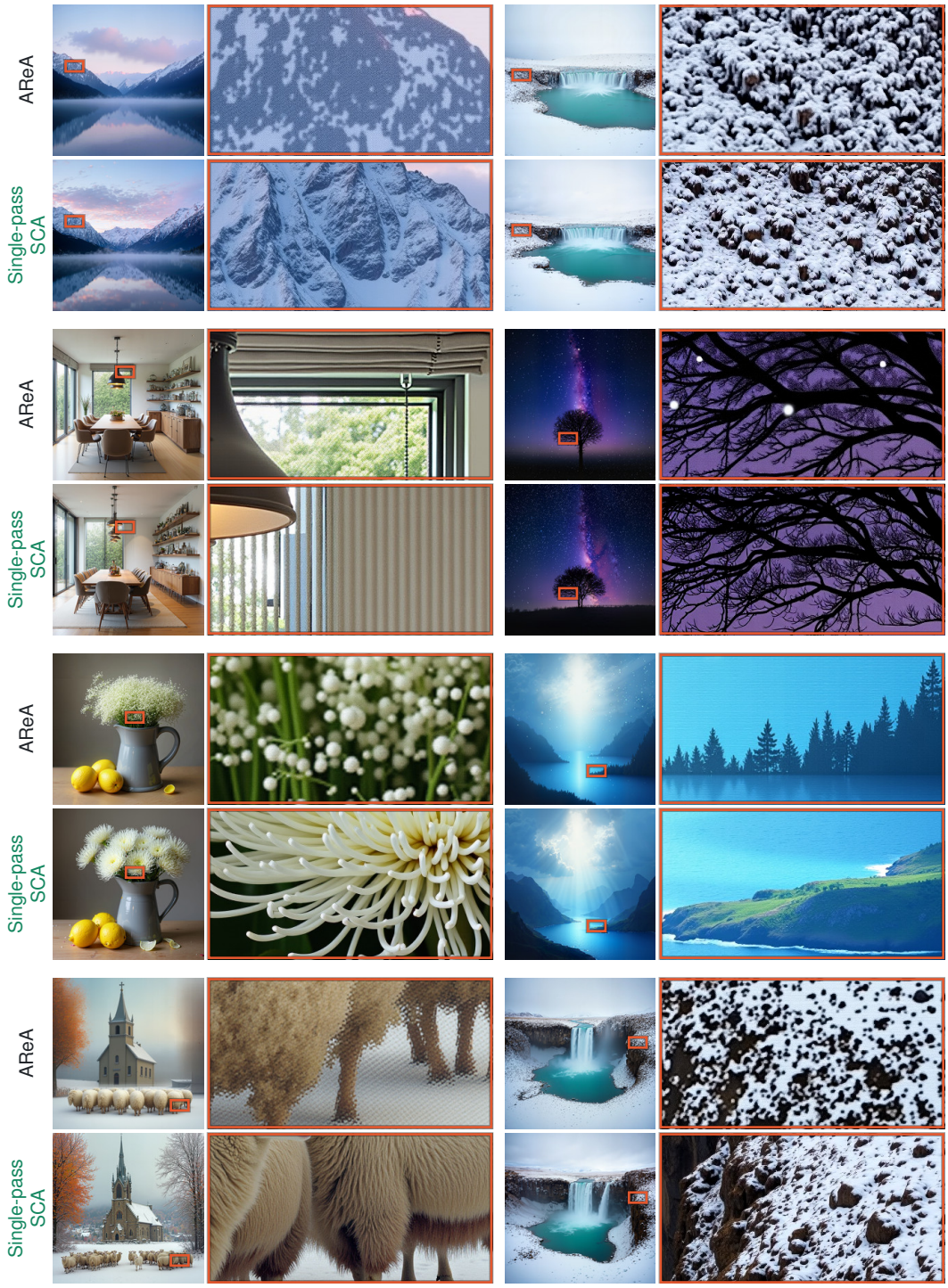


Figure L.7: **Single-pass SCA beside AReA at 4096^2, prompts 9 to 16 of 16.** Evaluation prompts that name no person (keyword rule) not shown in an earlier sample figure, chosen by a fixed seed (seed 0); seed 0 for every image. Each crop shows native pixels of the window that maximises the smallest edge energy among both methods.



Figure L.8: **At 2048 × 4096 both methods keep the layout.** Four evaluation prompts that name no person (keyword rule), chosen by a fixed seed (seed 33); seed 0 for every image. Each crop shows native pixels of the window that maximises the smallest edge energy among both methods. At this size single-pass SCA wins three of 15 metrics against AReA and loses three (Table G.3).



Figure L.9: **Single-pass SCA beside AReA at 2048 × 6144.** Four evaluation prompts that name no person (keyword rule), chosen by a fixed seed (seed 39); seed 0 for every image. Each crop shows native pixels of the window that maximises the smallest edge energy among both methods. At this size single-pass SCA wins seven of 15 metrics against AReA, both MUSIQ scores among them (Table G.3).